2026 | paper1&paper2

Paper 1 · Paper 2 — June 2026 · with mark scheme

Paper 1 — Multiple Choice

9702/11/M/J/26 · 9702/12/M/J/26 · 9702/13/M/J/26 · 9702/14/M/J/26

Paper 2 — AS Structured

9702/21/M/J/26 · 9702/22/M/J/26 · 9702/23/M/J/26 · 9702/24/M/J/26

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Cambridge International AS & A Level

PHYSICS **9702/11**

Paper 1 Multiple Choice **May/June 2026**

1 hour 15 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

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INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid or tape.
- Do **not** write on any bar codes.
- You may use a calculator.

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

This document has **16** pages. Any blank pages are indicated.

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2

Data

acceleration of free fall	$g = 9.81\,\text{m s}^{-2}$
speed of light in free space	$c = 3.00 \times 10^{8}\,\text{m s}^{-1}$
elementary charge	$e = 1.60 \times 10^{-19}\,\text{C}$
unified atomic mass unit	$1\,\text{u} = 1.66 \times 10^{-27}\,\text{kg}$
rest mass of proton	$m_\text{p} = 1.67 \times 10^{-27}\,\text{kg}$
rest mass of electron	$m_\text{e} = 9.11 \times 10^{-31}\,\text{kg}$
Avogadro constant	$N_\text{A} = 6.02 \times 10^{23}\,\text{mol}^{-1}$
molar gas constant	$R = 8.31\,\text{J K}^{-1}\,\text{mol}^{-1}$
Boltzmann constant	$k = 1.38 \times 10^{-23}\,\text{J K}^{-1}$
gravitational constant	$G = 6.67 \times 10^{-11}\,\text{N m}^2\,\text{kg}^{-2}$
permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\,\text{F m}^{-1}$ $\left(\frac{1}{4\pi\varepsilon_0} = 8.99 \times 10^{9}\,\text{m F}^{-1}\right)$
Planck constant	$h = 6.63 \times 10^{-34}\,\text{J s}$
Stefan–Boltzmann constant	$\sigma = 5.67 \times 10^{-8}\,\text{W m}^{-2}\,\text{K}^{-4}$

Formulae

uniformly accelerated motion	$s = ut + \frac{1}{2}at^2$ $v^2 = u^2 + 2as$
hydrostatic pressure	$\Delta p = \rho g \Delta h$
upthrust	$F = \rho g V$
Doppler effect for sound waves	$f_\text{o} = \frac{f_\text{s} v}{v \pm v_\text{s}}$
electric current	$I = Anvq$
resistors in series	$R = R_1 + R_2 + \ldots$
resistors in parallel	$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \ldots$

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3

1 What is an SI base quantity?

A ampere

B charge

C coulomb

D current

2 The surface area of one face of a cube is determined with an uncertainty of ± 1.0%.

This area is used to calculate the volume of the cube.

What is the percentage uncertainty in the calculated volume of the cube?

A ± 1.0% **B** ± 1.5% **C** ± 3.0% **D** ± 6.0%

3 What represents a physical quantity?

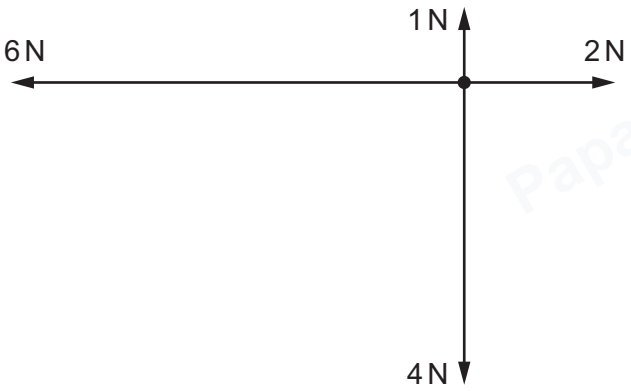
A 3.0

B kilogram

C 7 N

D 40%

4 The diagram shows four forces acting on a small object.



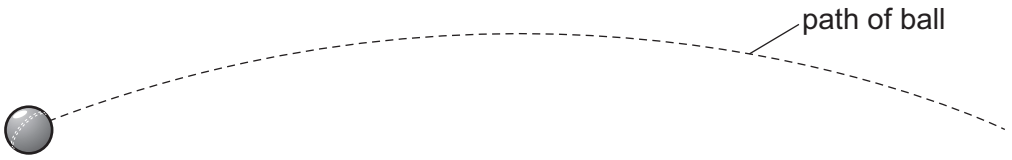
What is the magnitude of the resultant force?

A 1 N **B** 5 N **C** 7 N **D** 13 N

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4

5 A ball is thrown across a flat field.



Air resistance is negligible.

Which statement describes the motion of the ball?

A The ball lands with the same velocity at which it is thrown.

B The horizontal component of acceleration is constant throughout the motion.

C The horizontal and vertical components of acceleration are both zero at the highest point of the motion.

D The horizontal and vertical components of velocity are both zero at the highest point of the motion.

6 What is the definition of velocity?

A distance travelled per unit time

B product of acceleration and change in time

C product of frequency and wavelength

D rate of change of displacement

7 A ball is thrown horizontally with a speed of $10.0\,m\,s^{-1}$ above horizontal ground. The ball hits the ground at a time of 3.0 s after being thrown.

Air resistance is negligible.

What is the speed of the ball just before it hits the ground?

A $10\,m\,s^{-1}$ **B** $29\,m\,s^{-1}$ **C** $31\,m\,s^{-1}$ **D** $39\,m\,s^{-1}$

8 A parent and a child run from rest at the bottom of a stairway to the top of the stairway. Both take the same time.

The mass of the parent is greater than the mass of the child.

Which physical quantity for the parent and for the child has the same numerical value as a result of climbing the stairway?

A average kinetic energy

B average velocity

C gravitational potential energy gained

D power generated

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5

9 A book rests on a table. The book has weight *W*.

Newton's third law describes the relationship between pairs of forces.

Which force pairs with *W* to make a Newton's third law pair of forces?

A the contact force acting on the book due to the table

B the contact force acting on the table due to the book

C the friction force between the book and the table

D the gravitational force acting on the Earth due to the book

10 The mass of a fully grown African elephant is approximately four times the mass of an average car.

The maximum speed of an African elephant is 40 km per hour.

What is a reasonable estimate of the maximum momentum of a fully grown African elephant?

A $5 \times 10^3\,\text{kg m s}^{-1}$

B $7 \times 10^4\,\text{kg m s}^{-1}$

C $9 \times 10^5\,\text{kg m s}^{-1}$

D $2 \times 10^6\,\text{kg m s}^{-1}$

11 A ball of mass 0.20 kg is falling vertically through a liquid and has not reached terminal velocity. The upthrust on the ball is 0.65 N.

At one instant, the drag force on the ball is 0.50 N.

What is the magnitude of the resultant force on the ball at this instant?

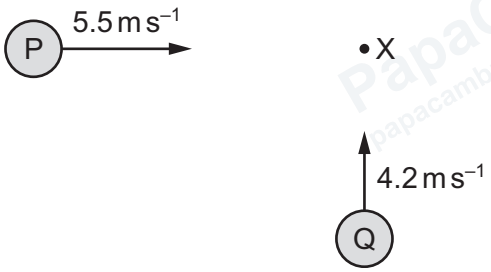
A 0.81 N **B** 1.5 N **C** 1.8 N **D** 2.0 N

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6

12 The diagram shows the view from above of two hard balls rolling along a horizontal table.



The balls each have a mass of 0.16 kg.

The initial velocity of ball P is 5.5 m s^{-1} and the initial velocity of ball Q is 4.2 m s^{-1}.

The initial velocity of ball P is perpendicular to the initial velocity of ball Q.

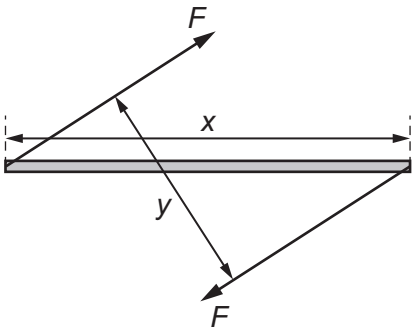
The balls collide at point X where ball Q stops immediately.

There are no resistive forces acting on the balls.

What is the speed of ball P after the collision?

A 1.3 m s^{-1} **B** 5.5 m s^{-1} **C** 6.9 m s^{-1} **D** 9.7 m s^{-1}

13 Two forces, each of magnitude *F*, act on a beam of length *x* as shown.



The perpendicular distance between the forces is *y* and the two forces form a couple.

What is the torque of the couple?

A *Fx* **B** 2*Fx* **C** *Fy* **D** 2*Fy*

14 An object is submerged in a liquid.

Which statement about the upthrust acting on the object is correct?

A It is proportional to the density of the liquid.

B It is proportional to the density of the object.

C It is proportional to the total depth of the liquid.

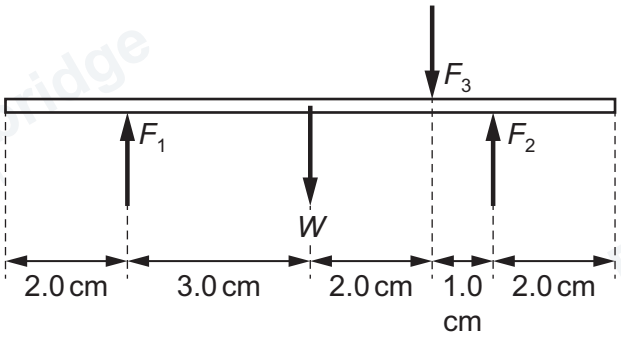
D It is proportional to the weight of the object.

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7

15 A horizontal uniform rod of length 10.0 cm is shown.



The rod is acted on by its weight W and by three forces F_1, F_2 and F_3.

The rod is in equilibrium. All the forces are vertical.

$$F_2 = \frac{4}{3}F_3$$

What is the ratio $\frac{F_1}{F_2}$?

A $\frac{1}{4}$ **B** $\frac{1}{2}$ **C** 1 **D** 2

16 What is a unit for density?

A $\mathrm{N\,m^{-3}}$ **B** $\mathrm{g\,mm^{-1}}$ **C** $\mathrm{kg\,cm^{-2}}$ **D** $\mathrm{\mu g\,mm^{-3}}$

17 What is meant by the efficiency of a system?

A the difference between the useful energy output from the system and the total energy input

B the difference between the useful energy output from the system and the wasted energy output

C the ratio of the useful energy output from the system to the total energy input

D the ratio of the useful energy output from the system to the wasted energy output

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8

18 Object X is at a small height h above the surface of the Earth and then falls from rest to the surface.

The speed of object X when it reaches the surface of the Earth is v_X.

Object Y is at a small height $2h$ above the surface of the Moon and then falls from rest to the surface.

The speed of object Y when it reaches the surface of the Moon is v_Y.

The acceleration of free fall on the Moon is $1.6\,\text{m}\,\text{s}^{-2}$.

The drag force acting on both objects is negligible.

What is the ratio $\frac{v_X}{v_Y}$?

A 1.0 **B** 1.8 **C** 2.5 **D** 3.1

19 Which equation is **not** required to derive the formula $P = Fv$?

A $W = Fs$ **B** $v = \frac{s}{t}$ **C** $F = ma$ **D** $P = \frac{W}{t}$

20 A metal wire is stretched by a tensile force.

The tensile force is increased.

The volume of the wire remains constant and the limit of proportionality is not exceeded.

Which quantity remains constant?

A the cross-sectional area of the wire

B the strain of the wire

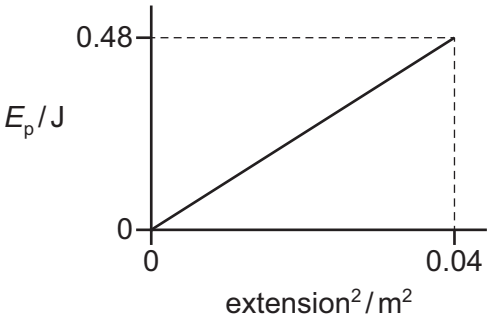
C the stress in the wire

D the Young modulus of the metal

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21 The graph shows the variation of the elastic potential energy E_p stored in a spring with the square of the extension of the spring.

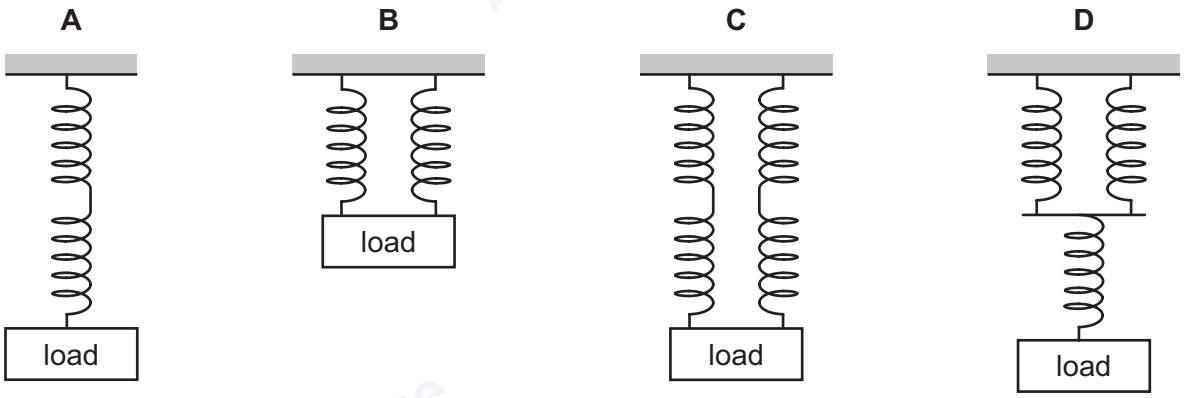


What is the spring constant of the spring?

A $0.0096\,\text{N m}^{-1}$ **B** $6.0\,\text{N m}^{-1}$ **C** $12\,\text{N m}^{-1}$ **D** $24\,\text{N m}^{-1}$

22 Identical springs are joined in four arrangements.

Which arrangement has the same spring constant as a single spring?



23 A vertically polarised electromagnetic wave has an intensity of $3.2 \times 10^4\,\text{W m}^{-2}$.

The wave is incident on a polarising filter that has a transmission axis at an angle θ to the vertical.

The transmitted wave has an intensity of $2.5 \times 10^4\,\text{W m}^{-2}$.

What is θ?

A 28° **B** 39° **C** 52° **D** 62°

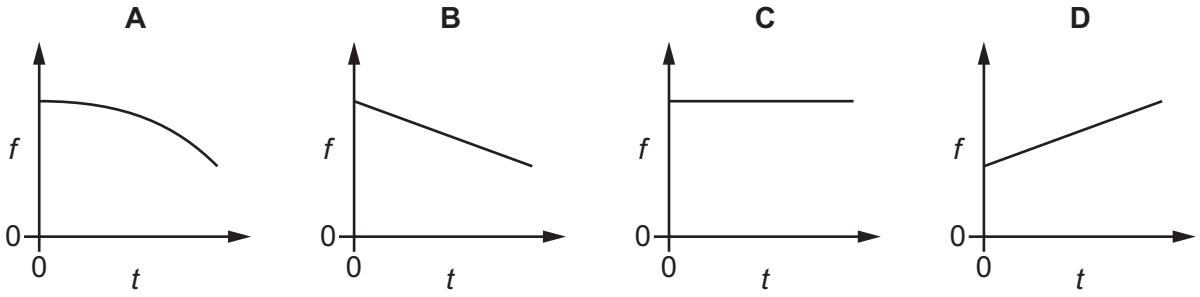
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24 A train is travelling with a constant velocity.

It operates its whistle as it moves directly towards a student.

The whistle emits a continuous sound of constant frequency.

Which graph shows the variation with time t of the frequency f of the sound heard by the student?



25 An electromagnetic wave in free space has a wavelength of 0.1 nm.

In which principal region of the electromagnetic spectrum does this wave lie?

A infrared

B ultraviolet

C visible

D X-ray

26 A wave is incident on a surface of area S placed perpendicular to the direction of travel of the wave.

The wave has amplitude R and the power incident on the surface is P.

The amplitude of the wave is increased to $3R$ and the wave is now incident normally on a surface of area $3S$.

What is the power incident on the surface of area $3S$?

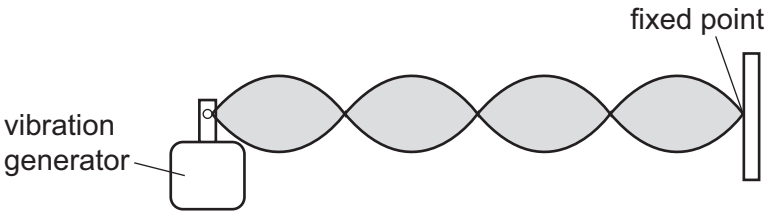
A P **B** $3P$ **C** $9P$ **D** $27P$

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11

27 A string is stretched between a vibration generator and a fixed point.

When the vibration generator is vibrating at a frequency *f*, a stationary wave with five nodes is created on the stretched string, as shown. There is a node at the end of the string that is attached to the vibration generator.

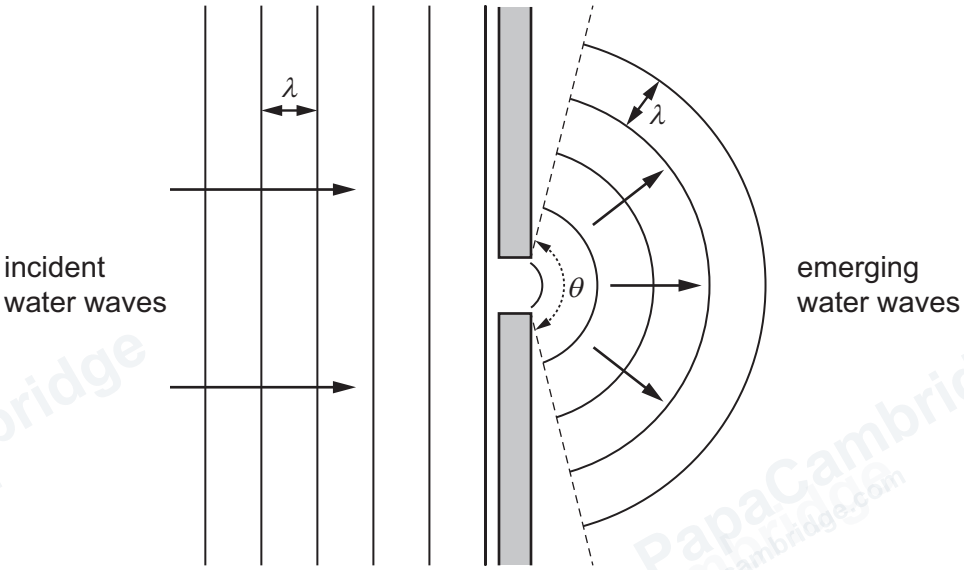


The frequency of vibration of the vibration generator is slowly increased.

What is the next frequency that produces a stationary wave on the string?

A 1.25*f* **B** 1.50*f* **C** 1.75*f* **D** 2.00*f*

28 Water waves of wavelength λ are incident normally on an obstacle with a narrow gap. The width of the gap is equal to λ. The waves from the gap emerge over an angle θ as shown.



The gap is slowly widened.

Which changes, if any, occur to θ and to the wavelength of the emerging waves?

	θ	wavelength
A	decreases	remains the same
B	increases	remains the same
C	remains the same	decreases
D	remains the same	increases

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12

29 What is a condition for two sources of waves to produce an interference pattern?

A The waves must be electromagnetic waves.

B The waves must be emitted with a constant phase difference.

C The waves must be emitted with the same amplitude.

D The waves must be emitted with the same intensity.

30 Green light of wavelength 550 nm passes through a double slit. This produces an interference pattern with a fringe spacing of 5.0 cm on a screen.

Which changes to the experimental arrangement result in a fringe spacing of 2.5 cm?

	slit separation	distance between slits and screen
A	doubled	unchanged
B	unchanged	doubled
C	halved	doubled
D	doubled	halved

31 What is the definition of the potential difference (p.d.) across a component?

A the energy transferred per unit charge

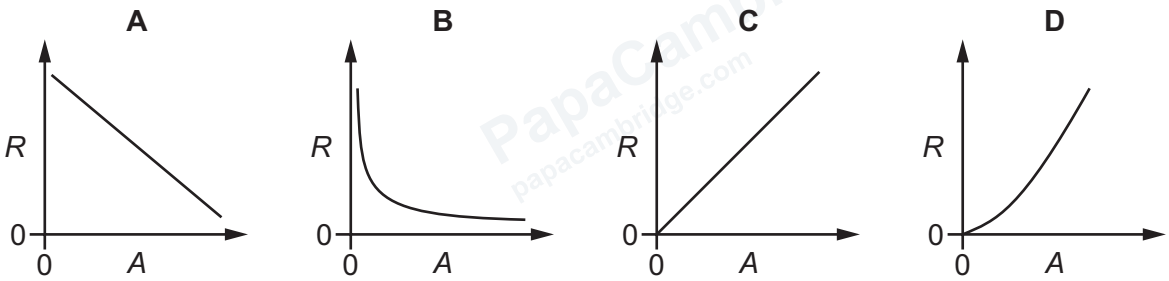
B the energy transferred per unit current

C the power transferred per unit charge

D the power transferred per unit current

32 Several wires of the same fixed length are made of metal of constant resistivity. The wires have different cross-sectional areas.

Which graph represents the variation of the resistance *R* of the wires with their cross-sectional area *A*?



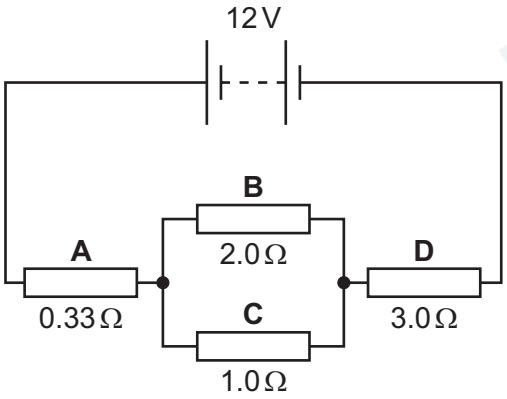
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13

33 The diagram shows a circuit.

Which resistor dissipates the least power?



34 There is an electric current of 10 mA in a conductor.

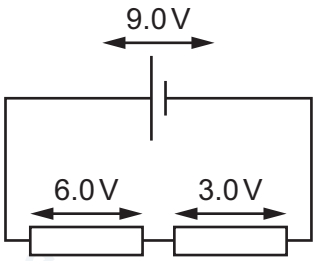
What is the charge that passes a fixed point in the conductor in a time of 50 s?

A 0.5 C **B** 5 C **C** 500 C **D** 5000 C

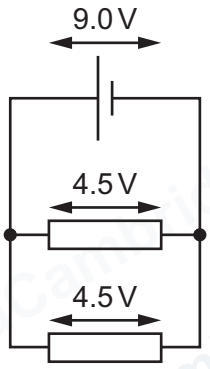
35 In the diagrams shown, the cells have negligible internal resistance and the wires have negligible resistance.

Which diagram demonstrates Kirchhoff's second law?

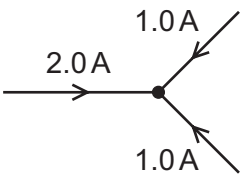
A



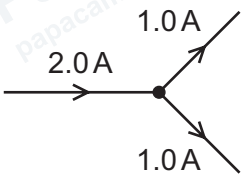
B



C



D

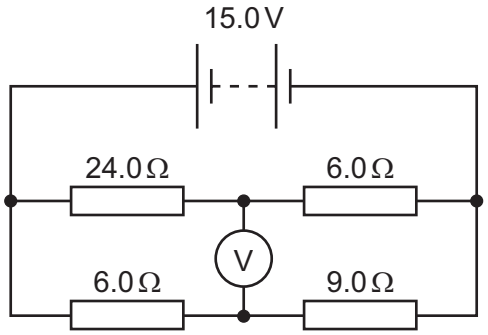


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14

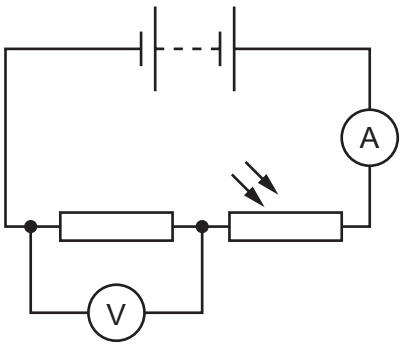
36 A circuit consists of a battery, a high-resistance voltmeter and four fixed resistors as shown. The battery has an electromotive force (e.m.f.) of 15.0 V and negligible internal resistance.



What is the reading on the voltmeter?

A 3.0 V **B** 6.0 V **C** 9.0 V **D** 12.0 V

37 The diagram shows a resistor connected in series with a light-dependent resistor (LDR) and a battery.



The light intensity incident on the LDR increases.

What is the effect of this change on the voltmeter and ammeter readings?

	voltmeter reading	ammeter reading
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

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15

38 A stream of identical particles flows between two points.

The stream of particles does **not** form an electric current.

What could the particles be?

A α-particles

B β^- particles

C neutrons

D protons

39 $^{10}_{6}C$ decays by positron emission.

Which nuclide is the product of the decay?

A $^{10}_{5}B$ **B** $^{11}_{5}B$ **C** $^{10}_{6}C$ **D** $^{11}_{6}C$

40 A positively charged meson consists of a quark and an antiquark.

What could be the quark and antiquark?

A charm and antiup

B down and antitop

C strange and antibottom

D up and antistrange

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CAMBRIDGE
International Education

Cambridge International AS & A Level

PHYSICS **9702/11**

Paper 1 Multiple Choice **May/June 2026**

MARK SCHEME

Maximum Mark: 40

Published

This mark scheme shows the correct answer to each question.

We cannot discuss the mark scheme with schools, and we recommend that schools read the mark scheme together with the question paper and the Principal Examiner Report for Teachers.

This document consists of **3** printed pages.

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[Turn over

Question	Answer	Marks
1	D	**1**
2	B	**1**
3	C	**1**
4	B	**1**
5	B	**1**
6	D	**1**
7	C	**1**
8	B	**1**
9	D	**1**
10	B	**1**
11	A	**1**
12	C	**1**
13	C	**1**
14	A	**1**
15	B	**1**
16	D	**1**
17	C	**1**
18	B	**1**
19	C	**1**
20	D	**1**
21	D	**1**
22	C	**1**
23	A	**1**
24	C	**1**
25	D	**1**
26	D	**1**
27	A	**1**
28	A	**1**

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Question	Answer	Marks
29	B	**1**
30	A	**1**
31	A	**1**
32	B	**1**
33	B	**1**
34	A	**1**
35	A	**1**
36	B	**1**
37	D	**1**
38	C	**1**
39	A	**1**
40	D	**1**

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PHYSICS **9702/12**

Paper 1 Multiple Choice **May/June 2026**

1 hour 15 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid or tape.
- Do **not** write on any bar codes.
- You may use a calculator.

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

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2

Data

acceleration of free fall	$g = 9.81\,\text{m s}^{-2}$
speed of light in free space	$c = 3.00 \times 10^{8}\,\text{m s}^{-1}$
elementary charge	$e = 1.60 \times 10^{-19}\,\text{C}$
unified atomic mass unit	$1\,\text{u} = 1.66 \times 10^{-27}\,\text{kg}$
rest mass of proton	$m_\text{p} = 1.67 \times 10^{-27}\,\text{kg}$
rest mass of electron	$m_\text{e} = 9.11 \times 10^{-31}\,\text{kg}$
Avogadro constant	$N_\text{A} = 6.02 \times 10^{23}\,\text{mol}^{-1}$
molar gas constant	$R = 8.31\,\text{J K}^{-1}\,\text{mol}^{-1}$
Boltzmann constant	$k = 1.38 \times 10^{-23}\,\text{J K}^{-1}$
gravitational constant	$G = 6.67 \times 10^{-11}\,\text{N m}^2\,\text{kg}^{-2}$
permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\,\text{F m}^{-1}$ $\left(\frac{1}{4\pi\varepsilon_0} = 8.99 \times 10^{9}\,\text{m F}^{-1}\right)$
Planck constant	$h = 6.63 \times 10^{-34}\,\text{J s}$
Stefan–Boltzmann constant	$\sigma = 5.67 \times 10^{-8}\,\text{W m}^{-2}\,\text{K}^{-4}$

Formulae

uniformly accelerated motion	$s = ut + \frac{1}{2}at^2$ $v^2 = u^2 + 2as$
hydrostatic pressure	$\Delta p = \rho g \Delta h$
upthrust	$F = \rho g V$
Doppler effect for sound waves	$f_\text{o} = \frac{f_\text{s} v}{v \pm v_\text{s}}$
electric current	$I = Anvq$
resistors in series	$R = R_1 + R_2 + \ldots$
resistors in parallel	$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \ldots$

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3

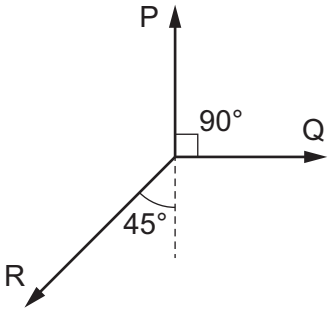
1 Which row correctly identifies scalar and vector quantities?

	scalar quantities	vector quantities
A	energy and work	momentum and power
B	energy and momentum	force and power
C	energy and power	force and momentum
D	momentum and power	force and velocity

2 What are the base units for intensity?

A $\mathrm{kg\,s^{-3}}$ **B** $\mathrm{kg\,s}$ **C** $\mathrm{kg\,m^2\,s^{-3}}$ **D** $\mathrm{kg\,m^4\,s^{-3}}$

3 Three coplanar forces, P, Q and R, act at a point.



The three forces are in equilibrium.

The magnitude of P is increased by 10%. The directions of the forces are unchanged.

Which percentage increase, if any, is required to the magnitudes of Q and R for the forces to be once again in equilibrium?

	% increase in magnitude of Q	% increase in magnitude of R
A	10	10
B	0	10
C	$10\sqrt{2}$	$10\sqrt{2}$
D	10	$10\sqrt{2}$

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4

4 A metal rod is exactly 62.00 cm long. The length of the rod is measured using two different devices, X and Y.

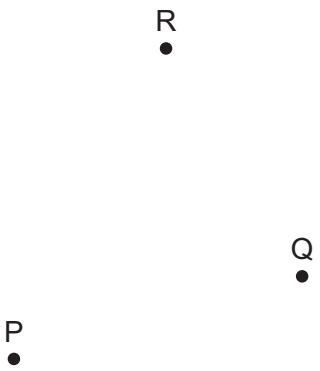
The reading on device X is 61.5 cm.

The reading on device Y is 62.65 cm.

Which device gives the more precise reading for the length, and which device gives the more accurate reading for the length?

	more precise reading	**more accurate reading**
A	X	X
B	X	Y
C	Y	X
D	Y	Y

5 One object moves directly from P to R.



In a shorter time, a second object moves from P to Q to R.

Which statement about the two objects is correct for the journey from P to R?

A They have the same average speed.

B They have the same average velocity.

C They have the same displacement.

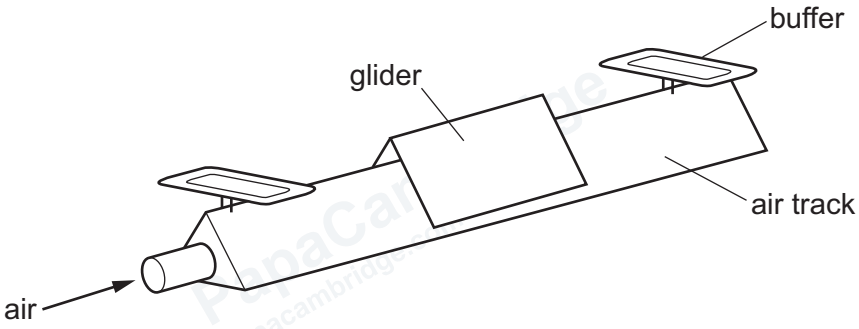
D They travel the same distance.

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5

6 A small glider moves along a horizontal air track as shown.

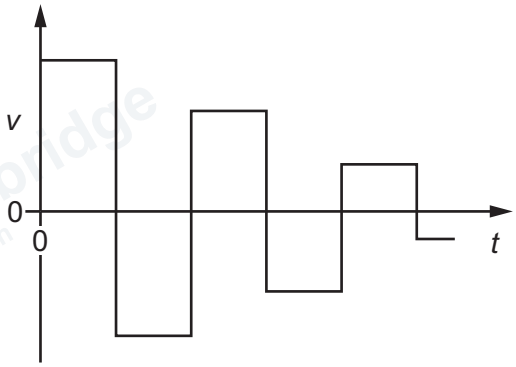


At each end of the air track, the glider has a perfectly elastic collision with a fixed buffer.

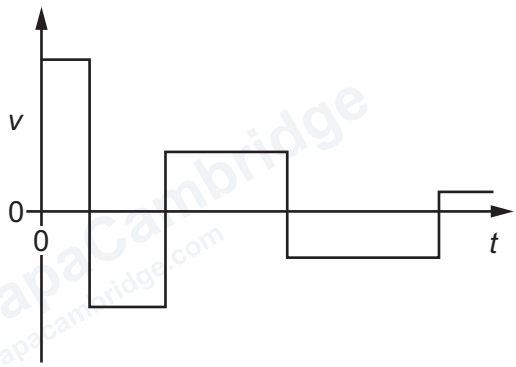
The glider moves at a constant speed between collisions.

Which graph represents the variation with time *t* of the velocity *v* of the glider as it moves between the two buffers?

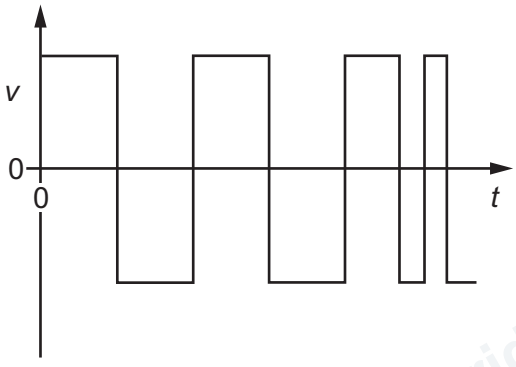
A



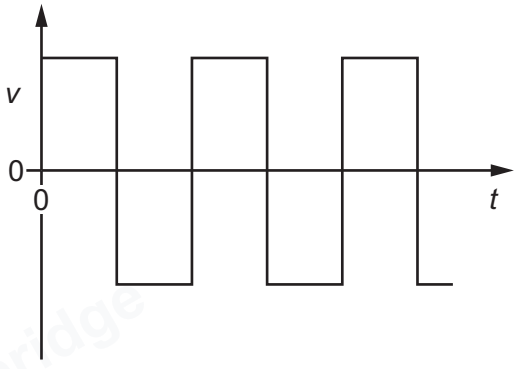
B



C



D



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6

7 The diagram shows an arrangement to stop trains that are travelling too fast.



Trains coming from the left travel at a speed of 50 m s^{-1}. At marker 1, the driver must apply the brakes so that the train decelerates uniformly in order to pass marker 2 at no more than 10 m s^{-1}.

The train carries a detector that notes the times when the train passes each marker and will apply an emergency brake if the time between passing marker 1 and marker 2 is less than 20 s.

How far from marker 2 should marker 1 be placed?

A 200 m **B** 400 m **C** 500 m **D** 600 m

8 An experiment is performed to measure the acceleration of free fall *g*. A body falls between two fixed points. Four measurements are taken.

Which measurement is **not** required for the calculation of *g*?

A the distance fallen by the body

B the initial velocity of the body

C the mass of the body

D the time taken for the body to fall

9 What is the definition of linear momentum?

A force per unit time

B mass per unit velocity

C product of force and time

D product of mass and velocity

10 A hollow steel object is dropped from rest in air. The object reaches a terminal velocity *v* before hitting the ground.

Which change results in an increase in the terminal velocity?

A Drop a solid steel object of the same dimensions from the same height.

B Drop the object from a greater height.

C Throw the object downwards from the same height with an initial velocity 0.5*v*.

D Throw the object upwards from the same height with an initial velocity 0.5*v*.

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11 A bowling ball of mass 4.50 kg falls a short distance vertically into a tray of sand.

The bowling ball decelerates uniformly in the sand.

The deceleration of the bowling ball when it is in contact with the sand is $8.00 \times 10^2\,\text{m s}^{-2}$.

What is the magnitude of the force exerted on the bowling ball by the sand as it decelerates?

A 44.1 N **B** 3560 N **C** 3600 N **D** 3640 N

12 Which statement is **not** a requirement of a pair of forces that obey Newton's third law of motion?

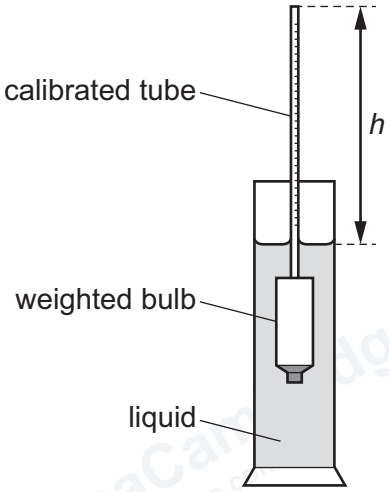
A The forces act in opposite directions.

B The forces act on different objects.

C The forces act on objects in contact.

D The forces are of equal magnitude.

13 A hydrometer is a device for measuring the density of liquids. It consists of a weighted bulb at the bottom of a calibrated tube.



The hydrometer is floating in a liquid at room temperature. The length of the tube above the level of the liquid is *h*.

Which change causes *h* to decrease?

A adding more liquid into the cylinder

B decreasing the density of the liquid

C increasing the air pressure above the liquid

D increasing the density of the liquid

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8

14 Torque is the result of a pair of forces acting to cause a turning effect. Torque is calculated as the product of force and distance.

Which row gives the correct description of the force and the distance used to calculate the torque of a couple?

	force	distance
A	the magnitude of one force	the perpendicular distance between the two forces
B	the sum of the magnitudes of both forces	the perpendicular distance between the two forces
C	the magnitude of one force	the perpendicular distance from one force to the mid-point between the forces
D	the resultant of the two forces	the perpendicular distance from one force to the mid-point between the forces

15 The density of paper is 800 kg m^{-3}. A typical sheet of paper has a width of 210 mm and a length of 300 mm.

The thickness of a pack of 500 sheets of paper is 50 mm.

What is the mass of a single sheet of paper?

A 0.5 g **B** 5 g **C** 50 g **D** 500 g

16 Which statement about equilibrium is correct?

A If a system is in equilibrium then there must be no forces acting on it.

B If a system is in equilibrium then there must be no resultant torque on it.

C If there are no moments due to forces acting on a system then it must be in equilibrium.

D If there is no resultant force acting on a system then it must be in equilibrium.

17 An object of weight 3.0 N is moved vertically at a constant velocity *v* by an electric motor.

The electric motor has an efficiency of 25%.

The object travels a vertical distance of 1.2 m in 3.0 s.

Assume that the useful output energy of the motor is equal to the increase in gravitational potential energy of the object.

What is the electric power supplied to the motor?

A 0.3 W **B** 1.2 W **C** 4.8 W **D** 30 W

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9

18 An alpha particle has 2.2×10^{-13} J of kinetic energy.

What is the speed of the alpha particle?

A $4.1 \times 10^{6}\,\text{m s}^{-1}$

B $5.8 \times 10^{6}\,\text{m s}^{-1}$

C $8.1 \times 10^{6}\,\text{m s}^{-1}$

D $1.2 \times 10^{7}\,\text{m s}^{-1}$

19 An initially unstretched spring is stretched within its limit of proportionality to a length of 80 mm.

The work done stretching the spring is 0.010 J.

The spring constant of the spring is $22.5\,\text{N m}^{-1}$.

What is the unstretched length of the spring?

A 30 mm **B** 50 mm **C** 60 mm **D** 110 mm

20 What is meant by elastic deformation of a wire under tension?

A deformation for which the elastic potential energy is proportional to the extension

B deformation for which the extension is proportional to the tensile force

C deformation for which the extension remains constant as the tensile force is removed

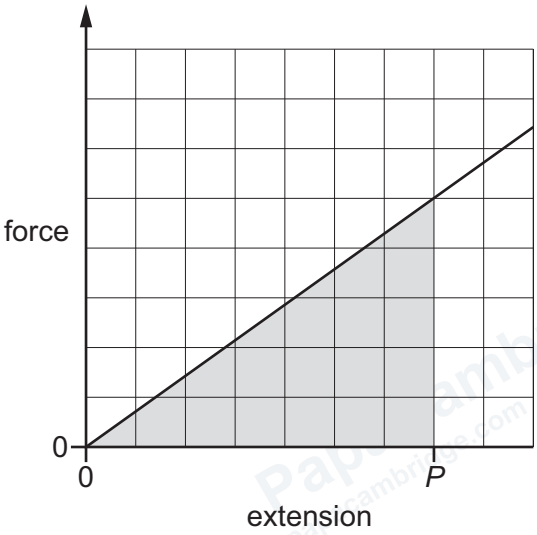
D deformation for which the extension returns to zero when the tensile force is removed

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10

21 The graph shows the variation of force with extension for a metal object deformed within its limit of proportionality.



What does the shaded area represent?

A the elastic limit of the object

B the elastic potential energy stored in the object at an extension of *P*

C the spring constant of the object

D the Young modulus of the metal at an extension of *P*

22 What is the order of these principal regions of the electromagnetic spectrum from shortest to longest wavelength?

A infrared radiation, visible light, ultraviolet radiation

B ultraviolet radiation, infrared radiation, visible light

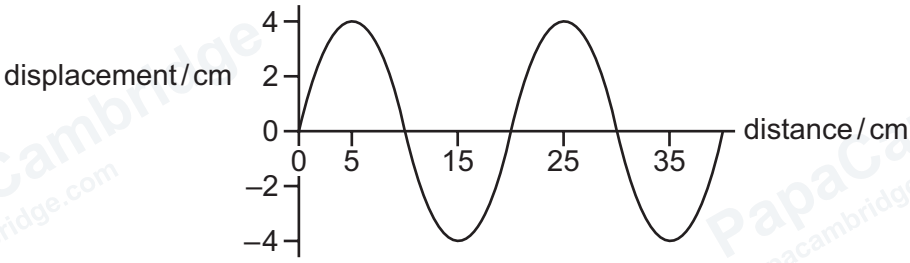
C ultraviolet radiation, visible light, infrared radiation

D visible light, ultraviolet radiation, infrared radiation

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23 The graph shows the displacement of each particle in a transverse wave against distance at one instant in time.



The speed of the wave is $1.6\,\text{m}\,\text{s}^{-1}$.

What is the frequency of the wave?

A 0.050 Hz **B** 0.080 Hz **C** 5.0 Hz **D** 8.0 Hz

24 The siren on an emergency vehicle emits a sound of constant frequency 1100 Hz.

The driver in a stationary car hears the sound at a frequency of 1000 Hz.

The speed of sound in air is $340\,\text{m}\,\text{s}^{-1}$.

What is the velocity of the emergency vehicle?

A $31\,\text{m}\,\text{s}^{-1}$ away from the stationary car

B $34\,\text{m}\,\text{s}^{-1}$ away from the stationary car

C $31\,\text{m}\,\text{s}^{-1}$ towards the stationary car

D $34\,\text{m}\,\text{s}^{-1}$ towards the stationary car

25 A source of sound of constant power P is situated in an open space. The intensity I of sound at distance r from this source is given by the equation shown.

$$I = \frac{P}{4\pi r^2}$$

How does the amplitude a of the vibrating air molecules vary with the distance r from the source?

A $a \propto \frac{1}{r}$ **B** $a \propto \frac{1}{r^2}$ **C** $a \propto r$ **D** $a \propto r^2$

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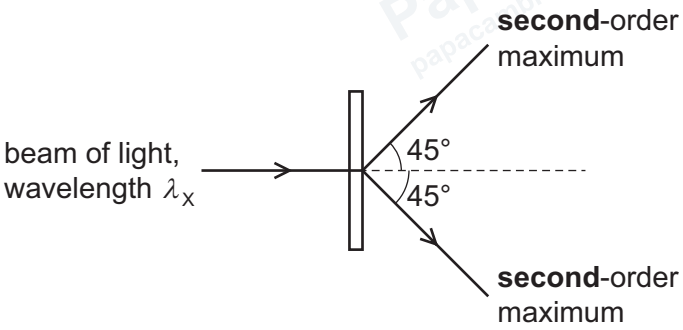
12

26 Water waves diffract as they pass through a gap.

Which combination produces the greatest amount of diffraction?

	wavelength / m	gap width / m
A	10^{-1}	10^{0}
B	10^{-1}	10^{-2}
C	10^{0}	10^{0}
D	10^{0}	10^{2}

27 The diagram shows a beam of light of wavelength λ_X that is incident normally on a diffraction grating.



The diagram shows only the **second**-order intensity maxima that are produced at an angle of 45° to the direction of the incident beam of light.

A replacement beam of light of a different wavelength λ_Y is now incident normally on the diffraction grating. This new wavelength also produces intensity maxima that are at an angle of 45° to the direction of the incident beam of light.

What is a possible value of the ratio $\frac{\lambda_X}{\lambda_Y}$?

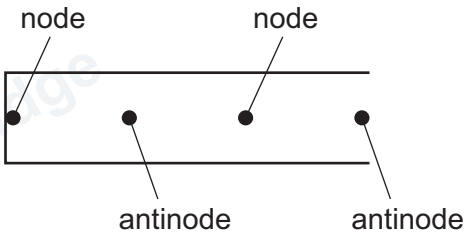
A 0.25 **B** 0.75 **C** 1.25 **D** 1.50

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13

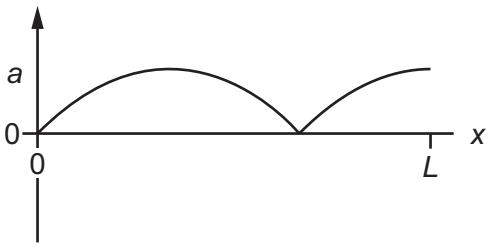
28 A stationary sound wave is formed in a tube of length *L* that is closed at one end.

The diagram shows the positions of the nodes and antinodes of the stationary wave.

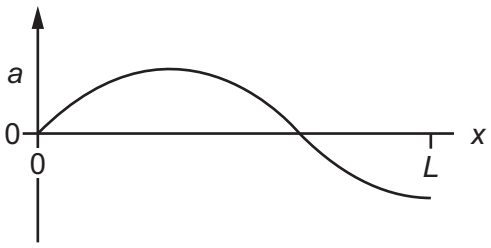


Which graph shows the variation of the amplitude *a* of the wave with distance *x* measured from the closed end of the tube?

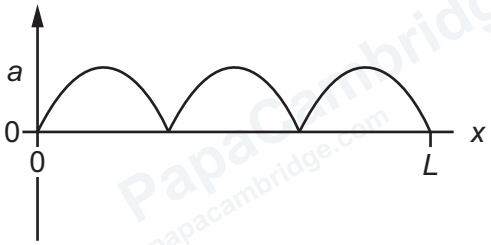
A



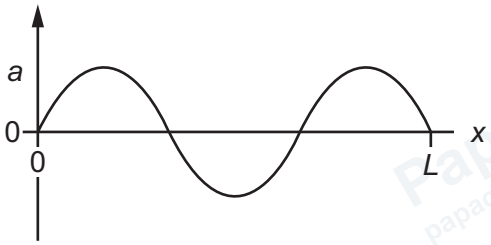
B



C



D



29 Waves are emitted by two coherent sources. The two waves meet at point X where they have a constant phase difference of 180°.

At point X, the amplitude of one of the waves is 0.50 m and the amplitude of the other wave is 0.30 m. The waves superpose.

What is the amplitude of the resultant wave at point X?

A 0 m **B** 0.20 m **C** 0.40 m **D** 0.80 m

30 There is a current in a resistor for an unknown time.

Which two quantities can be used to calculate the energy dissipated by the resistor?

A the current in the resistor and the potential difference across the resistor

B the resistance of the resistor and the current in the resistor

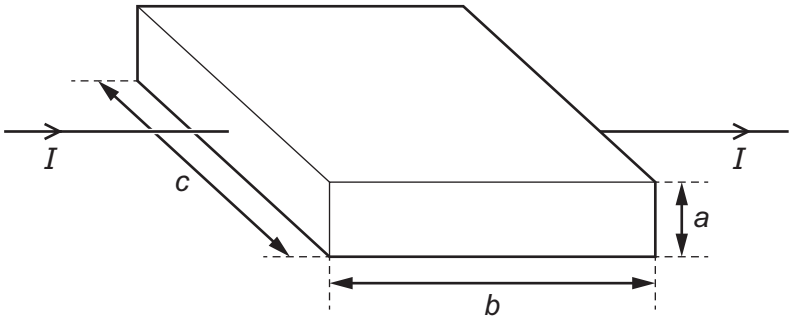
C the total charge passing through the resistor and the potential difference across the resistor

D the total charge passing through the resistor and the resistance of the resistor

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31 The diagram shows a metal block.



The block has sides of length *a*, *b* and *c* as shown, and its volume is *V*. Each charge carrier has a charge –*q* and the number density of the charge carriers in the metal is *n*. It takes each charge carrier an average time of *t* to pass through the block.

What is an expression for the current *I*?

A $I = nqabc$ **B** $I = \frac{nqV}{t}$ **C** $I = \frac{nqbc}{t}$ **D** $I = nqaV$

32 An electrical component is connected into a circuit.

Which component and which change causes an increase in the resistance of the component?

	component	change
A	filament lamp	increase in electric current
B	light-dependent resistor (LDR)	increase in light intensity
C	semiconductor diode	increase in electric current
D	thermistor	increase in temperature

33 A metal wire with circular cross-section has length *L* and diameter *d*.

The resistance of the wire is *R*.

The wire is now melted and reformed into a new wire of diameter 0.50*d*.

The volume of the wire is unchanged.

What is the resistance of the new wire?

A *R* **B** 2*R* **C** 4*R* **D** 16*R*

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34 A battery with internal resistance is connected by wires to an external resistor.

Which statement describes the electromotive force (e.m.f.) of the battery?

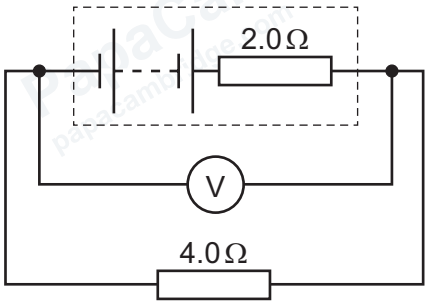
A It is the energy transferred per unit charge in driving charge through a connecting wire.

B It is the energy transferred per unit charge in driving charge through the complete circuit.

C It is the energy transferred per unit charge in driving charge through the external resistor.

D It is the energy transferred per unit charge in driving charge through the internal resistance.

35 The diagram shows a circuit with a battery connected to a resistor with a resistance of 4.0 Ω.



The battery has an internal resistance of 2.0 Ω.

The voltmeter reads 12 V.

A second resistor of resistance 4.0 Ω is connected in parallel to the first resistor.

What is the new reading on the voltmeter?

A 6.0 V **B** 9.0 V **C** 12 V **D** 24 V

36 Kirchhoff's first and second laws are a consequence of the conservation of different quantities.

What are those quantities?

	Kirchhoff's first law	Kirchhoff's second law
A	charge	energy
B	energy	current
C	current	charge
D	energy	charge

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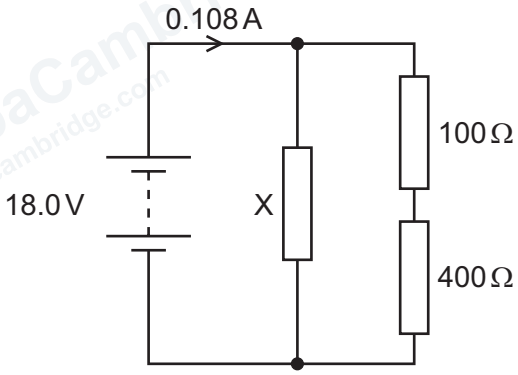
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37 A battery of electromotive force (e.m.f.) 18.0 V and negligible internal resistance is connected to a network of three fixed resistors as shown.



The current in the battery is 0.108 A.

What is the resistance of resistor X?

A 125 Ω **B** 167 Ω **C** 250 Ω **D** 500 Ω

38 A nucleus P undergoes β^+ decay, becoming nucleus Q and emitting two particles X and Y as shown.

$$P \rightarrow Q + X + Y$$

Which two particles are X and Y?

A electron and antineutrino

B electron and neutrino

C positron and antineutrino

D positron and neutrino

39 Which particle is a lepton?

A neutrino

B proton

C strange antiquark

D up quark

40 A nucleus decays in a series of stages.

At each stage, an α-particle or β^- particle is emitted.

Which sequence of emitted particles forms a nucleus that is an isotope of the original nucleus?

A $\alpha, \alpha, \beta^-, \beta^-$ **B** β^-, β^-, α **C** $\alpha, \alpha, \alpha, \alpha$ **D** $\beta^-, \beta^-, \beta^-$

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PHYSICS **9702/12**

Paper 1 Multiple Choice **May/June 2026**

MARK SCHEME

Maximum Mark: 40

Published

This mark scheme shows the correct answer to each question.

We cannot discuss the mark scheme with schools, and we recommend that schools read the mark scheme together with the question paper and the Principal Examiner Report for Teachers.

This document consists of **3** printed pages.

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Question	Answer	Marks
1	C	**1**
2	A	**1**
3	A	**1**
4	C	**1**
5	C	**1**
6	D	**1**
7	D	**1**
8	C	**1**
9	D	**1**
10	A	**1**
11	D	**1**
12	C	**1**
13	B	**1**
14	A	**1**
15	B	**1**
16	B	**1**
17	C	**1**
18	C	**1**
19	B	**1**
20	D	**1**
21	B	**1**
22	C	**1**
23	D	**1**
24	B	**1**
25	A	**1**
26	C	**1**
27	D	**1**
28	A	**1**

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Question	Answer	Marks
29	B	1
30	C	1
31	B	1
32	A	1
33	D	1
34	B	1
35	B	1
36	A	1
37	C	1
38	D	1
39	A	1
40	B	1

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CAMBRIDGE
International Education

Cambridge International AS & A Level

PHYSICS **9702/13**

Paper 1 Multiple Choice **May/June 2026**

1 hour 15 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid or tape.
- Do **not** write on any bar codes.
- You may use a calculator.

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

This document has **20** pages. Any blank pages are indicated.

IB26 06_9702_13/3RP
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[Turn over

Data

acceleration of free fall	$g = 9.81\,\text{m s}^{-2}$
speed of light in free space	$c = 3.00 \times 10^{8}\,\text{m s}^{-1}$
elementary charge	$e = 1.60 \times 10^{-19}\,\text{C}$
unified atomic mass unit	$1\,\text{u} = 1.66 \times 10^{-27}\,\text{kg}$
rest mass of proton	$m_\text{p} = 1.67 \times 10^{-27}\,\text{kg}$
rest mass of electron	$m_\text{e} = 9.11 \times 10^{-31}\,\text{kg}$
Avogadro constant	$N_\text{A} = 6.02 \times 10^{23}\,\text{mol}^{-1}$
molar gas constant	$R = 8.31\,\text{J K}^{-1}\,\text{mol}^{-1}$
Boltzmann constant	$k = 1.38 \times 10^{-23}\,\text{J K}^{-1}$
gravitational constant	$G = 6.67 \times 10^{-11}\,\text{N m}^2\,\text{kg}^{-2}$
permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\,\text{F m}^{-1}$ $\left(\frac{1}{4\pi\varepsilon_0} = 8.99 \times 10^{9}\,\text{m F}^{-1}\right)$
Planck constant	$h = 6.63 \times 10^{-34}\,\text{J s}$
Stefan–Boltzmann constant	$\sigma = 5.67 \times 10^{-8}\,\text{W m}^{-2}\,\text{K}^{-4}$

Formulae

uniformly accelerated motion	$s = ut + \frac{1}{2}at^2$ $v^2 = u^2 + 2as$
hydrostatic pressure	$\Delta p = \rho g \Delta h$
upthrust	$F = \rho g V$
Doppler effect for sound waves	$f_\text{o} = \frac{f_\text{s} v}{v \pm v_\text{s}}$
electric current	$I = Anvq$
resistors in series	$R = R_1 + R_2 + \ldots$
resistors in parallel	$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \ldots$

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3

1 Which quantity is a vector quantity?

A density

B mass

C pressure

D weight

2 What are the advantages of taking repeat measurements and calculating a mean in an experiment?

	taking repeat measurements	calculating a mean
A	shows if the results are accurate	reduces the effects of random errors
B	shows if the results are accurate	reduces the effects of systematic errors
C	shows if the results are precise	reduces the effects of random errors
D	shows if the results are precise	reduces the effects of systematic errors

3 The value of a quantity X is calculated from the values of quantities Y and Z by using the equation shown.

$$X = \frac{Y^2}{Z}$$

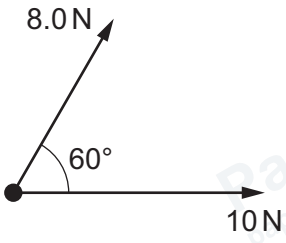
The percentage uncertainty in the value of Y is twice the percentage uncertainty in the value of Z.

What is the ratio $\frac{\text{percentage uncertainty in the value of } X}{\text{percentage uncertainty in the value of } Z}$?

A 1 **B** 3 **C** 4 **D** 5

4 Two coplanar forces act on a small object.

One force has magnitude 10 N. The other force has magnitude 8.0 N and acts at an angle of 60° to the first force.



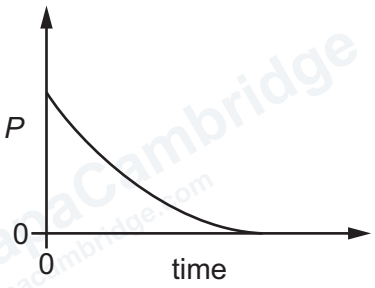
What is the magnitude of the resultant of the two forces?

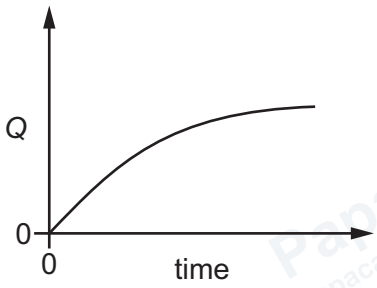
A 14 N **B** 16 N **C** 17 N **D** 18 N

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5 An object falls from a large height. The air resistance acting on the object is significant.

The graphs show the variation of two quantities *P* and *Q* with time.

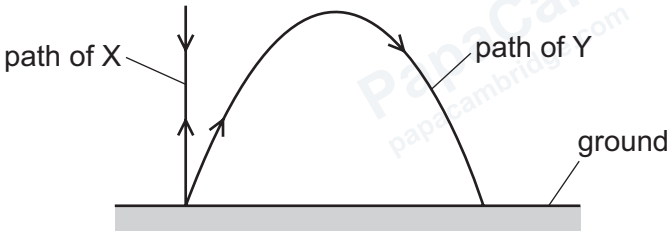




Which quantities could *P* and *Q* represent?

	P	***Q***
A	speed	acceleration
B	acceleration	speed
C	distance fallen	speed
D	speed	distance fallen

6 Two projectiles, X and Y, are fired into the air from the same place on level ground and reach the same maximum height, as shown.



Projectile X is fired vertically upwards and projectile Y is fired at an angle to the horizontal.

Air resistance is negligible.

Which statement is correct?

A X and Y are at rest at their maximum heights.

B X and Y are fired with the same speed.

C X and Y take the same time to return to the ground.

D X and Y travel the same distance.

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5

7 A lead sphere is released from rest at point X, a long way above the surface of a planet. The sphere falls in a vacuum. After a time of 4.0 s, it has fallen through a vertical distance of 3.0 m. Assume the acceleration of free fall is constant.

How far will the sphere have fallen from point X at a time of 20 s after its release?

A 15 m **B** 75 m **C** 80 m **D** 2000 m

8 The forces acting on an object are in equilibrium at one instant.

Which quantity related to the object's motion **must** have zero magnitude at this instant?

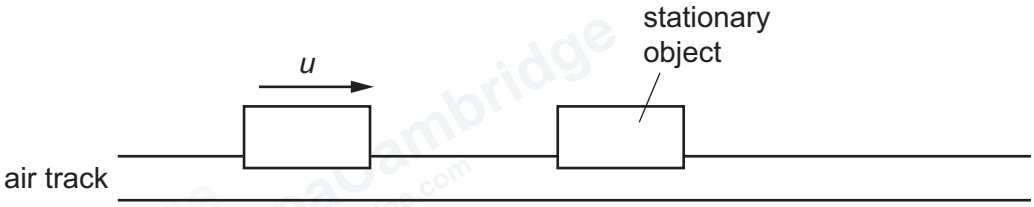
A acceleration

B displacement

C momentum

D velocity

9 The diagram shows an object on a horizontal air track moving with velocity u towards a stationary identical object.



The objects collide and stick together. The objects then move along the track with velocity v.

Assume there is no friction between the objects and the track.

What is the effect of the collision on the total momentum and the total kinetic energy of the objects?

	total momentum	**total kinetic energy**
A	conserved	changed
B	conserved	unchanged
C	not conserved	changed
D	not conserved	unchanged

10 A rock of mass $2m$, travelling in deep space at velocity v, explodes into two parts of equal mass, one of which is then stationary.

What is the kinetic energy of the moving part after the explosion?

A $\frac{1}{2}mv^2$ **B** mv^2 **C** $\frac{3}{2}mv^2$ **D** $2mv^2$

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11 Which statement describes the mass of an object?

A the force the object experiences due to gravity

B the momentum of the object before a collision

C the resistance of the object to changes in motion

D the weight of the object as measured by a balance

12 An object is in equilibrium.

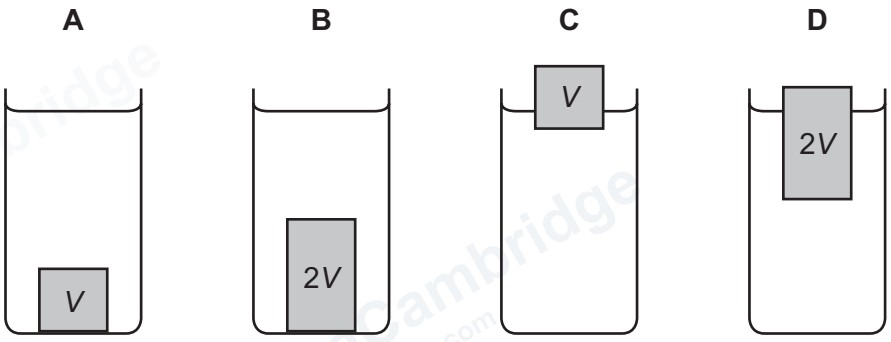
Which row about the resultant force and the resultant torque acting on the object must be correct?

	resultant force	**resultant torque**
A	zero	zero
B	zero	not zero
C	not zero	zero
D	not zero	not zero

13 Four different objects are in equilibrium in identical beakers of water as shown.

Two of the objects have volume V and the other two objects have volume $2V$.

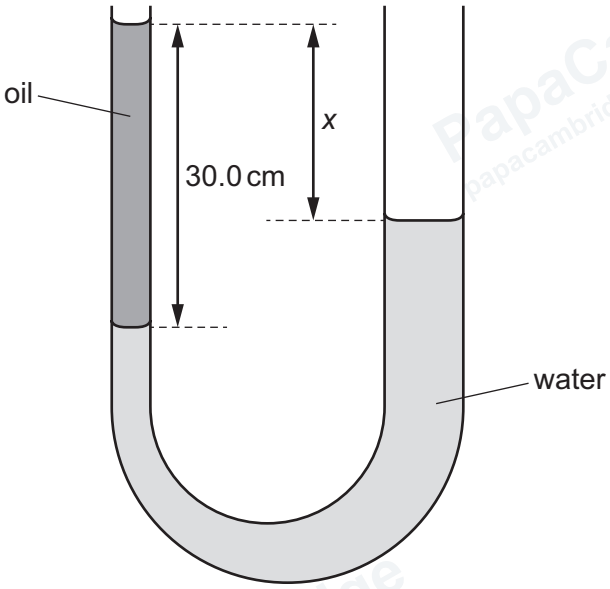
Which object is acted upon by the greatest upthrust?



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7

14 A U-tube has a left arm of cross-sectional area A and a right arm of cross-sectional area $4A$. The tube contains water of density $1000\,\text{kg}\,\text{m}^{-3}$ and oil of density $850\,\text{kg}\,\text{m}^{-3}$ as shown.

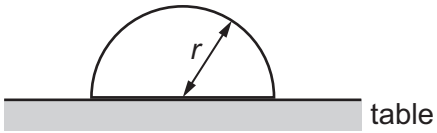


The column of oil on top of the water in the left-hand arm is of length 30.0 cm.

What is the difference in height x between the levels in the two arms of the tube?

A 4.5 cm **B** 6.2 cm **C** 23.8 cm **D** 25.5 cm

15 An object shaped as a hemisphere rests with its flat surface on a table. The object has radius r and density ρ.



The volume of a sphere is $\frac{4}{3}\pi r^3$.

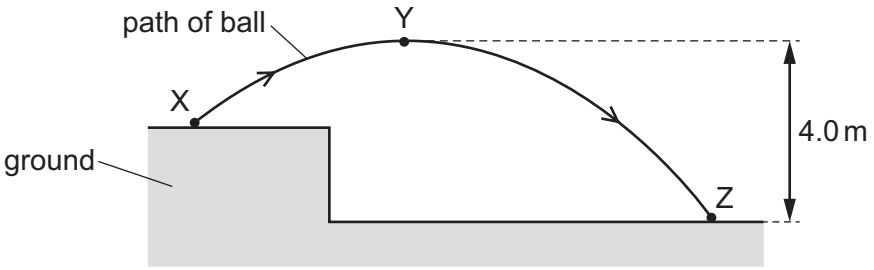
What is the average pressure that the object exerts on the table?

A $\frac{1}{3}\rho r^2$ **B** $\frac{1}{3}\rho r^2 g$ **C** $\frac{2}{3}\rho r$ **D** $\frac{2}{3}\rho r g$

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8

16 A ball is projected from the ground at point X. The ball then reaches its maximum height at point Y before landing on lower ground at point Z. The diagram shows the path of the ball.



not to scale

The ball leaves the ground at X with a kinetic energy of 5.2 J. The ball has a kinetic energy of 9.6 J when it hits the ground at Z.

The weight of the ball is 2.0 N. The vertical height of Y above Z is 4.0 m.

Air resistance is negligible.

What is the vertical height of Y above X?

A 1.8 m **B** 2.0 m **C** 2.2 m **D** 2.6 m

17 An object of mass m has a constant acceleration due to a constant resultant force F.

In a distance s, the speed of the object is increased from u to v.

What is the relationship between s, m, F, u and v?

A $Fs = mv^2 - mu^2$

B $2Fs = mv^2 - mu^2$

C $v^2u = u^2 - \left(\frac{F}{m}\right)s$

D $v^2 = u^2 - 2\left(\frac{F}{m}\right)s$

18 An object is displaced horizontally to the right in a uniform vertical gravitational field.

Which statement describes the change in the gravitational potential energy of the object?

A It decreases in direct proportion to the displacement.

B It does not change with the displacement.

C It increases in direct proportion to the displacement.

D It increases in direct proportion to the square of the displacement.

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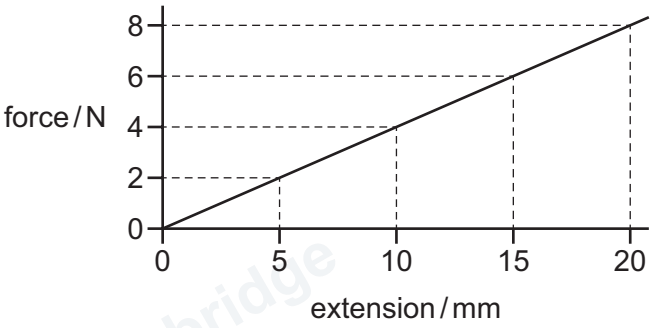
9

19 A small diesel engine uses a volume of $1.5 \times 10^4\,\text{cm}^3$ of fuel per hour to produce a useful power output of 40 kW. Assume that 34 kJ of energy is transferred to the engine when it uses $1.0\,\text{cm}^3$ of fuel.

What is the rate of transfer from the engine of energy that is wasted?

A 102 kW **B** 142 kW **C** 182 kW **D** 470 kW

20 The graph shows the relationship between the applied force and the extension for a spring.



X is the work done by the force in extending the spring from 0 to 10 mm. *Y* is the work done by the force in extending the spring from 10 mm to 20 mm.

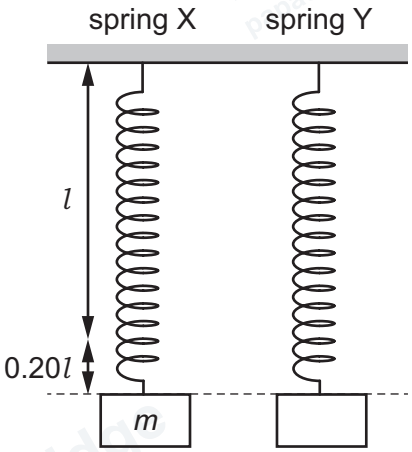
What is the ratio *X* : *Y*?

A 1 : 1 **B** 1 : 2 **C** 1 : 3 **D** 1 : 4

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21 Two springs X and Y are attached to a fixed surface. Blocks are attached to their free ends. The blocks hang vertically in equilibrium from the springs as shown.



The stretched lengths of X and Y are equal.

Spring X has spring constant k, unstretched length l and extension $0.20l$.

Spring Y has spring constant $3.0k$ and extension $0.10l$.

The mass of the block attached to spring X is m.

What is the mass of the block attached to spring Y?

A $0.67m$ **B** $1.0m$ **C** $1.5m$ **D** $3.0m$

22 A wire is being stretched by a tensile force.

Which statement about the elastic limit **must** be correct?

A The deformation is plastic after the elastic limit has been reached.

B The deformation is plastic until the elastic limit is reached.

C The extension is proportional to the tensile force after the elastic limit has been reached.

D The extension is proportional to the tensile force until the elastic limit is reached.

23 Which statement about waves is correct?

A All transverse waves are electromagnetic waves.

B Longitudinal waves include sound waves and microwaves.

C The equation $v = f\lambda$ applies only to transverse waves and **not** to longitudinal waves.

D Transverse waves and longitudinal waves can both be progressive waves.

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24 A loudspeaker moves with a constant speed relative to a stationary observer.

The loudspeaker emits a sound of constant frequency.

The frequency of sound observed depends on the speed and the direction of motion of the loudspeaker.

Which motion of the loudspeaker will cause the highest observed frequency?

	direction of motion of loudspeaker	speed / m s^{-1}
A	away from observer	10
B	away from observer	5
C	towards observer	10
D	towards observer	5

25 Plane polarised light of intensity I is incident normally on a set of polarising filters.

The transmission axis of the first filter is at 30° to the plane of polarisation of the incident light.

The transmission axis of each successive filter is at 30° to the transmission axis of the previous filter.

The intensity of the light emerging from the final filter is $0.24I$.

How many polarising filters are in the set?

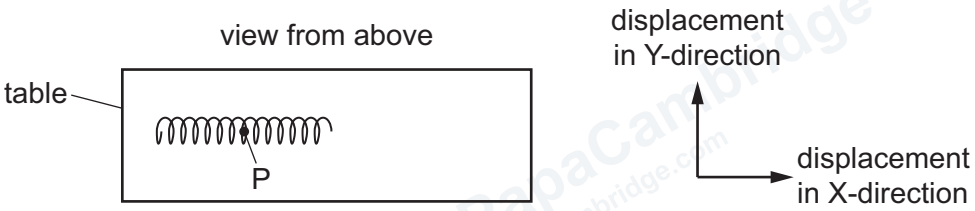
A 3 **B** 5 **C** 6 **D** 10

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12

26 A student demonstrates a longitudinal wave using a spring that lies on a table.

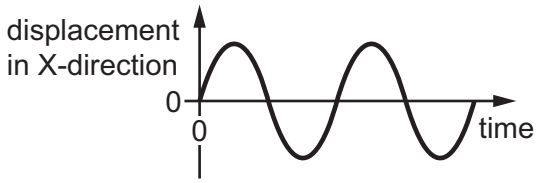


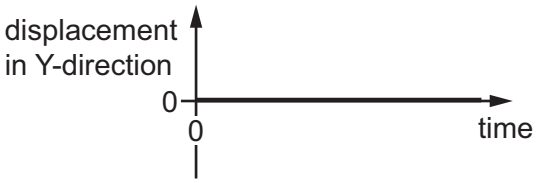
The X- and Y-directions are shown.

P is a point on the spring.

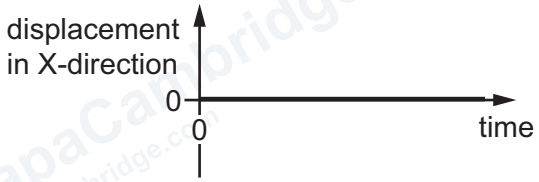
Which pair of graphs could show the variation with time of the displacement of point P in the X- and Y-directions?

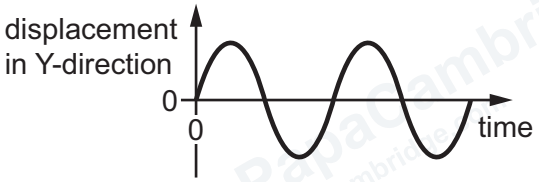
A



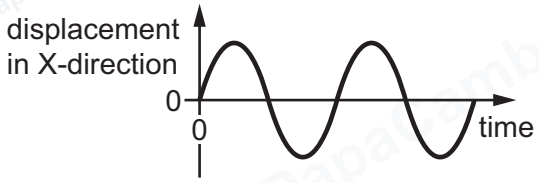


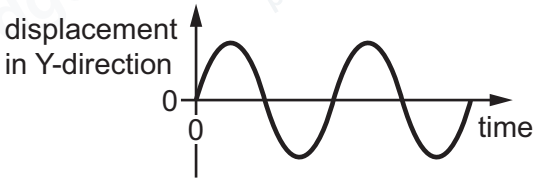
B



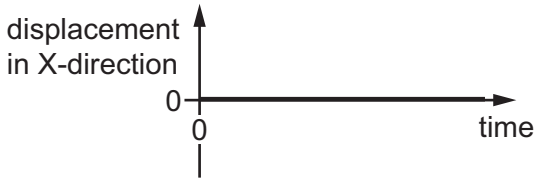


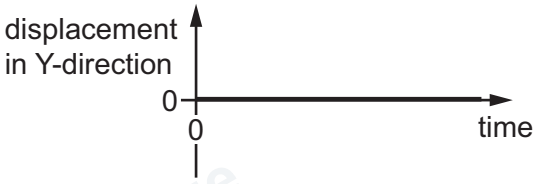
C





D





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13

27 Light from a laser is incident normally on a double slit. A wave of light emerges from each slit.

Bright and dark interference fringes are observed on a screen that is placed parallel to the double slit.

Which statement explains the formation of a dark fringe?

A A dark fringe is formed where none of the light reaches the screen because it was blocked at the double slit.

B A dark fringe is formed where the waves of light from the two slits do **not** overlap and so **cannot** superpose.

C A dark fringe is formed where a trough from one wave of light meets a trough from the other wave of light.

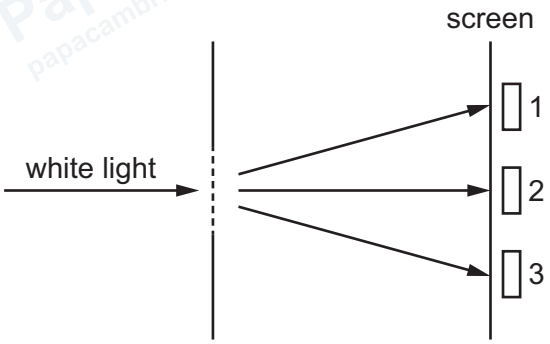
D A dark fringe is formed where the two waves of light meet and have a phase difference of 180°.

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14

28 A beam of white light is incident normally on a diffraction grating.

Three bands of bright fringes are formed on a screen, labelled 1, 2 and 3 as shown.

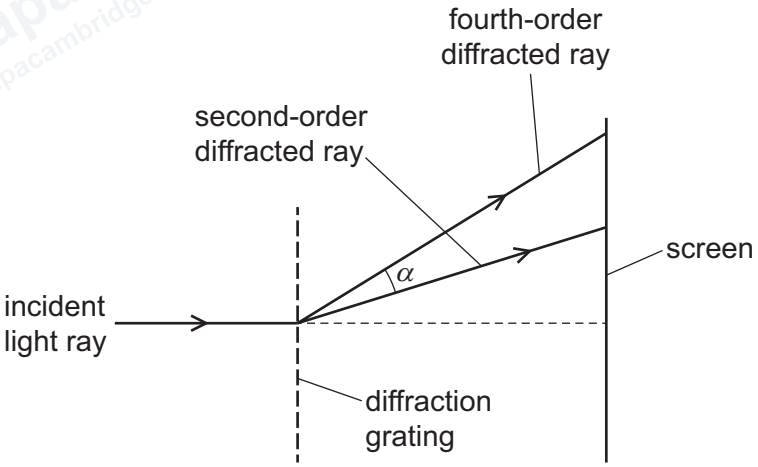


What are the colours of the fringes seen on the screen?

	1	2	3
A	continuous spectrum red ⟷ violet	white	continuous spectrum violet ⟷ red
B	continuous spectrum violet ⟷ red	white	continuous spectrum red ⟷ violet
C	continuous spectrum red ⟷ violet	continuous spectrum red – violet – red	continuous spectrum violet ⟷ red
D	continuous spectrum violet ⟷ red	continuous spectrum violet – red – violet	continuous spectrum red ⟷ violet

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29 Light of wavelength 400 nm passes normally through a diffraction grating with a line spacing of 2.0×10^{-6} m and forms a diffraction pattern on a screen.



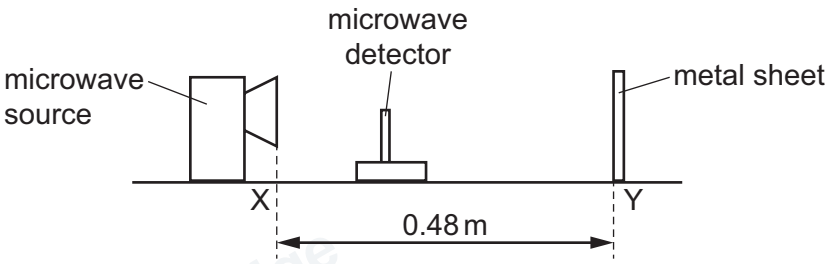
not to scale

What is the angle α between the second-order and the fourth-order diffracted rays?

A 24° **B** 30° **C** 53° **D** 77°

30 A microwave source is placed on a horizontal surface at position X.

A shiny metal sheet is placed at position Y that is a distance of 0.48 m from X.



The microwaves are reflected back from the metal sheet towards the source.

A microwave detector is moved from X to Y and all positions of minimum microwave intensity are recorded.

Intensity minima were detected at X and Y and also at 15 positions in between X and Y.

The speed of the microwaves is 3.0×10^8 m s^{-1}.

What is the frequency of the microwaves emitted by the source?

A 4.7×10^9 Hz **B** 5.0×10^9 Hz **C** 9.4×10^9 Hz **D** 1.0×10^{10} Hz

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31 A resistor with a resistance R is connected to a cell.

The current in the resistor is I and the potential difference (p.d.) across the resistor is V.

Which equations can be used to calculate the power dissipated in the resistor?

1 $P = I^2R$

2 $P = VI$

3 $P = \frac{V^2}{R}$

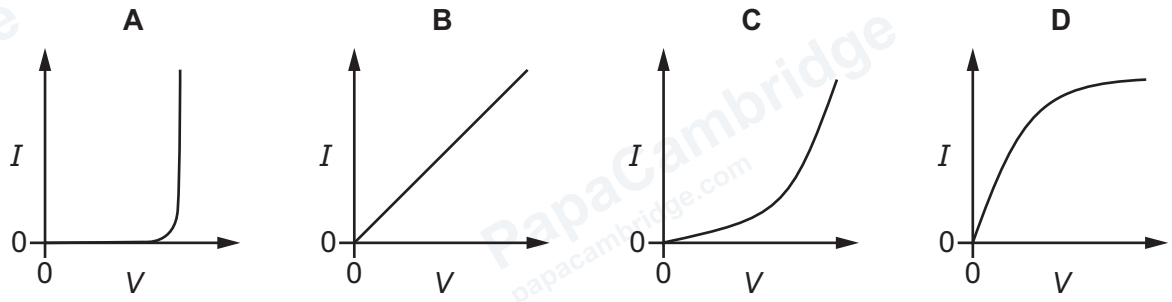
A 1, 2 or 3 **B** 1 or 2 only **C** 1 only **D** 2 only

32 A wire carries a current of 2.0 A for 1.0 hour.

How many electrons pass a point in the wire in this time?

A 1.2×10^{-15} **B** 7.2×10^{3} **C** 1.3×10^{19} **D** 4.5×10^{22}

33 Which graph shows the I–V characteristic of a filament lamp?



34 A straight copper wire, of diameter 4.2×10^{-4} m, has a number density of free electrons of 8.5×10^{28} m^{-3}. In a given time interval, a charge of 0.15 C moves through the wire.

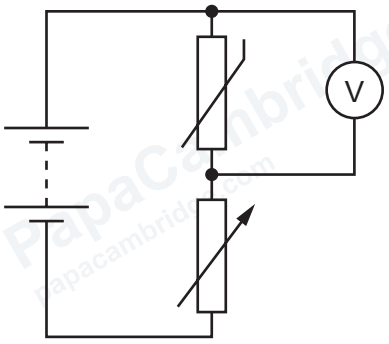
What is the average displacement of the free electrons along the wire in this time interval?

A 3.3×10^{-8} m **B** 2.0×10^{-5} m **C** 8.0×10^{-5} m **D** 2.5×10^{-4} m

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35 The diagram shows a thermistor and a variable resistor connected to a battery.

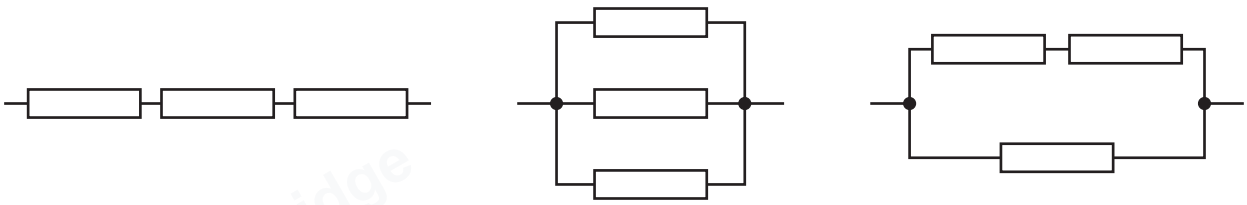


The temperature of the thermistor decreases.

What happens to the reading on the voltmeter, and which change to the resistance of the variable resistor is needed to cause the voltmeter reading to return to its initial value?

	voltmeter reading	change needed to resistance of variable resistor
A	decreases	decrease
B	decreases	increase
C	increases	decrease
D	increases	increase

36 Three resistor networks, X, Y and Z, are formed from identical resistors as shown.



X Y Z

What is the order of the networks from smallest total resistance to largest total resistance?

A X → Z → Y **B** Y → X → Z **C** Y → Z → X **D** Z → Y → X

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37 Which piece of apparatus is represented by the circuit symbol shown?



A galvanometer

B generator

C microphone

D oscilloscope

38 What is **not** a fundamental particle?

A electron

B neutrino

C neutron

D positron

39 A nucleus undergoes β^+ decay.

A neutrino and two other particles are produced in this decay.

Which other two particles are produced?

	positron	electron	proton	neutron
A	✓	✗	✗	✓
B	✗	✓	✓	✗
C	✗	✓	✗	✓
D	✓	✗	✓	✗

key

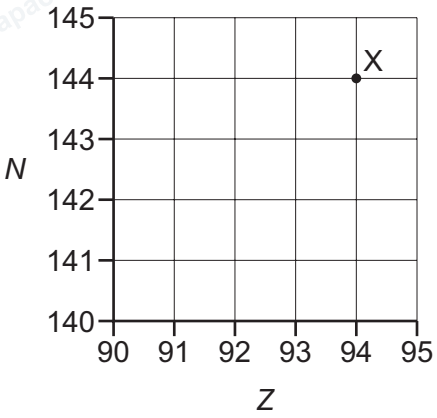
✓ = produced

✗ = not produced

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19

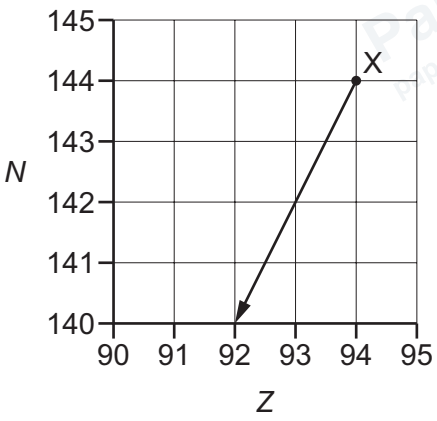
40 A nucleus X is plotted on a graph of neutron number N against proton number Z.



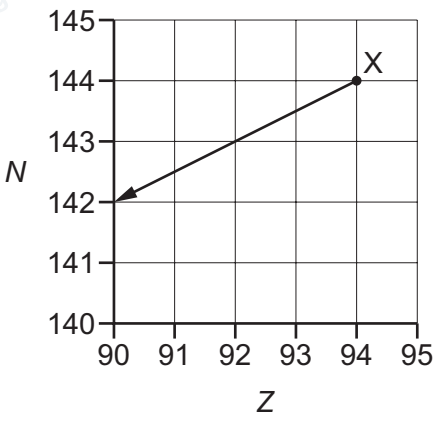
The nucleus undergoes α-decay.

Which arrow shows the changes in N and Z?

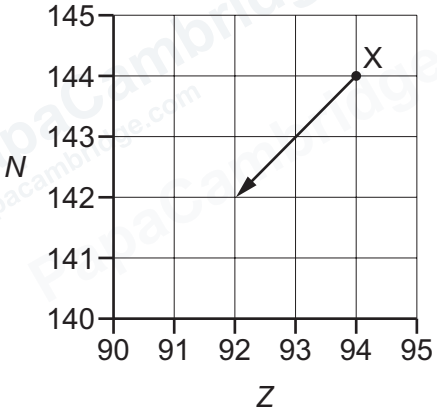
A



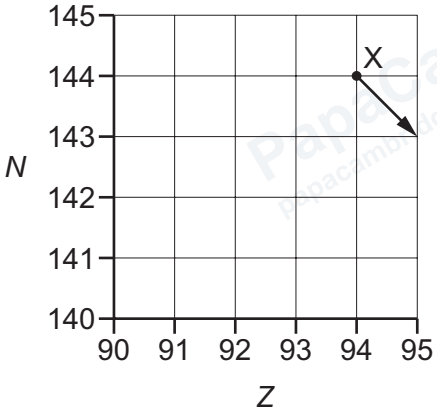
B



C



D



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20

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CAMBRIDGE
International Education

Cambridge International AS & A Level

PHYSICS **9702/13**

Paper 1 Multiple Choice **May/June 2026**

MARK SCHEME

Maximum Mark: 40

Published

This mark scheme shows the correct answer to each question.

We cannot discuss the mark scheme with schools, and we recommend that schools read the mark scheme together with the question paper and the Principal Examiner Report for Teachers.

This document consists of **3** printed pages.

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[Turn over

Question	Answer	Marks
1	D	**1**
2	C	**1**
3	D	**1**
4	B	**1**
5	B	**1**
6	C	**1**
7	B	**1**
8	A	**1**
9	A	**1**
10	D	**1**
11	C	**1**
12	A	**1**
13	B	**1**
14	A	**1**
15	D	**1**
16	A	**1**
17	B	**1**
18	B	**1**
19	A	**1**
20	C	**1**
21	C	**1**
22	A	**1**
23	D	**1**
24	C	**1**
25	B	**1**
26	A	**1**
27	D	**1**
28	A	**1**

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Question	Answer	Marks
29	B	**1**
30	B	**1**
31	A	**1**
32	D	**1**
33	D	**1**
34	C	**1**
35	D	**1**
36	C	**1**
37	D	**1**
38	C	**1**
39	A	**1**
40	C	**1**

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International Education

Cambridge International AS & A Level

PHYSICS **9702/14**

Paper 1 Multiple Choice **May/June 2026**

1 hour 15 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid or tape.
- Do **not** write on any bar codes.
- You may use a calculator.

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

This document has **20** pages. Any blank pages are indicated.

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[Turn over

Data

acceleration of free fall	$g = 9.81\,\text{m s}^{-2}$
speed of light in free space	$c = 3.00 \times 10^{8}\,\text{m s}^{-1}$
elementary charge	$e = 1.60 \times 10^{-19}\,\text{C}$
unified atomic mass unit	$1\,\text{u} = 1.66 \times 10^{-27}\,\text{kg}$
rest mass of proton	$m_\text{p} = 1.67 \times 10^{-27}\,\text{kg}$
rest mass of electron	$m_\text{e} = 9.11 \times 10^{-31}\,\text{kg}$
Avogadro constant	$N_\text{A} = 6.02 \times 10^{23}\,\text{mol}^{-1}$
molar gas constant	$R = 8.31\,\text{J K}^{-1}\,\text{mol}^{-1}$
Boltzmann constant	$k = 1.38 \times 10^{-23}\,\text{J K}^{-1}$
gravitational constant	$G = 6.67 \times 10^{-11}\,\text{N m}^2\,\text{kg}^{-2}$
permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\,\text{F m}^{-1}$ $\left(\frac{1}{4\pi\varepsilon_0} = 8.99 \times 10^{9}\,\text{m F}^{-1}\right)$
Planck constant	$h = 6.63 \times 10^{-34}\,\text{J s}$
Stefan–Boltzmann constant	$\sigma = 5.67 \times 10^{-8}\,\text{W m}^{-2}\,\text{K}^{-4}$

Formulae

uniformly accelerated motion	$s = ut + \frac{1}{2}at^2$ $v^2 = u^2 + 2as$
hydrostatic pressure	$\Delta p = \rho g \Delta h$
upthrust	$F = \rho g V$
Doppler effect for sound waves	$f_\text{o} = \frac{f_\text{s} v}{v \pm v_\text{s}}$
electric current	$I = Anvq$
resistors in series	$R = R_1 + R_2 + \ldots$
resistors in parallel	$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \ldots$

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3

1 What is an example of a measurement of a physical quantity?

A newton

B power

C 6.4%

D 9.6 J

2 An object has a mass of (10.00 ± 0.04) g.

What is the percentage uncertainty in the value of the mass?

A 0.004% **B** 0.04% **C** 0.4% **D** 4%

3 What is a reasonable estimate for the volume of a fully-inflated spherical football?

A $6000\,\text{mm}^3$ **B** $60\,\text{cm}^3$ **C** $6000\,\text{cm}^3$ **D** $0.60\,\text{m}^3$

4 At temperatures close to 0 K, the specific heat capacity c of a particular solid is given by $c = bT^3$, where T is the temperature and b is a constant, characteristic of the solid.

The SI unit of specific heat capacity is $\text{J kg}^{-1}\text{ K}^{-1}$.

What is the unit of constant b, expressed in SI base units?

A $\text{m}^2\text{ s}^{-2}\text{ K}^{-3}$

B $\text{m}^2\text{ s}^{-2}\text{ K}^{-4}$

C $\text{kg m}^2\text{ s}^{-2}\text{ K}^{-3}$

D $\text{kg m}^2\text{ s}^{-2}\text{ K}^{-4}$

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4

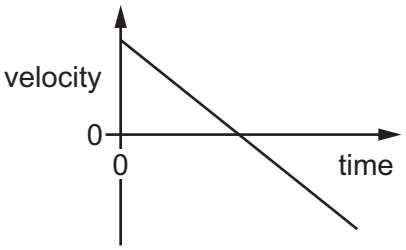
5 An athlete runs at a constant speed in a straight line to a point X.

At X, the athlete changes direction and runs back in a straight line to the starting point at the same constant speed.

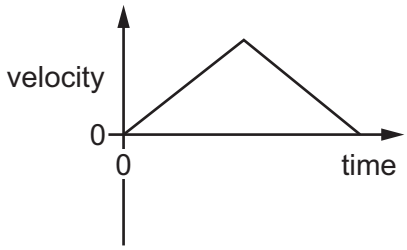
Assume that the athlete changes direction instantaneously.

Which velocity–time graph represents the motion of the athlete?

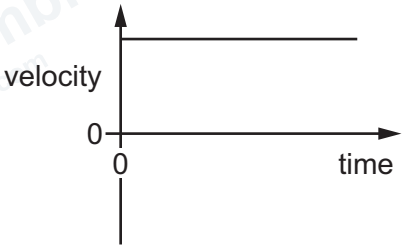
A



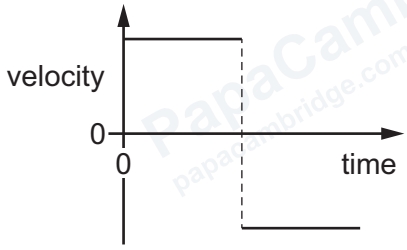
B



C



D



6 A stone is thrown vertically upwards from a point that is 12 m above the sea. It then falls into the sea below after 3.4 s.

Air resistance is negligible.

At which speed is the stone released when it is thrown?

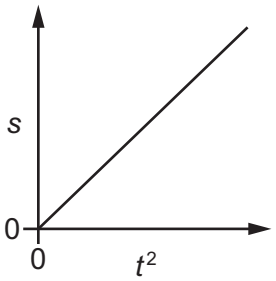
A $3.5\,m\,s^{-1}$ **B** $6.6\,m\,s^{-1}$ **C** $13\,m\,s^{-1}$ **D** $20\,m\,s^{-1}$

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5

7 At time $t = 0$, an object moves from rest with constant acceleration in a straight line. At time t, the object is distance s from its rest position.

A graph is drawn of s against t^2 as shown.



Which statement describes the acceleration of the object?

A It is equal to half the value of the gradient of the graph.

B It is equal to the value of the gradient of the graph.

C It is equal to twice the value of the gradient of the graph.

D It is equal to the reciprocal of the gradient of the graph.

8 A skydiver is falling at terminal velocity.

The skydiver opens her parachute.

Which quantity increases in magnitude immediately after the skydiver opens her parachute?

A the resultant force acting on the skydiver

B the velocity of the skydiver

C the distance between the skydiver and the ground

D the weight of the skydiver

9 A car of mass m is travelling with a velocity v.

The velocity of the car is increased by Δv due to a constant resultant force F acting for a time t.

What is a correct expression for t?

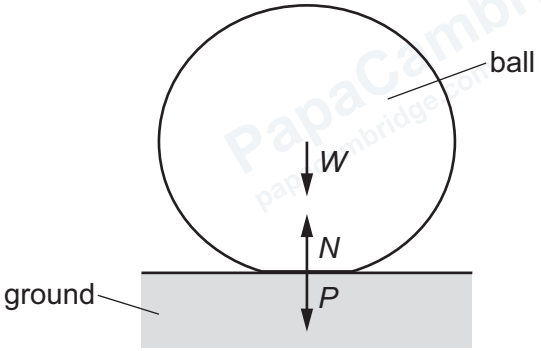
A $t = \frac{mv + m\Delta v}{F}$ **B** $t = \frac{m\Delta v}{F}$ **C** $t = \frac{F}{m\Delta v}$ **D** $t = \frac{F}{mv + m\Delta v}$

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10 A ball is dropped onto horizontal ground and bounces vertically upwards. When the ball is in contact with the ground, the following forces act:

- the weight W of the ball
- the contact force P exerted on the ground by the ball
- the contact force N exerted on the ball by the ground.



When the ball is in contact with the ground, the ball is momentarily stationary.

At this instant, which relationship is correct?

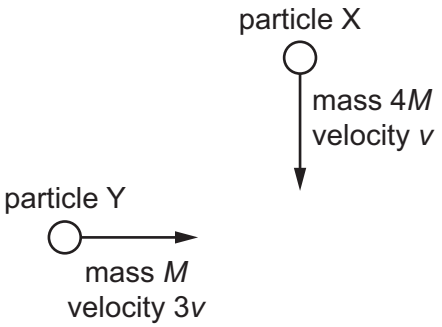
A $N = P + W$ **B** $N > P$ **C** $N = W$ **D** $N > W$

11 The diagram shows two particles in motion.

Particle X has mass $4M$ and is travelling with velocity v.

Particle Y has mass M and is travelling with velocity $3v$.

The paths of the particles are perpendicular.



The two particles collide and stick together to form a combined particle. No external forces act on the particles.

What is the magnitude of the momentum of the combined particle after the collision?

A Mv **B** $4Mv$ **C** $5Mv$ **D** $7Mv$

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12 An object is maintained in equilibrium by three coplanar forces.

Which statement is **not** correct?

A The acceleration of the object must be zero.

B The resultant force acting on the object must be zero.

C The resultant moment of any two forces about the point of action of the third force must be zero.

D The velocity of the object must be zero.

13 Which row describes an object that is floating in equilibrium in water?

	upthrust on object	hydrostatic pressure on object
A	equal to volume of water displaced	not equal at all points
B	equal to volume of water displaced	equal at all points
C	equal to weight of water displaced	not equal at all points
D	equal to weight of water displaced	equal at all points

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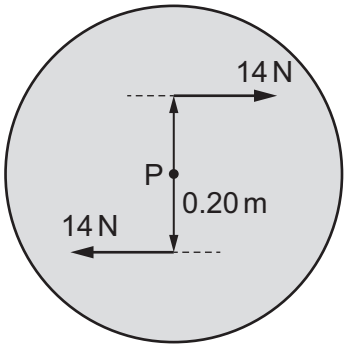
8

14 A pair of forces acts on a uniform disc pivoted at its centre of gravity P.

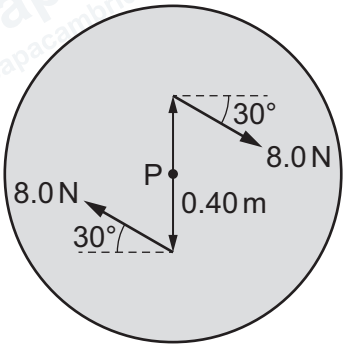
The diagrams show pairs of forces and the distances between their points of action.

Which pair of forces forms a couple with a torque of 1.4 N m?

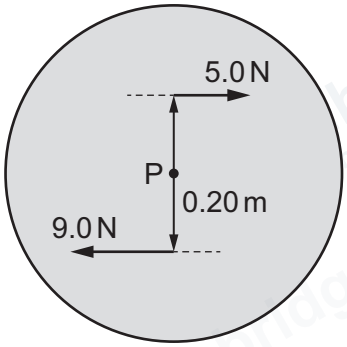
A



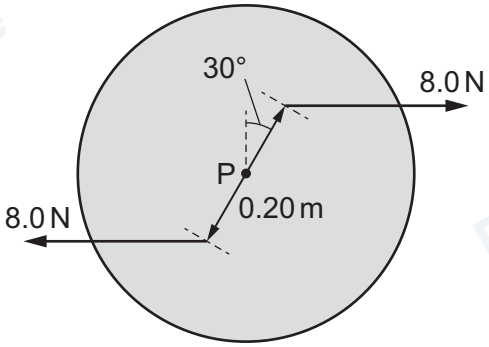
B



C



D



15 Concrete has a density of 2400 kg m^{-3}.

What is the mass of concrete that fills a rectangular space of dimensions 0.080 m × 0.90 m × 1.10 m?

A 79 kg **B** 190 kg **C** 790 kg **D** 1900 kg

16 An electric motor raises an object of mass m at a constant speed v through a height h in time t.

The acceleration of free fall is g.

What is the change in energy of the object in time t?

A mgh **B** $mgh + \frac{1}{2}mv^2$ **C** mgv **D** $\frac{mgv}{t}$

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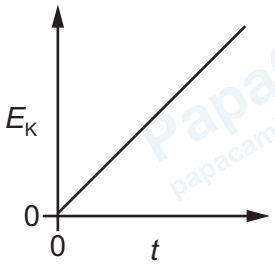
9

17 An object is released from rest at time $t = 0$ and falls downwards through a long distance.

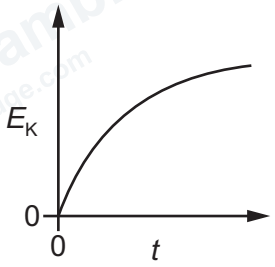
Air resistance is negligible.

Which graph shows the variation of the kinetic energy E_K of the object with time t?

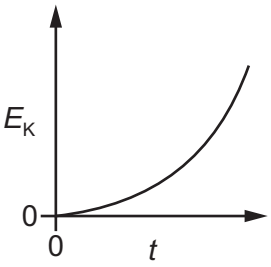
A



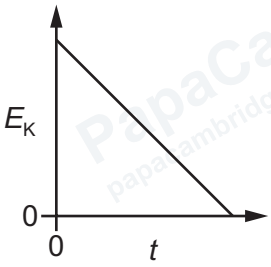
B



C



D



18 Which product of two quantities is equal to power?

A force × distance

B force × velocity

C work done × time

D work done × velocity

19 A student tests two wires X and Y by hanging a series of masses on them and measuring the extensions of the wires.

The student concludes that X has a smaller spring constant than Y.

Which result supports this conclusion?

A X has a greater extension for the same mass when obeying Hooke's law than Y.

B X supports a greater maximum mass before breaking than Y.

C Y has a greater extension at its elastic limit than X.

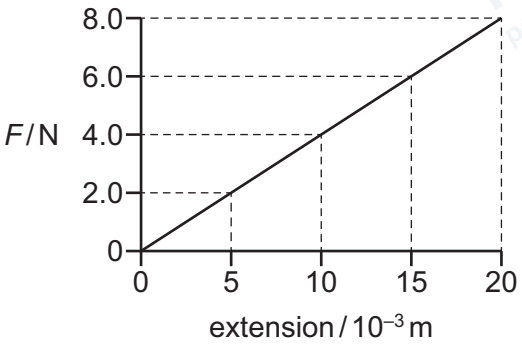
D Y stops obeying Hooke's law for a smaller mass than X.

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10

20 The diagram shows a force–extension graph for a spring.

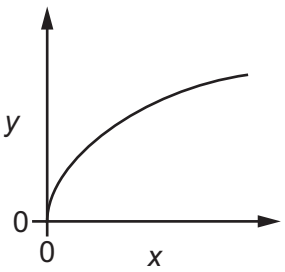


What is the value of the work done to increase the extension of the spring from 10×10^{-3} m to 20×10^{-3} m?

A 0.02 J **B** 0.06 J **C** 0.08 J **D** 0.12 J

21 A wire is extended by different forces. The wire obeys Hooke's law.

A graph is plotted to show the variation of a quantity *y* with a quantity *x*.



What could quantities *x* and *y* represent?

	x	***y***
A	elastic potential energy	extension
B	extension	force
C	force	extension
D	extension	elastic potential energy

22 What is the definition of the Young modulus of a material?

A force per unit extension

B product of force and extension

C product of stress and strain

D stress per unit strain

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11

23 A police car has a two-tone siren emitting sound of frequencies of 700 Hz and 1000 Hz.

The police car is travelling at a speed of 40.0 m s^{-1} towards a stationary observer. The speed of sound in the air is 340 m s^{-1}.

What is the difference between the two frequencies of the sound that is heard by the observer?

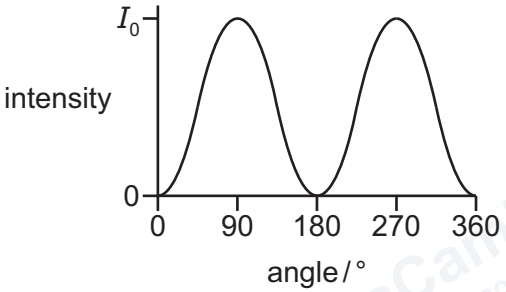
A 268 Hz **B** 300 Hz **C** 335 Hz **D** 340 Hz

24 A beam of plane-polarised light of intensity I_0 is incident normally on a polarising filter. The initial intensity of the transmitted light is zero.

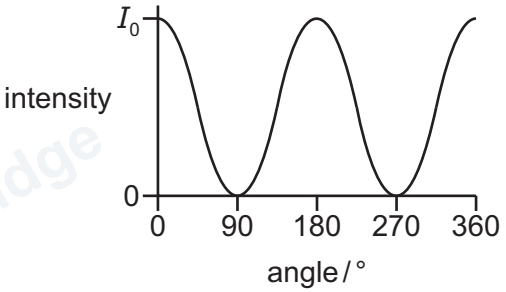
The filter is then rotated through 360° whilst remaining normal to the beam.

Which graph shows how the intensity of the transmitted light varies with the angle of the filter to its original position?

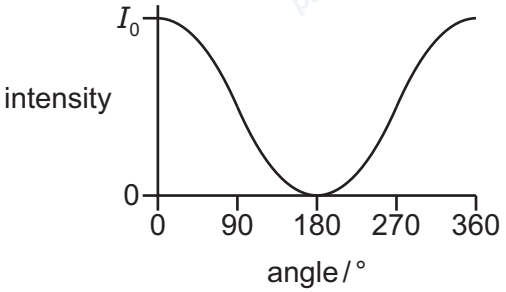
A



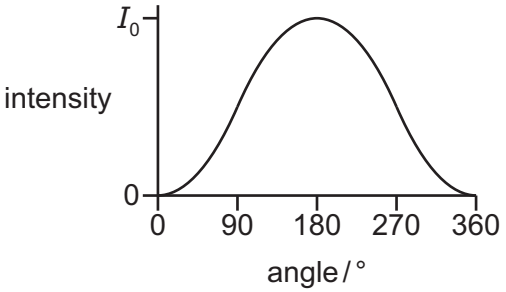
B



C



D



25 Which statement about transverse waves is correct?

A Transverse waves can diffract.

B All transverse waves have nodes and antinodes.

C Sound in air is an example of a transverse wave.

D Transverse waves **cannot** be formed on a spring.

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12

26 Bees can see electromagnetic radiation in free space with wavelength in the range 300 nm to 650 nm.

Which range of wavelength of electromagnetic radiation in free space is visible to both humans and bees?

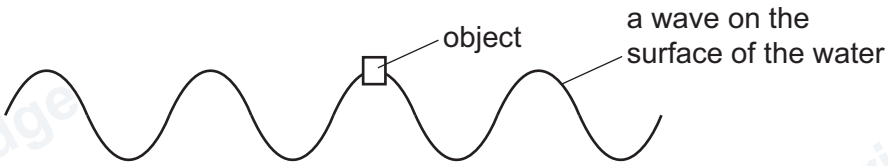
A 300 nm to 700 nm

B 300 nm to 650 nm

C 400 nm to 650 nm

D 400 nm to 700 nm

27 The diagram shows an object floating in water through which a progressive wave is passing.



In time *t*, the object moves from maximum displacement at a peak of the wave to minimum displacement at a trough and then back to maximum displacement again.

Through which vertical distance does the object move in time *t*?

A one wavelength

B twice the wavelength

C twice the amplitude

D four times the amplitude

28 Two waves of the same type travelling in a medium meet at a point.

Which statement describes the principle of superposition?

A The resultant amplitude of the two waves at that point is equal to the sum of the amplitudes of the individual waves.

B The resultant displacement of the two waves at that point is equal to the sum of the displacements of the individual waves.

C The resultant frequency of the two waves at that point is equal to the sum of the frequencies of the individual waves.

D The resultant intensity of the two waves at that point is equal to the sum of the intensities of the individual waves.

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29 Light of wavelength 500 nm is incident normally on a double slit.

A flat screen is placed parallel to the plane of the double slit. The screen is at a distance of 3.5 m from the double slit.

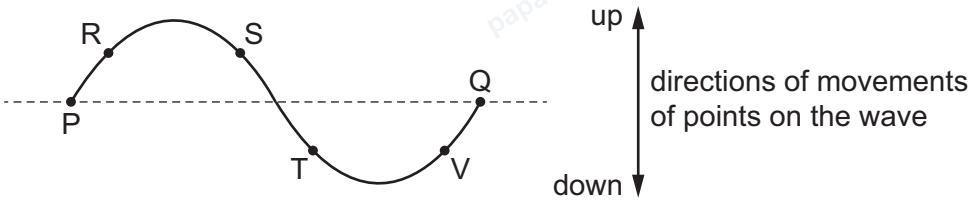
A series of bright and dark fringes is formed on the screen.

The distance from the centre of the first bright fringe to the centre of the tenth bright fringe is 22 mm.

What is the separation of the two slits?

A 0.080 mm **B** 0.35 mm **C** 0.72 mm **D** 0.80 mm

30 The diagram shows part of a stationary wave between two nodes P and Q.



R, S, T and V are points on the wave.

The wave is shown at one instant in time when point V is moving in a downwards direction.

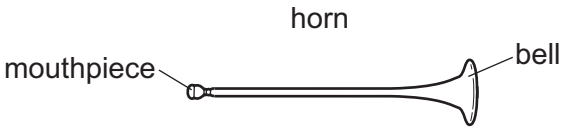
What are the directions of movement of points R, S and T at this instant?

	direction of movement of R	direction of movement of S	direction of movement of T
A	down	down	up
B	down	up	up
C	up	down	down
D	up	up	down

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31 The basic principle of note production in a horn is to set up a stationary wave in an air column.



For any note produced by the horn, a node is formed at the mouthpiece, and an antinode is formed at the bell. The frequency of the lowest note is 75 Hz.

What are the frequencies of the next two higher notes for this air column?

	first higher note / Hz	second higher note / Hz
A	113	150
B	150	225
C	150	300
D	225	375

32 Power P is supplied to a component of resistance R in an electrical circuit. When charge Q passes through the component, the energy transferred to the component is E.

Which equation defines the potential difference (p.d.) V across the component?

A $V = EQ$ **B** $V = \frac{E}{Q}$ **C** $V = PR$ **D** $V = \sqrt{PR}$

33 A beam of electrons strikes a screen. The current at the screen is 0.200 μA.

How many electrons strike the screen in a time of 5.00 s?

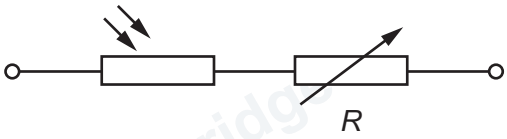
A 1.25×10^{12} **B** 6.25×10^{12} **C** 1.25×10^{15} **D** 6.25×10^{15}

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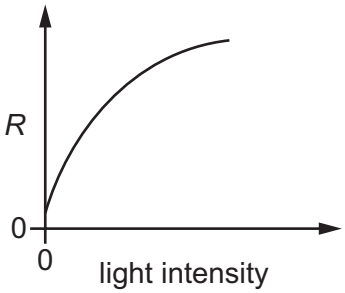
34 A variable resistor and a light-dependent resistor (LDR) are connected in series.



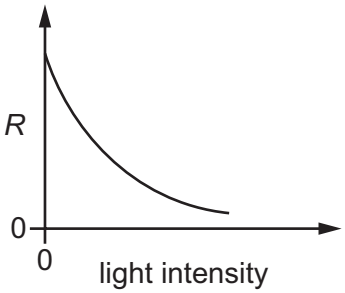
The resistance *R* of the variable resistor is adjusted so that, when light falls on the LDR, the combined resistance of the two resistors remains the same.

Which graph could represent the variation of *R* with the intensity of light that is incident on the LDR?

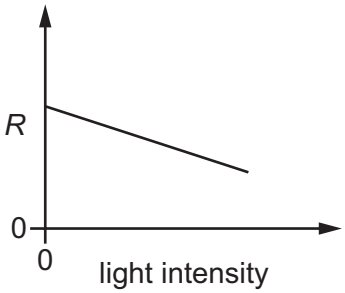
A



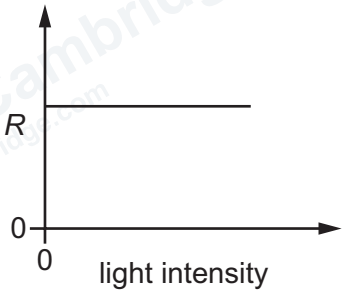
B



C



D



35 Two resistors connected in parallel have a combined resistance of 8.0 Ω.

One of the resistors has a resistance of 24 Ω.

The two resistors are now connected in series.

What is their combined resistance?

A 20 Ω **B** 30 Ω **C** 32 Ω **D** 36 Ω

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36 Which statement about the terminal potential difference (p.d.) of a power supply in a series circuit is correct?

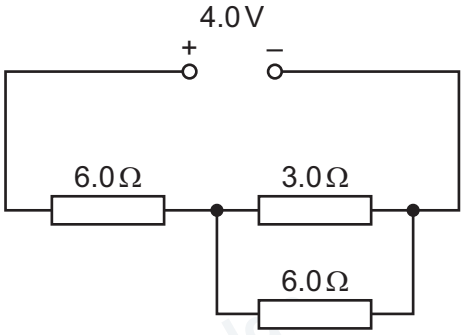
A The difference between the electromotive force (e.m.f.) and the terminal p.d. of the power supply is equal to the total energy dissipated inside the power supply due to its internal resistance.

B The terminal p.d. of the power supply is equal to the e.m.f. when there is no current in the supply.

C The terminal p.d. of the power supply is the energy transferred in moving a unit charge around the complete circuit.

D The terminal p.d. of the power supply is always more than the e.m.f. due to the internal resistance of the supply.

37 A network consists of a 3.0 Ω resistor and two 6.0 Ω resistors as shown.



The potential difference (p.d.) across the network is 4.0 V.

What is the current through the 3.0 Ω resistor?

A 0.17 A **B** 0.25 A **C** 0.33 A **D** 1.3 A

38 Which conclusion about the nucleus of an atom is **not** a result of observations from the α-particle scattering experiment?

A The diameter of the nucleus is very small compared to the diameter of the atom.

B The majority of the mass of the atom is concentrated in the nucleus.

C The nucleus of the atom contains protons and neutrons.

D The nucleus of the atom is charged.

39 An unstable nucleus $^{11}_{6}C$ decays in a single step to form a stable nucleus.

During the decay, a nucleon of the unstable nucleus is changed when an up quark in the nucleon becomes a down quark.

What is the stable nucleus that is formed?

A $^{11}_{5}B$ **B** $^{11}_{7}N$ **C** $^{12}_{6}C$ **D** $^{10}_{6}C$

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40 The nuclei of the isotopes of an element all contain the same number of a certain particle.

What is this particle?

A electron

B neutron

C nucleon

D proton

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Cambridge International AS & A Level

PHYSICS **9702/14**

Paper 1 Multiple Choice **May/June 2026**

MARK SCHEME

Maximum Mark: 40

Published

This mark scheme shows the correct answer to each question.

We cannot discuss the mark scheme with schools, and we recommend that schools read the mark scheme together with the question paper and the Principal Examiner Report for Teachers.

This document consists of **3** printed pages.

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[Turn over

Question	Answer	Marks
1	D	1
2	C	1
3	C	1
4	B	1
5	D	1
6	C	1
7	C	1
8	A	1
9	B	1
10	D	1
11	C	1
12	D	1
13	C	1
14	D	1
15	B	1
16	A	1
17	C	1
18	B	1
19	A	1
20	B	1
21	A	1
22	D	1
23	D	1
24	A	1
25	A	1
26	C	1
27	D	1
28	B	1

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Question	Answer	Marks
29	C	**1**
30	D	**1**
31	D	**1**
32	B	**1**
33	B	**1**
34	A	**1**
35	D	**1**
36	B	**1**
37	C	**1**
38	C	**1**
39	A	**1**
40	D	**1**

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Cambridge International AS & A Level

CANDIDATE NAME

CENTRE NUMBER

CANDIDATE NUMBER

4497179093

PHYSICS **9702/21**

Paper 2 AS Level Structured Questions **May/June 2026**

1 hour 15 minutes

You must answer on the question paper.

No additional materials are needed.

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen. Do **not** use correction fluid or tape.
- Do **not** write on any bar codes.
- You may use a calculator.
- You should show all your working and use appropriate units.

INFORMATION

- The total mark for this paper is 60.
- The number of marks for each question or part question is shown in brackets [ ].

This document has **16** pages. Any blank pages are indicated.

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Data

acceleration of free fall	g	=	$9.81\ \text{m s}^{-2}$
speed of light in free space	c	=	$3.00 \times 10^{8}\ \text{m s}^{-1}$
elementary charge	e	=	$1.60 \times 10^{-19}\ \text{C}$
unified atomic mass unit	1 u	=	$1.66 \times 10^{-27}\ \text{kg}$
rest mass of proton	m_p	=	$1.67 \times 10^{-27}\ \text{kg}$
rest mass of electron	m_e	=	$9.11 \times 10^{-31}\ \text{kg}$
Avogadro constant	N_A	=	$6.02 \times 10^{23}\ \text{mol}^{-1}$
molar gas constant	R	=	$8.31\ \text{J K}^{-1}\ \text{mol}^{-1}$
Boltzmann constant	k	=	$1.38 \times 10^{-23}\ \text{J K}^{-1}$
gravitational constant	G	=	$6.67 \times 10^{-11}\ \text{N m}^2\ \text{kg}^{-2}$
permittivity of free space	ε_0	=	$8.85 \times 10^{-12}\ \text{F m}^{-1}$
	$\left(\frac{1}{4\pi\varepsilon_0}\right.$	=	$\left.8.99 \times 10^{9}\ \text{m F}^{-1}\right)$
Planck constant	h	=	$6.63 \times 10^{-34}\ \text{J s}$
Stefan–Boltzmann constant	σ	=	$5.67 \times 10^{-8}\ \text{W m}^{-2}\ \text{K}^{-4}$

Formulae

uniformly accelerated motion	s	=	$ut + \frac{1}{2}at^2$
	v^2	=	$u^2 + 2as$
hydrostatic pressure	Δp	=	$\rho g \Delta h$
upthrust	F	=	$\rho g V$
Doppler effect for sound waves	f_o	=	$\frac{f_\text{s} v}{v \pm v_\text{s}}$
electric current	I	=	$Anvq$
resistors in series	R	=	$R_1 + R_2 + \ldots$
resistors in parallel	$\frac{1}{R}$	=	$\frac{1}{R_1} + \frac{1}{R_2} + \ldots$

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1 **(a)** Define force.

.....

..... [1]

(b) A rocket is launched vertically upwards from a launch pad. The initial mass of the rocket, with all of its fuel, is 2.83×10^6 kg. For the first 60 s after lift-off, the rocket engine burns fuel at a rate of 2.0×10^4 kg s^{-1} to produce a constant thrust force.

Figure 1.1 shows the variation of the speed of the rocket with time for the first 60 s after lift-off.

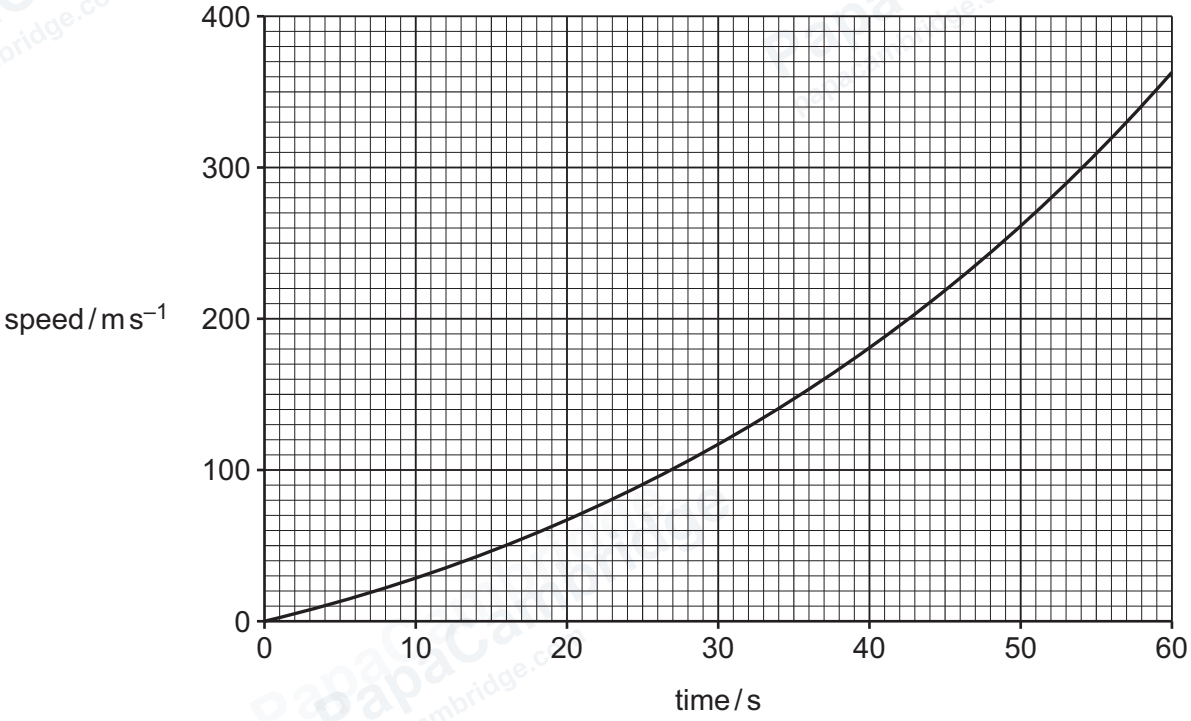


Figure 1.1

Assume that the drag force acting on the rocket is negligible over this time.

(i) By considering the mass of unburnt fuel in the rocket, explain why the speed of the rocket does **not** increase linearly with time, despite the thrust force from the engine being constant.

.....

.....

.....

..... [2]

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(ii) Describe how Figure 1.1 may be used to determine the acceleration of the rocket at lift-off. No calculations are required.

.....

.....

..... [2]

(iii) The acceleration of the rocket at lift-off is $2.5\,m\,s^{-2}$.

Determine a value for the thrust force T produced by the rocket engine.

T = ... N [2]

(iv) Determine the kinetic energy of the rocket at time 60 s after lift-off.

kinetic energy = ... J [3]

(v) Use Figure 1.1 to estimate the height of the rocket above the launch pad at time 60 s after lift-off. Explain your reasoning.

height = ... m [2]

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(c) The mass of the emitted exhaust gases is the same as the mass of fuel burnt by the engine.

Use your answer in **1(b)(iii)** to calculate the speed at which the exhaust gases are emitted from the rocket in **1(b)**.

speed = ... $m\,s^{-1}$ [2]

(d) In practice, the drag force on the rocket in **1(b)** is not negligible.

By placing **one** tick (✓) in each row, complete Table 1.1 to show the effect of the drag force on the acceleration at lift-off and on the kinetic energy at time 60 s, compared with the values in **1(b)** that assumed negligible drag force.

Table 1.1

	less	**same**	**greater**
acceleration at lift-off			
kinetic energy at time 60 s			

[2]

[Total: 16]

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2 **(a)** State what is meant by the spring constant of a spring.

.....

..... [1]

(b) A block of mass 0.086 kg is suspended in equilibrium from a spring that obeys Hooke's law. In this position, the extension of the spring is 0.026 m.

The block is pulled vertically down a distance of 0.015 m from the equilibrium position and then released from rest so that it starts moving upwards. Air resistance is negligible.

(i) Identify **one** possible source of random error in determining experimentally the extension of the spring using a ruler.

.....

..... [1]

(ii) Calculate the spring constant k of the spring.

k = ... N m^{-1} [2]

(iii) Show that, in the equilibrium position, the elastic potential energy of the spring is 0.011 J.

[2]

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(iv) Assume that the spring has not exceeded its elastic limit when the block is in the position from which it is released.

Calculate the additional elastic potential energy stored in the spring when the block moves from its equilibrium position to the position from which it is released.

additional elastic potential energy = ... J [2]

(v) Use the principle of conservation of energy to determine the kinetic energy of the block as it passes the equilibrium position moving upwards.

kinetic energy = ... J [2]

(c) Suggest, with a reason, how the answer in **2(b)(iv)** is affected if the spring in **2(b)** has exceeded its elastic limit when the block is at the position from which it is released.

.....

.....

..... [2]

[Total: 12]

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3 **(a)** State what is meant by a progressive wave.

.....

..... [1]

(b) A transverse progressive wave travels along a long string PQ from P to Q.

Figure 3.1 shows the string at time $t = 0$.

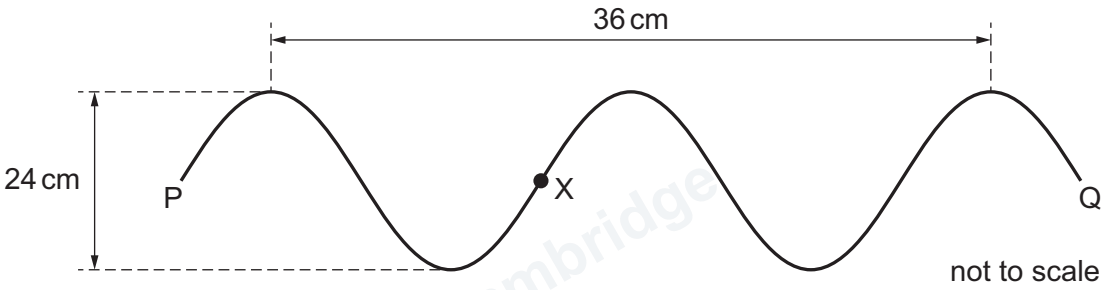


Figure 3.1

A point on the string is labelled X. At $t = 0$, point X is at its equilibrium position.
The speed of the wave on the string is $75\,\text{cm}\,\text{s}^{-1}$.

(i) Show that the period of the wave is 0.24 s.

[2]

(ii) On Figure 3.2, sketch the variation of the displacement from the equilibrium position of point X with t between $t = 0$ and $t = 0.48$ s.

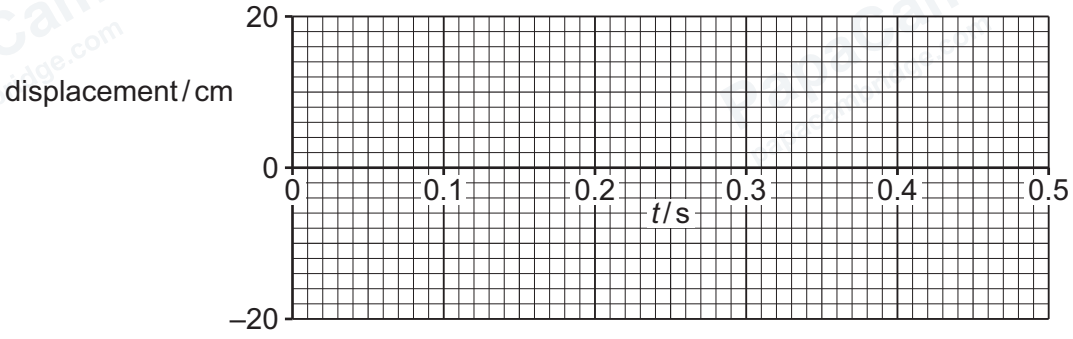


Figure 3.2

[3]

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(c) The string in **3(b)** is now fixed to walls at ends P and Q. An oscillator causes transverse oscillations of the string of period 0.24 s. The speed of waves on the string is still 75 cm s^{-1}.

The distance between P and Q is 45 cm.

(i) Explain how a stationary wave is formed on the string.

.....

.....

.....

..... [2]

(ii) Figure 3.3 shows the string at one instant in time.

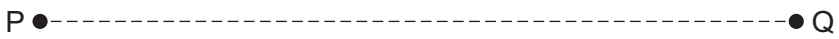


Figure 3.3

On Figure 3.3, draw a solid line to show a possible appearance of the string at a time 0.060 s later. [2]

(iii) On Figure 3.3, draw:

- a cross (×) labelled with the letter N at **one** point on the string that is at a node
- a cross (×) labelled with the letter A at **one** point on the string that is at an antinode.

[2]

[Total: 12]

4 **(a)** Define resistance.

.....

..... [1]

(b) Figure 4.1 shows the variation with temperature of the resistivity of copper.

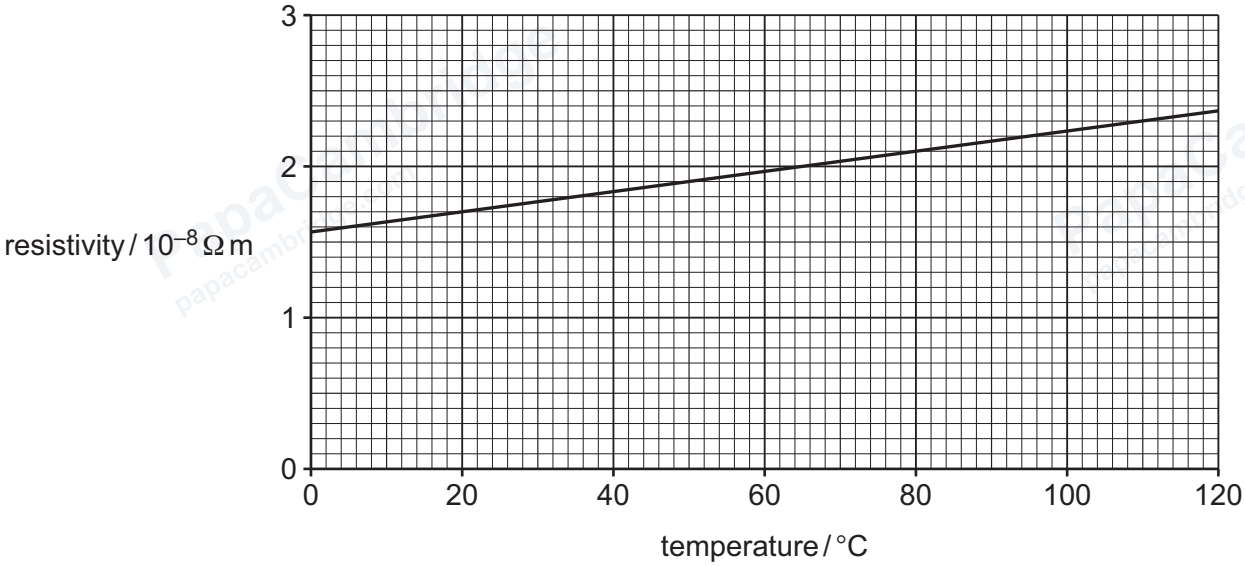


Figure 4.1

A copper wire has a length of 2.0 m and a diameter of 0.60 mm.

(i) Show that the cross-sectional area of the wire is 2.8×10^{-7} m^2.

[1]

(ii) Determine the resistance of the wire at 20 °C.

resistance = ... Ω [2]

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(iii) A different copper wire has the same resistance as the answer in **4(b)(ii)**. This wire also has a diameter of 0.60 mm but has a length that is greater than 2.0 m.

With reference to Figure 4.1, state and explain whether the temperature of this wire is less than, equal to or greater than 20 °C.

.....

.....

..... [2]

(iv) State **one** way in which the variation of resistance with temperature for a thermistor is different from that of a copper wire.

..... [1]

(c) Each metal has a property known as its temperature coefficient of resistivity α. This is given by

$$\alpha = \frac{\text{change in resistivity}}{\text{initial resistivity} \times \text{change in temperature}}.$$

(i) Deduce the SI unit for α.

unit ..... [1]

(ii) Use Figure 4.1 to determine a value of α, in its SI unit, for copper at a temperature of 0 °C.

α = ... SI unit [2]

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(d) There is a current of 5.6A in the wire in **4(b)**.
The number density of charge carriers (free electrons) for copper is $8.5 \times 10^{28}\,m^{-3}$.

Calculate the mean drift speed v of the charge carriers in the wire.

v = ... $m\,s^{-1}$ [2]

[Total: 12]

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5 **(a)** The general quark composition of a meson is a quark and an antiquark.

State the general quark composition of a baryon.

.....

..... [1]

(b) **(i)** Explain, in terms of quark composition, how a meson may have an overall charge of +1*e*, where *e* is the elementary charge.

.....

.....

..... [2]

(ii) State and explain the **two** other possible overall charges for a meson.

.....

.....

.....

..... [2]

(c) State the names of **two** particles that are classed as leptons.

1 .....

2 .....

[2]

(d) A similarity between leptons and mesons is that they both have the same possible overall charges as those described in **5(b)(i)** and **5(b)(ii)**.

State **one** difference between leptons and mesons.

.....

..... [1]

[Total: 8]

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Cambridge International AS & A Level

CANDIDATE NAME

CENTRE NUMBER

CANDIDATE NUMBER

9190438800

PHYSICS **9702/22**

Paper 2 AS Level Structured Questions **May/June 2026**

1 hour 15 minutes

You must answer on the question paper.

No additional materials are needed.

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen. Do **not** use correction fluid or tape.
- Do **not** write on any bar codes.
- You may use a calculator.
- You should show all your working and use appropriate units.

INFORMATION

- The total mark for this paper is 60.
- The number of marks for each question or part question is shown in brackets [ ].

This document has **20** pages. Any blank pages are indicated.

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Data

acceleration of free fall	g	=	$9.81\ \text{m s}^{-2}$
speed of light in free space	c	=	$3.00 \times 10^{8}\ \text{m s}^{-1}$
elementary charge	e	=	$1.60 \times 10^{-19}\ \text{C}$
unified atomic mass unit	1 u	=	$1.66 \times 10^{-27}\ \text{kg}$
rest mass of proton	m_p	=	$1.67 \times 10^{-27}\ \text{kg}$
rest mass of electron	m_e	=	$9.11 \times 10^{-31}\ \text{kg}$
Avogadro constant	N_A	=	$6.02 \times 10^{23}\ \text{mol}^{-1}$
molar gas constant	R	=	$8.31\ \text{J K}^{-1}\ \text{mol}^{-1}$
Boltzmann constant	k	=	$1.38 \times 10^{-23}\ \text{J K}^{-1}$
gravitational constant	G	=	$6.67 \times 10^{-11}\ \text{N m}^2\ \text{kg}^{-2}$
permittivity of free space	ε_0	=	$8.85 \times 10^{-12}\ \text{F m}^{-1}$
	$(\frac{1}{4\pi\varepsilon_0}$	=	$8.99 \times 10^{9}\ \text{m F}^{-1})$
Planck constant	h	=	$6.63 \times 10^{-34}\ \text{J s}$
Stefan–Boltzmann constant	σ	=	$5.67 \times 10^{-8}\ \text{W m}^{-2}\ \text{K}^{-4}$

Formulae

uniformly accelerated motion	s	=	$ut + \frac{1}{2}at^2$
	v^2	=	$u^2 + 2as$
hydrostatic pressure	Δp	=	$\rho g \Delta h$
upthrust	F	=	$\rho g V$
Doppler effect for sound waves	f_o	=	$\frac{f_\text{s} v}{v \pm v_\text{s}}$
electric current	I	=	$Anvq$
resistors in series	R	=	$R_1 + R_2 + \ldots$
resistors in parallel	$\frac{1}{R}$	=	$\frac{1}{R_1} + \frac{1}{R_2} + \ldots$

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1 **(a)** By placing **one** tick (✓) in each row, complete Table 1.1 to identify whether each of the listed quantities is a scalar or a vector.

Table 1.1

quantity	scalar	vector
amplitude		
force		
momentum		
power		

[2]

(b) A car moves in a straight line along a road. Figure 1.1 shows the variation with time t of the velocity v of the car.

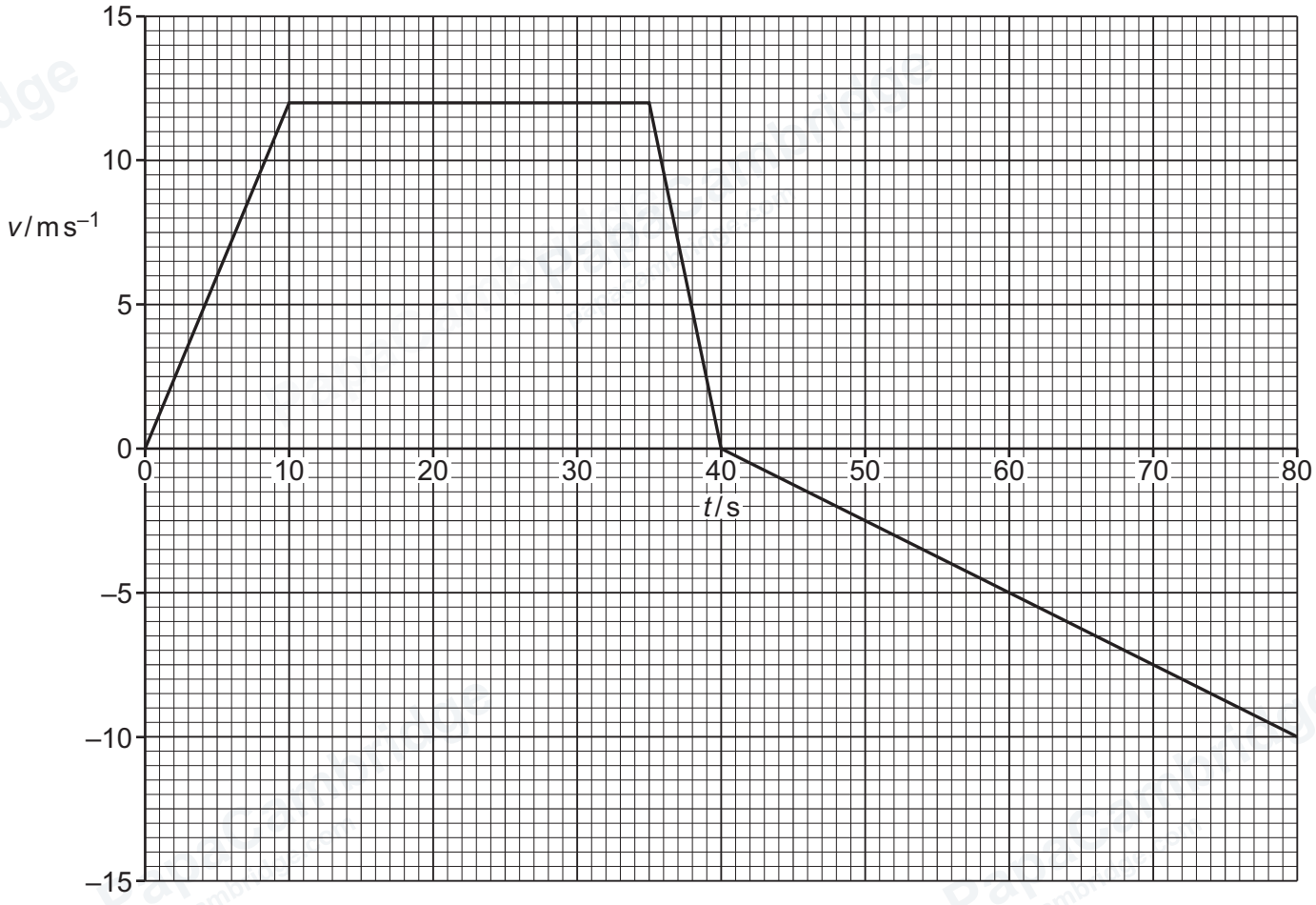


Figure 1.1

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(i) State the feature of the graph that represents the distance travelled by the car between $t = 0$ and $t = 40\,s$.

..... [1]

(ii) Calculate the distance travelled by the car between $t = 0$ and $t = 40\,s$.

distance = ... m [1]

(iii) Determine the maximum magnitude of the acceleration of the car.

acceleration = ... $m\,s^{-2}$ [2]

(iv) At $t = 25\,s$, the car is at a point P.
At $t = 40\,s$, the car is at a distance of 150 m from P.
After $t = 40\,s$, the car travels with a constant acceleration of $-0.25\,m\,s^{-2}$ until the car is again at point P.

Calculate the speed of the car when it is at point P for the second time.

speed = ... $m\,s^{-1}$ [2]

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(v) At $t = 0$, the displacement of the car is zero.

On Figure 1.2, sketch the variation of the displacement of the car with t from $t = 0$ to $t = 80$ s. Values for displacement are **not** required.

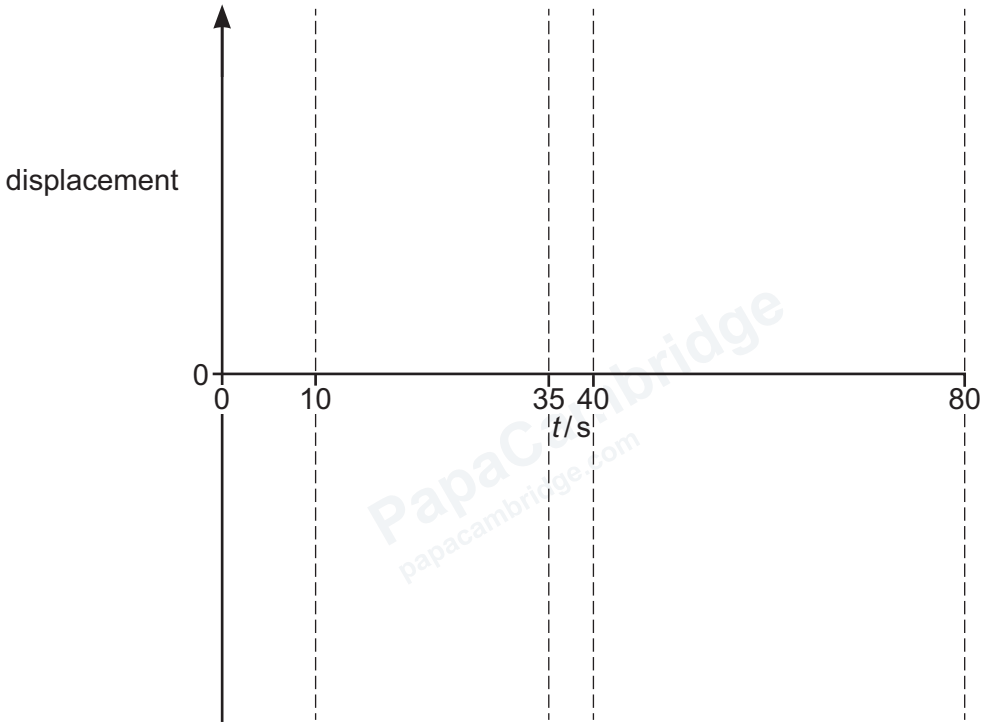


Figure 1.2

[3]

[Total: 11]

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2 **(a)** State the conditions required for an object to be in equilibrium.

.....

.....

..... [2]

(b) A block rests in equilibrium on the centre of a uniform horizontal plank. The block is attached to two ropes, each of which passes over a fixed pulley and then attaches to an end of the plank as shown in Figure 2.1.

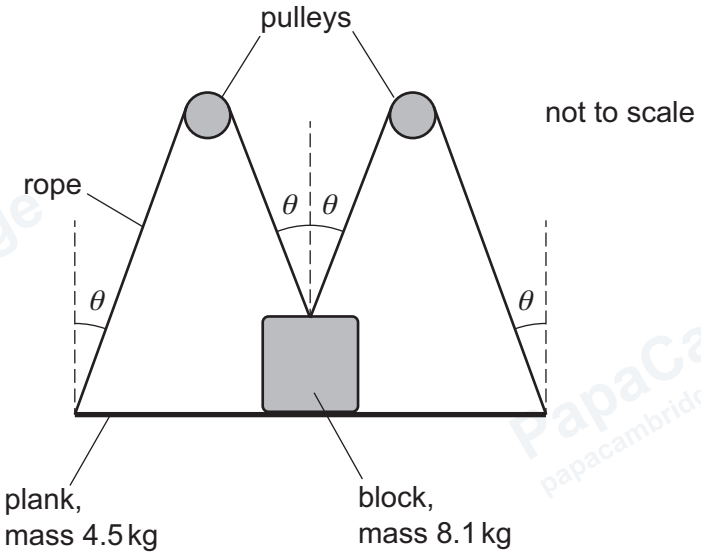


Figure 2.1

Where the ropes attach to the block and to the plank, the angle between the ropes and the vertical is θ. There is negligible friction between the pulleys and the ropes.

The mass of the block is 8.1 kg. The mass of the plank is 4.5 kg.

Figure 2.2 shows the directions of the forces acting on the plank and the block.
The weight of the plank is W_P, the weight of the block is W_B, the tension in each rope is T and the forces that the plank and the block exert on each other each have magnitude R.

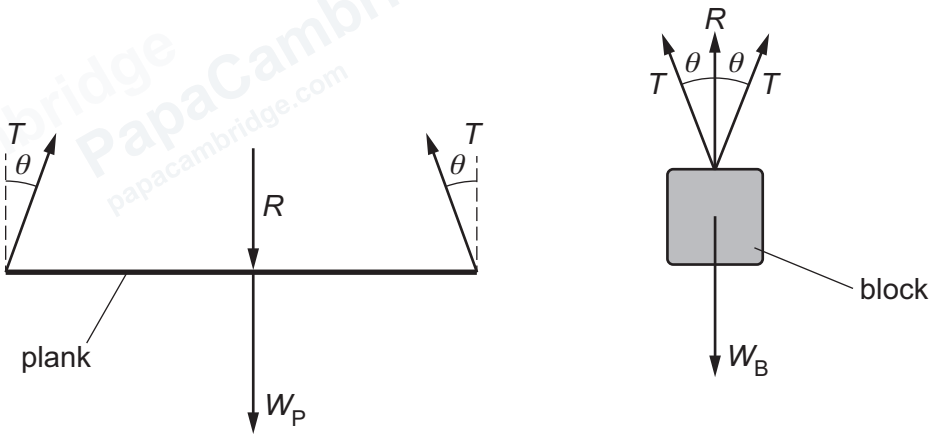


Figure 2.2

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(i) Use Figure 2.2 to show that

$$R + W_P = W_B - R.$$

[2]

(ii) Use the equation in **2(b)(i)** to calculate the magnitude of R.

R = ... N [2]

(c) The extension of each rope in **2(b)** is proportional to the force applied to it.

(i) State the name of the law that describes this behaviour.

..... [1]

(ii) The ropes have elastic deformation.

State what is meant by elastic deformation.

.....

..... [1]

[Total: 8]

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3 Two oscillating dippers are used to create progressive transverse water waves on the surface of a lake. The waves are shown from above in Figure 3.1.

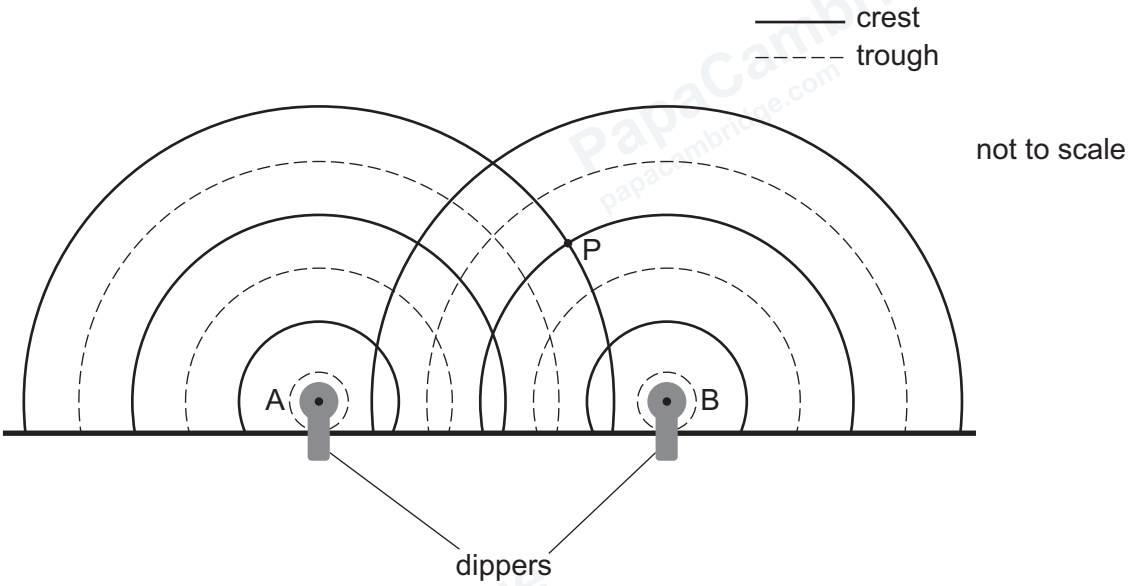


Figure 3.1

Wave crests are represented by a solid line. Wave troughs are represented by a dashed line. The waves superpose and an interference pattern is formed on the surface of the water.

The dippers oscillate in phase with each other and with a frequency of 3.0 Hz.

(a) State what is meant by a progressive wave.

.....

..... [1]

(b) **(i)** Calculate the period of the waves.

period = ... s [1]

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(ii) Points A and B are at the centre of the dippers. Point P is shown on Figure 3.1. The distance from A to P is 3.9 m. The distance from B to P is 2.4 m.

Determine the speed of the waves.

speed = ... $m\,s^{-1}$ [3]

(c) A ball of mass 0.44 kg is placed at P, where it floats on the surface of the lake. The amplitude of the vertical oscillation of the ball is 15 cm.

(i) Calculate the difference in the gravitational potential energy ΔE of the ball between its highest and lowest points.

ΔE = ... J [3]

(ii) The frequency of oscillation of the dippers is increased to 4.0 Hz without changing their amplitude of oscillation. The ball remains at the same position on the surface of the lake.

State and explain whether the amplitude of the oscillation of the ball increases, decreases or remains the same.

.....

.....

.....

..... [2]

[Total: 10]

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4 An electric train of mass 1.5×10^4 kg travels up a slope at a constant speed as shown in Figure 4.1.

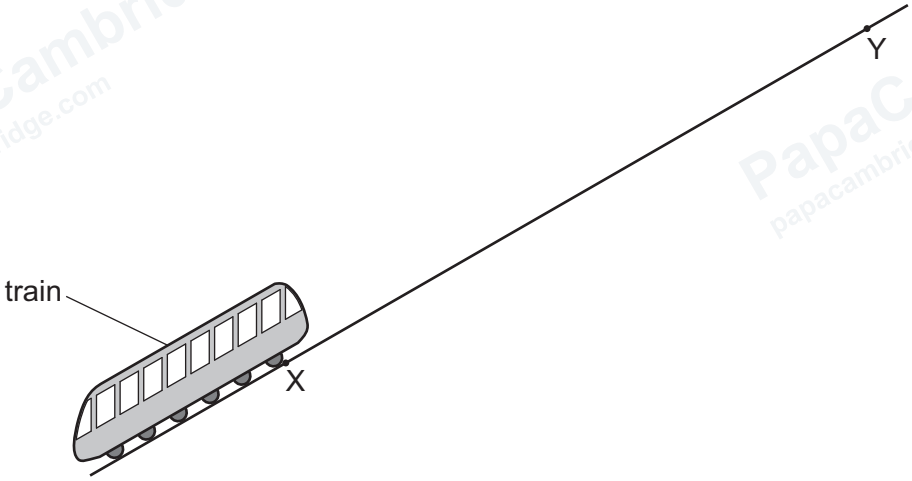


Figure 4.1

At point X, the train has gravitational potential energy 7.4 MJ.
At point Y, the train has gravitational potential energy 13.0 MJ.
The train takes a time of 52 s to move from X to Y.
The kinetic energy of the train is 1.3×10^5 J.

(a) Calculate the speed of the train.

speed = ... $m\,s^{-1}$ [3]

(b) The train is powered by an electric motor. The useful output power of the motor is 1.1×10^5 W.

(i) Define power.

..... [1]

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(ii) Determine the resistive force *F* acting on the train as it moves from point X to point Y.

F = ... N [3]

(iii) The current in the motor is 95A. The potential difference (p.d.) across the motor is 1400V.

Calculate the percentage efficiency of the motor.

percentage efficiency = ...% [3]

[Total: 10]

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5 **(a)** Three resistors of resistances R_1, R_2 and R_3 are connected in a network in parallel as shown in Figure 5.1.

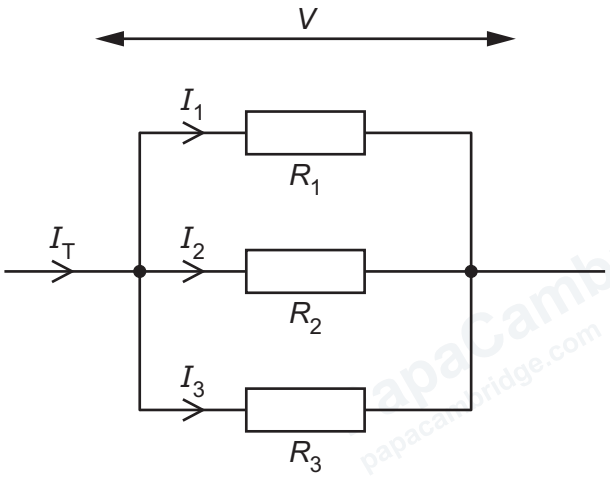


Figure 5.1

The currents in the resistors are I_1, I_2 and I_3 respectively. The total current entering the network is I_T. The potential difference (p.d.) across the resistors is V.

Derive the equation

$$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

where R_T is the total resistance of the network.

[2]

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(b) Two identical resistance wires, each of resistance 2.8 Ω, are connected in parallel to a battery with negligible internal resistance and electromotive force (e.m.f.) 12 V as shown in Figure 5.2.

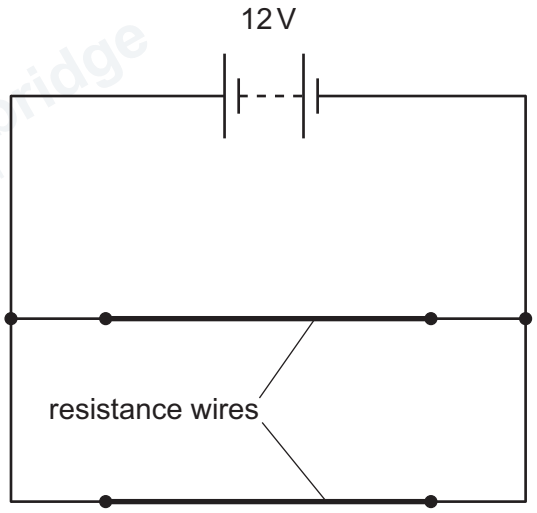


Figure 5.2

(i) Calculate the current in the battery.

current = ...A [2]

(ii) Each resistance wire has a cross-sectional area of $6.1 \times 10^{-6}\,m^2$.

The wires are made from a metal with a number density of charge carriers (free electrons) of $8.5 \times 10^{28}\,m^{-3}$.

Calculate the average drift speed of the charge carriers in the wires.

speed = ... $m\,s^{-1}$ [2]

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(c) A component X is added to the circuit in **5(b)** as shown in Figure 5.3.

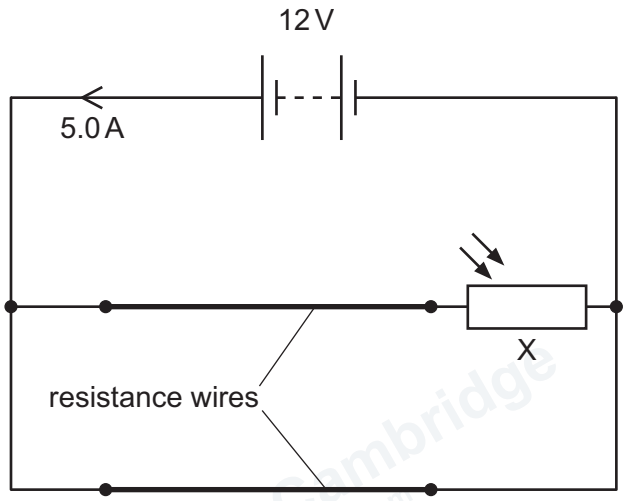


Figure 5.3

The current in the battery is now 5.0 A.
The resistance of each resistance wire remains 2.8 Ω.

(i) State the name of component X.

..... [1]

(ii) Determine the resistance of component X.

resistance = ... Ω [3]

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(iii) A change in the environment of component X causes the current in the battery to decrease. The resistance of the resistance wires remains constant.

State and explain how the environment of X has changed.

.....

.....

.....

..... [2]

[Total: 12]

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6 **(a)** One of the conclusions from the α-particle scattering experiment was that the atom is mostly empty space.

State the observation from the experiment that provides evidence for this conclusion.

.....

..... [1]

(b) An α-particle travelling with a speed of $5.60 \times 10^6\,m\,s^{-1}$ collides with a stationary gold nucleus of mass 197 u. After the collision, the α-particle is travelling with a speed of $5.38 \times 10^6\,m\,s^{-1}$ in the opposite direction to its initial motion.
No external forces act on this system.

(i) Show that the momentum of the α-particle before the collision is $3.72 \times 10^{-20}\,kg\,m\,s^{-1}$.

[2]

(ii) Use conservation of momentum to calculate the speed of the gold nucleus after the collision.

speed = ... $m\,s^{-1}$ [2]

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(iii) The absolute uncertainty in each of the measured speeds of the α-particle is $\pm 5 \times 10^4\,\text{m s}^{-1}$.

Calculate the percentage uncertainty in the change in velocity of the α-particle.

percentage uncertainty = ...% [2]

(c) Determine the number of up quarks in a nucleus of ${}^{2}_{1}\text{H}$.

number = ... [1]

(d) Describe the general quark composition of a meson.

.....

..... [1]

[Total: 9]

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PHYSICS **9702/22**

Paper 2 AS Level Structured Questions **May/June 2026**

MARK SCHEME

Maximum Mark: 60

Published

This mark scheme contains the instructions and marking guidance that our examiners used to mark candidate responses for this component.

Before they start marking, examiners complete a standardisation process. They review a range of candidate responses and discuss the mark scheme in detail to agree which answers can and cannot be accepted. Examiners award marks where it is clear that candidate responses have met the requirements of the mark scheme. These requirements may vary between different components or different versions of the same component, depending on the exact requirements of the question and the candidate responses seen during the standardisation process.

Sometimes, our mark schemes may allow marks to be awarded to responses which contain elements that are factually incorrect or for content not covered by the syllabus. We do this in response to the demands of a specific question to support candidates and make sure that our assessments are fair. However, these allowances should not be used as a guide for teaching and learning.

We cannot discuss the mark scheme with schools, and we recommend that schools read the mark scheme together with the assessment material and the Principal Examiner Report for Teachers.

This document consists of **21** printed pages.

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General marking principles

All examiners must apply these marking principles when marking candidate responses.

1	Examiners must award marks for each response in line with: • the specific **content** defined in the mark scheme • the specific **skills** defined in the mark scheme or in the level descriptions • the **standard of response** required as exemplified by the standardisation scripts.
2	Examiners must award **whole marks** (not half marks, or other fractions).
3	Examiners must award marks **positively**. This means that: • marks are not deducted for errors or omissions • marks are awarded for correct/valid answers as defined in the mark scheme and also for **other valid answers** which go beyond the scope of the syllabus and mark scheme referring to your team leader as appropriate • the quality of spelling, punctuation and grammar are judged **only** when the mark scheme indicates that these features are **specifically assessed** in the question. However, the meaning of all responses must be unambiguous.
4	Examiners must **consistently** apply the requirements of the mark scheme and any rules for what to do when candidates have not followed instructions correctly.
5	Examiners must use the **full range of marks** defined in the mark scheme, unless limited by the quality of the candidate responses seen.
6	Examiners must award marks according to the mark scheme and **must not** award marks with grade thresholds or grade descriptions in mind.

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Science-Specific Marking Principles

1	Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.
2	The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.
3	Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).
4	The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.
5	<u>'List rule' guidance</u> For questions that require ***n*** responses (e.g. State **two** reasons …): • The response should be read as continuous prose, even when numbered answer spaces are provided. • Any response marked *ignore* in the mark scheme should not count towards ***n***. • Incorrect responses should not be awarded credit but will still count towards ***n***. • Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should **not** be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response. • Non-contradictory responses after the first ***n*** responses may be ignored even if they include incorrect science.

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6 Calculation specific guidance

Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'.

For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values.

For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme.

Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.

7 Guidance for chemical equations

Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme.

State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

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Annotations guidance for centres

Examiners use a system of annotations as a shorthand for communicating their marking decisions to one another. Examiners are trained during the standardisation process on how and when to use annotations. The purpose of annotations is to inform the standardisation and monitoring processes and guide the supervising examiners when they are checking the work of examiners within their team. The meaning of annotations and how they are used is specific to each component and is understood by all examiners who mark the component.

We publish annotations in our mark schemes to help centres understand the annotations they may see on copies of scripts. Note that there may not be a direct correlation between the number of annotations on a script and the mark awarded. Similarly, the use of an annotation may not be an indication of the quality of the response.

The annotations listed below were available to examiners marking this component in this series.

Annotations

Annotation	Meaning
A	Information missing or insufficient for credit
AE	Arithmetic error
BOD	Benefit of the doubt given
CON	Contradiction in response, mark not awarded
	Incorrect point or mark not awarded
ECF	Error carried forward applied
I	Ignore the response
MO	Mandatory mark not awarded
POT	Power of ten error
SEEN	Blank page seen
SF	Error in number of significant figures

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Annotation	Meaning
TE	Transcription error
	Correct point or mark awarded
XP	Incorrect physics
SC	Special Case

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Abbreviations

/	Alternative and acceptable answers for the same marking point.
( )	Bracketed content indicates words which do not need to be explicitly seen to gain credit but which indicate the **context** for an answer. The context does not need to be seen but if a context is given that is incorrect then the mark should not be awarded.
___	Underlined content must be present in answer to award the mark. This means either the exact word or another word that has the same technical meaning.

Mark categories

B marks	These are independent marks, which do not depend on other marks. For a **B** mark to be awarded, the point to which it refers must be seen specifically in the candidate's answer.
M marks	These are mandatory marks upon which **A** marks later depend. For an **M** mark to be awarded, the point to which it refers must be seen specifically in the candidate's answer. If a candidate is not awarded an **M** mark, then the later **A** mark cannot be awarded either.
C marks	These are compensatory marks which can be awarded even if the points to which they refer are not written down by the candidate, providing subsequent working gives evidence that they must have known them. For example, if an equation carries a **C** mark and the candidate does not write down the actual equation but does correct working which shows the candidate knew the equation, then the **C** mark is awarded. If a correct answer is given to a numerical question, all of the preceding **C** marks are awarded automatically. It is only necessary to consider each of the **C** marks in turn when the numerical answer is not correct.
A marks	These are answer marks. They may depend on an **M** mark or allow a **C** mark to be awarded by implication.

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Question	Answer	Marks	Guidance
1(a)	amplitude and power identified as scalars	**B1**	Allow methods for marking the table other than a tick (e.g. ×) provided no ticks are seen
	force and momentum identified as vectors	**B1**	
1(b)(i)	area (under graph line)	**B1**	
1(b)(ii)	distance = 390 m	**A1**	
1(b)(iii)	$a = (v - u) / t$ or $a = \Delta v / (\Delta)t$ or gradient (of graph) e.g. = (0 – 12) / 5	**C1**	Allow $a = (u - v) / t$ or $a = (\Delta)v / (\Delta)t$ or a = –gradient Allow $a = (y_2 - y_1) / (x_2 - x_1)$ Allow a numerical expression equivalent to one of the three non-zero gradients on the graph, e.g. a = 12 / 5 or a = 12 / 10 or a = 10 / 40 No ECF of incorrect velocity from **1(b)(ii)**
	magnitude of acceleration = 2.4 m s^{-2}	**A1**	Allow –2.4 m s^{-2}
1(b)(iv)	$v^2 = 2 \times -0.25 \times -150$ = 75	**C1**	No mark for symbol formulae. For each method, the C1 mark is given for the full substitution of data into the formula. Allow $v^2 = 2 \times 0.25 \times 150$
	speed = 8.7 m s^{-1}	**A1**	Allow speed = –8.7 m s^{-1} for all methods
	Alternative methods: (–150 = ½ × –0.25 × t^2 so t = 34.6) v = –0.25 × 34.6 = –8.7	**(C1)**	Allow v = 0.25 × 34.6 Allow use of t = 35 to imply the C1 mark, but do not credit the A1 mark if leads to incorrect speed (8.8 m s^{-1})
	speed = 8.7 m s^{-1}	**(A1)**	

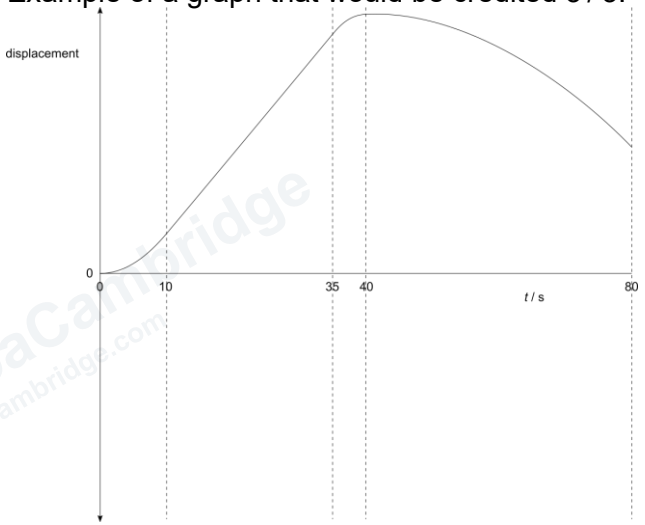
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Question	Answer	Marks	Guidance
1(b)(iv)	OR $-150 = ½ \times v \times 34.6$ $v = -8.7$	**(C1)**	Allow $150 = ½ \times v \times 34.6$ Allow use of $t = 35$ to imply the C1 mark, but do not credit the A1 mark if leads to incorrect speed (8.6 m s^{-1})
	speed = 8.7 m s^{-1}	**(A1)**	
	OR $-150 = (v \times 34.6) - (½ \times -0.25 \times 34.6^2)$ $v = -8.7$	**(C1)**	Allow $150 = (v \times 34.6) - (½ \times 0.25 \times 34.6^2)$
	speed = 8.7 m s^{-1}	**(A1)**	
	OR $t = 34.6 + 40$ $= 74.6$ At $t = 74.6$, $v = -8.7$ (from graph)	**(C1)**	No tolerance (for consistency with other methods)
	speed = 8.7 m s^{-1}	**(A1)**	

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Question	Answer	Marks	Guidance
1(b)(v)	A line from 0,0 to t = 80 s with increasing displacement then decreasing displacement that does not cross the t-axis	**B1**	For all marking points ignore line after t = 80 s Allow line to have discontinuous gradient at t = 10, 35, 40 Do not allow step change in displacement Allow graph to end up to 5 mm before t = 80 s Allow line to end at the t-axis Do not allow displacement that increases, decreases and then increases again
	A line with a maximum displacement at t = 40 s	**B1**	Allow centre of peak ±3 mm horizontally from t = 40 s
	A line with • increasing gradient from t = 0 to t = 10 s • constant positive gradient between t = 10 s and t = 35 s • decreasing gradient between t = 35 s and t = 40 s • decreasing gradient between t = 40 s and t = 80 s	**B1**	Allow line to have discontinuous gradient at t = 10, 35, 40 Do not allow step change in displacement If unclear whether a section of the line is straight or curved give BOD Allow graph to end up to 5 mm before t = 80 s Example of a graph that would be credited 3 / 3: 

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Question	Answer	Marks	Guidance
2(a)	resultant force (in any direction) is zero	**B1**	Allow 'net force is zero' Allow 'sum of forces is zero' or 'total force is zero' or '$\Sigma F = 0$' Allow 'sum of vertical forces is zero and sum of horizontal forces is zero'. Allow 'resultant force about any point is zero' Ignore 'forces balance' or 'forces cancel' or 'forces are equal' Ignore just 'no (external) forces act' or just 'force is zero' Ignore object stationary / constant velocity / zero acceleration etc
	resultant torque / moment (about any point) is zero	**B1**	Allow 'net moment is zero' Allow 'sum of moments is zero' or 'total moment is zero' Allow 'sum of CW moments = sum of ACW moments' Allow 'sum of CW and ACW moments is zero' Allow 'resultant moment at a point is zero' Allow 'torque' instead of 'moment' throughout Ignore 'moments balance' or 'moments cancel' Ignore 'no moments' or 'moment is zero' Not 'momentum' instead of 'moment'
2(b)(i)	one equation for forces on **either** block **or** plank: $2T\cos\theta = R + W_P$ or $2T\cos\theta + R = W_B$	**M1**	A full equation for the forces on either object must be seen Allow $0 = R + W_P - 2T\cos\theta$ Allow $0 = W_B - R - 2T\cos\theta$ Ignore just $R + W_P - 2T\cos\theta = W_B - R - 2T\cos\theta$ Allow $\sin(90 - \theta)$ for $\cos\theta$ Not $\sin\theta$ for $\cos\theta$
	both equations for forces on block **and** plank **and** ($2T\cos\theta =$) $R + W_P = W_B - R$	**A1**	Second mark for both separate equations seen and 'show that' answer given. Special case: $2T\sin\theta = R + W_P$ and $2T\sin\theta + R = W_B$ and ($2T\sin\theta =$) $R + W_P = W_B - R$ credit 1 / 2

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Question	Answer	Marks	Guidance
2(b)(ii)	$2R = 8.1 \times 9.81 - 4.5 \times 9.81$	**C1**	Allow $2R = 8.1g - 4.5g$ Allow $g = 10$
	$R = 18$ N	**A1**	
2(c)(i)	Hooke's (law)	**A1**	Allow phonetic misspellings of 'Hooke' e.g. Hook, Huk, Hookie
2(c)(ii)	When the tension is removed, the ropes have original length / extension is zero OR when the tension is removed object is not deformed	**A1**	Allow 'it' to mean ropes Allow 'force' / 'load' / 'T' / 'stress' / weight' for tension Ignore 'mass' for tension Allow 'unloaded' for tension is removed Allow 'tension is released' (BOD) but ignore 'object released' Allow 'unstretched' / 'initial' for original Allow 'shape and size' for length Allow 'dimensions' for length Ignore 'state' / 'form' for length

Question	Answer	Marks	Guidance
3(a)	(A wave which) transfers energy (in the direction of propagation)	**B1**	Allow 'it' to mean 'a wave' provided not clearly referring to something else Allow 'propagates / carries / moves' for transfers Ignore references to medium / matter Ignore references to transverse or longitudinal waves
3(b)(i)	$T = 0.33$ s	**A1**	
3(b)(ii)	$\lambda = 3.9 - 2.4$ $= 1.5$	**C1**	The C1 marks may be credited independently of each other
	$v = f\lambda$ or $v = \lambda / T$ $= 3 \times 1.5$ $= 1.5 / 0.33$	**C1**	Allow $c = f\lambda$ ECF T from **3(b)(i)**
	$= 4.5$ m s^{-1}	**A1**	

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Question	Answer	Marks	Guidance
3(c)(i)	$(\Delta)E = mg(\Delta)h$	**C1**	
	$= 0.44 \times 9.81 \times (0.15 \times 2)$	**C1**	Allow $g = 10$ as leads to same answer to 2SF Allow $0.44 \times 9.81 \times (-0.15 \times 2)$ $0.44 \times 9.81 \times 0.15$ implies 1st C1 mark but not 2nd C1 mark
	= 1.3 J	**A1**	Allow –1.3 J
3(c)(ii)	The path difference (from A and B) is not a whole number of wavelengths / (The waves at P are) not in phase	**B1**	Allow '(waves are) out of phase' / 'phase difference not zero' Not '(waves are) 180° out of phase / antiphase' Allow 'not (complete) constructive interference (at P)' (BOD)
	(The amplitude) decreases	**B1**	Allow 'it' to mean amplitude (unless it clearly refers to something else) Allow 'less' / 'smaller' for decreases

Question	Answer	Marks	Guidance
4(a)	$E_{(K)} = ½ mv^2$	**C1**	Allow any value of energy equated to $½ mv^2$
	$v^2 = (2 \times 1.3 \times 10^5 / 1.5 \times 10^4)$	**C1**	When candidates initially state the correct symbol equation, but then at the substitution stage write v instead of v^2, treat this as TE in the 2nd mark and allow ECF to the 3rd mark (max 2/3 marks). When candidates do not initially state a symbol equation, and then at the substitution stage write v instead of v^2, this implies an incorrect symbol equation and so treat as XP (no credit 0/3).
	$v = 4.2\ \text{m s}^{-1}$	**A1**	

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Question	Answer	Marks	Guidance
4(b)(i)	work (done) / time (taken)	**B1**	Allow 'work (done) per (unit) time' Allow 'ratio of work and time' Allow 'rate of (doing) work' Allow 'work over time' (BOD) Not 'work in unit time' Not 'rate of (doing) work per unit time' Not 'change of work (done) per unit time' Ignore 'work (done) by time' Ignore 'work (done) into time' Allow 'energy transferred / time' or 'energy converted / time' Allow 'change in energy / time' (BOD) Allow 'rate of energy transferred' Ignore just 'energy / time' Ignore symbols unless symbols are explained Ignore any units e.g. ignore 'work done per second' or 'Joules per time'

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Question	Answer	Marks	Guidance
4(b)(ii)	$W = Fs$ or $W = Fvt$ or $W = Pt$ useful work done = change in E_P + Fs	**C1**	Allow $(\Delta)E$ for W and allow 'd' or 'x' for 's' Allow 'WD' for work done (BOD) Allow any value of energy equated to Fs for 1st C1 mark
	$W = (1.1 \times 10^5 \times 52) - (13.0 \times 10^6 - 7.4 \times 10^6)$ $= 1.2 \times 10^5$	**C1**	Allow Fs or Fvt or Pt for W Treat not converting or incorrect conversion of both 13.0 MJ and 7.4 MJ to J as a single POT error ECF speed from **4(a)**
	$F = 1.2 \times 10^5 / (4.2 \times 52)$ = 550 N	**A1**	Allow 555 N (correct to 3SF because use of v=4.16 gives 554.7 N) but do not allow any other value that rounds to 560
	OR $P = Fv$ useful output power = power to raise weight + Fv	**(C1)**	Allow any value of power equated to Fv for this mark
	$P = 1.1 \times 10^5 - [(13.0 \times 10^6 - 7.4 \times 10^6) / 52]$ $= 2.3 \times 10^3$	**(C1)**	Allow Fv for P Treat not converting or incorrect conversion of both 13.0 MJ and 7.4 MJ to J as a single POT error ECF speed from **4(a)**
	$F = (2.3 \times 10^3) / 4.2$ = 550 N	**(A1)**	Allow 555 N (3SF) but Not 560 N
	OR $P = Fv$ driving force = component of weight parallel to slope + F	**(C1)**	

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Question	Answer	Marks	Guidance
4(b)(ii)	$1.1 \times 10^5 / 4.2 = F + 1.5 \times 10^4 \times 9.81 \sin\theta$ and $\sin\theta = (13.0 \times 10^6 - 7.4 \times 10^6) / (1.5 \times 10^4 \times 9.81 \times 4.2 \times 52)$ $1.1 \times 10^5 / 4.2 = F + (13.0 \times 10^6 - 7.4 \times 10^6) / (4.2 \times 52)$]	**(C1)**	ECF speed from **4(a)** Treat not converting or incorrect conversion of both 13.0 MJ and 7.4 MJ to J as a single POT error
	$F = 2.619 \times 10^4 - 2.564 \times 10^4$ $= 550$ N	**(A1)**	Allow 555 N (3SF) but not 560 N
4(b)(iii)	$P = IV$ $= 95 \times 1400$ $= 1.33 \times 10^5$	**C1**	For both methods C1 marks may be credited independently Allow $V = IR$ and either $P = V^2 / R$ or $P = I^2R$
	efficiency = useful output power / (total) input power (× 100)	**C1**	Ignore 'useful output/total input' but allow 'power' to be implied by substituted values of power Candidates may use useful output power = $(13.0 \times 10^6 - 7.4 \times 10^6) / 52 = 1.07692 \times 10^5$ so Allow $(1.08 \times 10^5 / (95 \times 1400))$ (× 100) to imply both C1 marks (but will lead to incorrect final answer 81%)
	$= [1.1 \times 10^5 / (95 \times 1400)]$ (× 100) = 83%	**A1**	
	OR $E = VIt$ $= 95 \times 1400 \times 52$ $= 6.92 \times 10^6$	**(C1)**	
	efficiency = useful output energy / (total) input energy (× 100)	**(C1)**	Ignore 'useful output/total input' but allow 'energy' to be implied by substituted values of energy Candidates may use useful output energy = $(13.0 \times 10^6 - 7.4 \times 10^6) = 5.6 \times 10^6$ so Allow $(5.6 \times 10^6 / (95 \times 1400 \times 52))$ (× 100) to imply both C1 marks (but will lead to incorrect final answer 81%)
	$= [(1.1 \times 10^5 \times 52) / (1400 \times 95 \times 52)]$ (× 100) = 83 %	**(A1)**	

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Question	Answer	Marks	Guidance
5(a)	$I_T = I_1 + I_2 + I_3$ either $(V / R_T) = (V / R_1) + (V / R_2) + (V / R_3)$ or $(I_T / V) = (I_1 / V) + (I_2 / V) + (I_3 / V)$	**B1**	Throughout allow I and R without subscript to mean I_T and R_T respectively, and allow V_T for V
	and $1 / R_T = 1 / R_1 + 1 / R_2 + 1 / R_3$	**B1**	If uses V_1, V_2, V_3 must also explicitly state that $(V =) V_1 = V_2 = V_3$. For the second mark, the associated working and final answer must both be shown A fully correct answer in terms of two rather than three resistors credit 1 / 2
5(b)(i)	$R_{(T)} = (1 / 2.8 + 1 / 2.8)^{-1}$ $= 1.4$ current in battery $= V / R_{(T)}$ $= 12 / 1.4$	**C1**	Allow any correct expression for the parallel combination e.g. $1 / R_{(T)} = 1 / 2.8 + 1 / 2.8$ or $R_{(T)} = (2.8 \times 2.8) / (2.8 + 2.8)$
	$= 8.6$ A	**A1**	
	OR $I = 12 / 2.8$ $= 4.3$	**(C1)**	
	current in battery $= 4.3 + 4.3$ $= 8.6$ A	**(A1)**	Special case: A final answer of 4.3 A without any supporting working credit 1 / 2

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Question	Answer	Marks	Guidance
5(b)(ii)	$v = I / Anq$ $= (8.6 / 2) / (6.1 \times 10^{-6} \times 8.5 \times 10^{28} \times 1.6 \times 10^{-19})$	**C1**	no mark for symbol formula as given in the formula sheet ECF current from **5(b)(i)** Allow $(12 / 1.4) / (2 \times 6.1 \times 10^{-6} \times 8.5 \times 10^{28} \times 1.6 \times 10^{-19})$ if treats as a single wire of double area
	$= 5.2 \times 10^{-5}$ m s^{-1}	**A1**	Special case: candidates that do not half their current from **5(b)(i)** can be credited 1 / 2 if working is shown e.g. $= 8.6 / (6.1 \times 10^{-6} \times 8.5 \times 10^{28} \times 1.6 \times 10^{-19})$ $= 1.0 \times 10^{-4}$ m s^{-1}
5(c)(i)	light-dependent resistor	**B1**	Allow LDR
5(c)(ii)	$I_{(X)} = 5 – 12 / 2.8$ $= 0.71$	**C1**	Allow $I_{(X)} = 5 – (8.6 / 2)$ (= 0.70) and allow ECF of current from **5(b)(i)**
	$R_{(\text{top branch})} = 12 / 0.71$ $= 16.8$	**C1**	
	$R_{(X)} = 16.8 – 2.8$ $= 14\ \Omega$	**A1**	
	OR $I_{(X)} = 5 – 12 / 2.8$ $= 0.71$	**(C1)**	Allow $I_{(X)} = 5 – (8.6 / 2)$ (= 0.70) and allow ECF of current from **5(b)(i)**
	$V_{(\text{resistance wire})} = 0.71 \times 2.8$ $= 2.0$ $V_{(X)} = 12 – 2.0$ $=10$	**(C1)**	
	$R_{(X)} = 10 / 0.71$ $= 14\ \Omega$	**(A1)**	

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Question	Answer	Marks	Guidance
5(c)(ii)	OR $R_{(T)}$ = 12 / 5 = 2.4	**(C1)**	
	1 / 2.4 = 1 / 2.8 + 1 / $R_{(top\ branch)}$ $R_{(top\ branch)}$ = 16.8	**(C1)**	Allow 1 / 2.4 = 1 / 2.8 + 1 / (2.8 + $R_{(X)}$)
	$R_{(X)}$ = 16.8 – 2.8 = 14 Ω	**(A1)**	
5(c)(iii)	(current in battery decreased, total resistance increased,) resistance (of X) increased	**M1**	Allow 'LDR' instead of X Allow *V*, *I*, *R* to mean potential difference, current and resistance respectively Allow ↓ for decreases and ↑ for increases (BOD) Ignore all comments about p.d. and emf
	light intensity (incident on X) decreased	**A1**	Allow '(environment is) darker / dimmer / less light / less bright' Allow 'temperature (of X/LDR) has decreased'

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Question	Answer	Marks	Guidance
6(a)	Most of the α-particles pass straight through (the atom) OR Most of the α-particles have (very) small deviations	**B1**	Allow 'majority' / 'nearly all' / 'large proportion' for 'most' Allow 'particles' / 'them' to mean α-particles (BOD) Allow 'no deviation' / 'undeviated' / 'deviate by small / insignificant / minimal angles' instead of small deviations Allow 'deflection' instead of deviation Allow 'gold / leaf' for atom Ignore 'some' instead of 'most' Ignore 'a large number' as this is not comparative Ignore just 'pass through' for 'pass straight through' Ignore 'only a (very) small proportion of alpha particles had (very) large deviations' as this does not necessarily mean the rest had little or no deviation
6(b)(i)	(p =) mv	**C1**	Allow any mass $\times$ any velocity e.g. 4(u) $\times$ 5.60 $\times$ 10^6 or 1.66 $\times$ 10^{-27} $\times$ 5.60 $\times$ 10^6 or 1.67 $\times$ 10^{-27} $\times$ 5.60 $\times$ 10^6
	= 4 $\times$ 1.66 $\times$ 10^{-27} $\times$ 5.60 $\times$ 10^6 = 3.72 $\times$ 10^{-20} (kg m s^{-1})	**A1**	As this is a 'show that' question the full working and answer must be seen Not 1.67 $\times$ 10^{-27} or 1.6 $\times$ 10^{-27} for A1 mark

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Question	Answer	Marks	Guidance
6(b)(ii)	$4(u) \times 5.60 \times 10^6 = 197\,(u)v - 4(u) \times 5.38 \times 10^6$ $3.72 \times 10^{-20} = (197 \times 1.66 \times 10^{-27} v) -$ $(4 \times 1.66 \times 10^{-27} \times 5.38 \times 10^6)$	**C1**	If u is present in equation it must be present in all terms Allow 1.67 instead of 1.66 as gives correct answer to 2SF
	speed = 2.2×10^5 m s^{-1}	**A1**	Special case: $4(u) \times 5.60 \times 10^6 = 197(u)v + (4(u) \times 5.38 \times 10^6)$ leading to 4500 m s^{-1} credit 1 / 2 working must be seen Alternative method if candidates (correctly) assume elastic collision and use speed of approach = speed of separation speed = $5.60 \times 10^6 - 5.38 \times 10^6$ (C1) $= 2.2 \times 10^5$ (A1)
6(b)(iii)	(absolute uncertainty =) $5 \times 10^4 + 5 \times 10^4$ $= 1 \times 10^5$	**C1**	Allow $2 \times 5 \times 10^4 = 1 \times 10^5$ Allow Δv to mean absolute uncertainty in v
	percentage uncertainty = $(1 \times 10^5 / (5.60 \times 10^6 + 5.38 \times 10^6)) \times 100$ = 0.9%	**A1**	Allow answers to 2 or more SF that round to 0.9
6(c)	3	**A1**	Allow '3 up' or '3u'
6(d)	One quark and one antiquark	**B1**	Allow 'quark and antiquark' (assume 1 of each) Allow $q\bar{q}$ BOD Ignore named quarks whether correct or incorrect e.g. 'up quark and antidown quark' as general composition required Ignore '2 quarks'

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CANDIDATE NAME

CENTRE NUMBER

CANDIDATE NUMBER

6466626864

PHYSICS **9702/23**

Paper 2 AS Level Structured Questions **May/June 2026**

1 hour 15 minutes

You must answer on the question paper.

No additional materials are needed.

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen. Do **not** use correction fluid or tape.
- Do **not** write on any bar codes.
- You may use a calculator.
- You should show all your working and use appropriate units.

INFORMATION

- The total mark for this paper is 60.
- The number of marks for each question or part question is shown in brackets [ ].

This document has **16** pages. Any blank pages are indicated.

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[Turn over

Data

acceleration of free fall	g	=	$9.81\ \text{m s}^{-2}$
speed of light in free space	c	=	$3.00 \times 10^{8}\ \text{m s}^{-1}$
elementary charge	e	=	$1.60 \times 10^{-19}\ \text{C}$
unified atomic mass unit	1 u	=	$1.66 \times 10^{-27}\ \text{kg}$
rest mass of proton	m_p	=	$1.67 \times 10^{-27}\ \text{kg}$
rest mass of electron	m_e	=	$9.11 \times 10^{-31}\ \text{kg}$
Avogadro constant	N_A	=	$6.02 \times 10^{23}\ \text{mol}^{-1}$
molar gas constant	R	=	$8.31\ \text{J K}^{-1}\ \text{mol}^{-1}$
Boltzmann constant	k	=	$1.38 \times 10^{-23}\ \text{J K}^{-1}$
gravitational constant	G	=	$6.67 \times 10^{-11}\ \text{N m}^2\ \text{kg}^{-2}$
permittivity of free space	ε_0	=	$8.85 \times 10^{-12}\ \text{F m}^{-1}$
	$\left(\frac{1}{4\pi\varepsilon_0}\right.$	=	$\left.8.99 \times 10^{9}\ \text{m F}^{-1}\right)$
Planck constant	h	=	$6.63 \times 10^{-34}\ \text{J s}$
Stefan–Boltzmann constant	σ	=	$5.67 \times 10^{-8}\ \text{W m}^{-2}\ \text{K}^{-4}$

Formulae

uniformly accelerated motion	s	=	$ut + \frac{1}{2}at^2$
	v^2	=	$u^2 + 2as$
hydrostatic pressure	Δp	=	$\rho g \Delta h$
upthrust	F	=	$\rho g V$
Doppler effect for sound waves	f_o	=	$\frac{f_\text{s} v}{v \pm v_\text{s}}$
electric current	I	=	$Anvq$
resistors in series	R	=	$R_1 + R_2 + \ldots$
resistors in parallel	$\frac{1}{R}$	=	$\frac{1}{R_1} + \frac{1}{R_2} + \ldots$

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1 **(a)** Define momentum.

.....

..... [1]

(b) A ball X of mass 0.15 kg moves along a horizontal frictionless surface. The ball collides elastically with ball Y that is initially stationary. Balls X and Y have equal diameters but different masses.

Figure 1.1 shows the variation with time of the velocity of ball X before, during and after the collision.

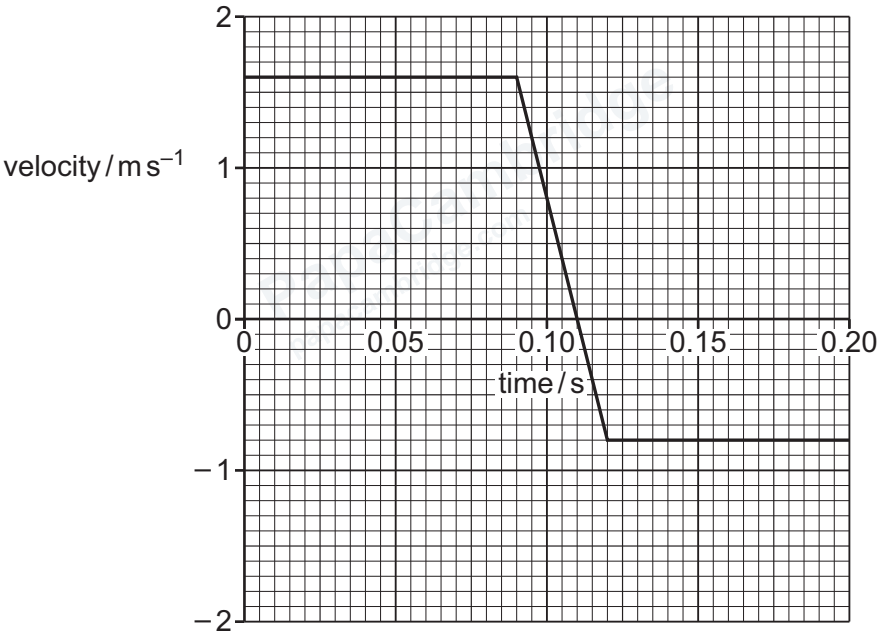


Figure 1.1

(i) Determine the magnitude of the acceleration of ball X during the collision.

acceleration = ... $m\,s^{-2}$ [2]

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(ii) Determine the magnitude of the change in momentum of ball X due to the collision.

change in momentum = ... kg m s^{-1} [2]

(iii) Determine the total distance travelled by ball X during the collision.

distance = ... m [2]

(c) (i) By considering relative speeds, explain why the velocity of ball Y after the collision must be +0.80 m s^{-1}.

.....

.....

..... [2]

(ii) On Figure 1.1, sketch the variation of the velocity of ball Y with time between time = 0 and time = 0.20 s. [2]

(iii) Calculate the mass of ball Y.

mass = ... kg [2]

[Total: 13]

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2 **(a)** Define the Young modulus of a material.

.....

..... [1]

(b) A long, thin thread of high-density polythene has a uniform cross-section. The thread is suspended from one end so that it hangs vertically.

The thread has a mass of 0.018 kg and a cross-sectional area of $2.6 \times 10^{-6}\,m^2$.
High-density polythene has a density of $960\,kg\,m^{-3}$ and a Young modulus of $8.0 \times 10^{8}\,Pa$.

(i) Show that the length of the thread is 7.2 m.

[2]

(ii) Show that the stress at the top of the thread due to its own weight is $6.8 \times 10^{4}\,Pa$.

[2]

(iii) Figure 2.1 shows the variation with distance from the top of the thread of the stress in the thread due to its own weight.

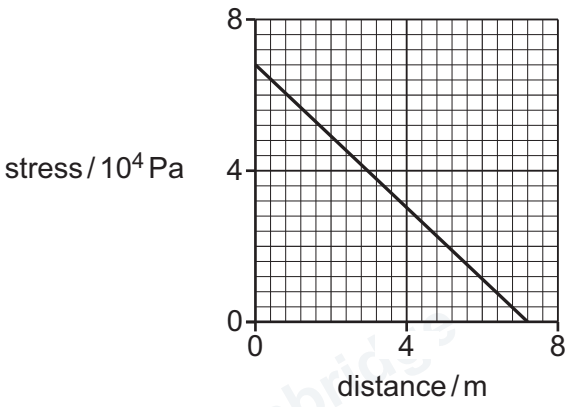


Figure 2.1

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Suggest an explanation for the fact that the variation shown in Figure 2.1 is linear.

.....

.....

.....

..... [2]

(c) The thread in **2(b)** is now laid horizontally on a frictionless bench. A tensile force is applied to the ends of the thread so that there is a uniform stress throughout the thread of 6.8×10^4 Pa.

(i) Calculate the extension of the thread.

extension = ... m [2]

(ii) Calculate the elastic potential energy of the thread.

elastic potential energy = ... J [2]

(d) Suggest, with a reason, whether the total extension of the vertically suspended thread in **2(b)** is less than, equal to or greater than the answer in **2(c)(i)**.

.....

.....

..... [1]

[Total: 12]

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3 **(a)** State the principle of superposition.

.....

.....

..... [2]

(b) Parallel rays of coherent light of wavelength 560 nm are incident normally in free space on two narrow slits as shown in Figure 3.1.

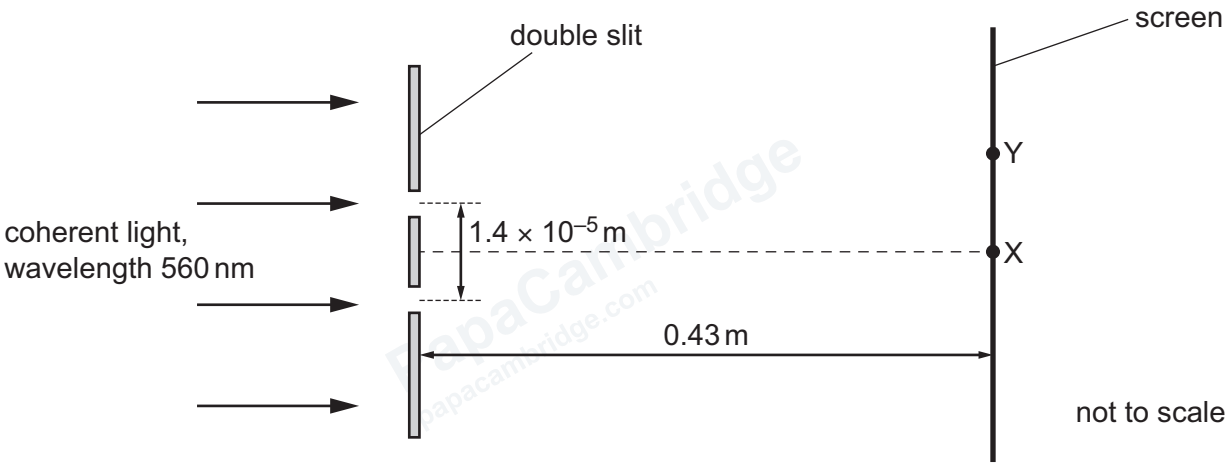


Figure 3.1

Light passing through the double slit reaches a screen that is parallel to the plane of the slits.
The distance between the centres of the slits is 1.4×10^{-5} m.
The distance between the slits and the screen is 0.43 m.

(i) State what is meant by coherent.

.....

..... [1]

(ii) State what happens to the light at the point where it passes through the slits.

.....

..... [1]

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(iii) At point X on the screen, there is a central bright fringe.

Explain how this bright fringe is formed.

.....

.....

.....

.....

..... [3]

(iv) At point Y on the screen, the dark fringe closest to point X is formed.

Calculate the distance XY.

distance = ... m [3]

(c) On the screen in Figure 3.1:

(i) draw a cross (×) labelled with the letter B at the position of another bright fringe [1]

(ii) draw a cross (×) labelled with the letter D at the position of another dark fringe. [1]

(d) Calculate the frequency of the light in **3(b)**.

frequency = ... Hz [2]

[Total: 14]

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4 A light-dependent resistor (LDR) is connected across the terminals of a cell that has internal resistance. A voltmeter is connected across the LDR.

(a) Draw a circuit diagram to show this arrangement.

[3]

(b) The cell has an electromotive force (e.m.f.) of 4.90 V.

At a particular light intensity, the LDR has a resistance of 230 Ω and a potential difference (p.d.) across it of 4.78 V.

(i) Identify **one** possible source of systematic error in measuring the p.d. across the LDR using a digital voltmeter.

.....

..... [1]

(ii) Calculate the power dissipated in the LDR.

power = ... W [2]

(iii) Determine the internal resistance of the cell.

internal resistance = ... Ω [2]

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(iv) The intensity of the light illuminating the LDR is gradually decreased.

By placing **one** tick (✓) in each row, complete Table 4.1 to show what happens to the quantities indicated as the light intensity decreases.

Table 4.1

	decreases	**unchanged**	**increases**
resistance of LDR			
current in cell			
p.d. across LDR			

[3]

[Total: 11]

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5 **(a)** State what is meant by a meson.

.....

..... [1]

(b) The elementary charge is *e*.

Explain, in terms of quark composition, how a baryon may have an overall charge of:

- –1*e* .....

  .....

- +2*e* .....

  .....

[3]

(c) The nuclide ${}^{18}_{8}\text{X}$ undergoes β^+ decay to produce nuclide ${}^{P}_{Q}\text{Y}$ and one other particle Z, as shown in the equation

$$ {}^{18}_{8}\text{X} \rightarrow {}^{P}_{Q}\text{Y} + {}^{R}_{S}\beta^+ + {}^{0}_{0}\text{Z}. $$

(i) State the values of *P*, *Q*, *R* and *S*.

P = ...

Q = ...

R = ...

S = ...

[2]

(ii) State the name of particle Z.

..... [1]

(iii) State the net change in quark composition as nuclide X turns into nuclide Y.

..... [1]

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(iv) Explain why the emission of particle Z results in the β^+ particles from the decays of different nuclei of ${}^{18}_{8}X$ having a continuous range of kinetic energies.

.....

.....

.....

..... [2]

[Total: 10]

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PHYSICS **9702/23**

Paper 2 AS Level Structured Questions **May/June 2026**

MARK SCHEME

Maximum Mark: 60

Published

This mark scheme contains the instructions and marking guidance that our examiners used to mark candidate responses for this component.

Before they start marking, examiners complete a standardisation process. They review a range of candidate responses and discuss the mark scheme in detail to agree which answers can and cannot be accepted. Examiners award marks where it is clear that candidate responses have met the requirements of the mark scheme. These requirements may vary between different components or different versions of the same component, depending on the exact requirements of the question and the candidate responses seen during the standardisation process.

Sometimes, our mark schemes may allow marks to be awarded to responses which contain elements that are factually incorrect or for content not covered by the syllabus. We do this in response to the demands of a specific question to support candidates and make sure that our assessments are fair. However, these allowances should not be used as a guide for teaching and learning.

We cannot discuss the mark scheme with schools, and we recommend that schools read the mark scheme together with the assessment material and the Principal Examiner Report for Teachers.

This document consists of **16** printed pages.

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[Turn over

General marking principles

All examiners must apply these marking principles when marking candidate responses.

1	Examiners must award marks for each response in line with: • the specific **content** defined in the mark scheme • the specific **skills** defined in the mark scheme or in the level descriptions • the **standard of response** required as exemplified by the standardisation scripts.
2	Examiners must award **whole marks** (not half marks, or other fractions).
3	Examiners must award marks **positively**. This means that: • marks are not deducted for errors or omissions • marks are awarded for correct/valid answers as defined in the mark scheme and also for **other valid answers** which go beyond the scope of the syllabus and mark scheme referring to your team leader as appropriate • the quality of spelling, punctuation and grammar are judged **only** when the mark scheme indicates that these features are **specifically assessed** in the question. However, the meaning of all responses must be unambiguous.
4	Examiners must **consistently** apply the requirements of the mark scheme and any rules for what to do when candidates have not followed instructions correctly.
5	Examiners must use the **full range of marks** defined in the mark scheme, unless limited by the quality of the candidate responses seen.
6	Examiners must award marks according to the mark scheme and **must not** award marks with grade thresholds or grade descriptions in mind.

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Science-Specific Marking Principles

1	Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.
2	The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.
3	Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).
4	The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.
5	‘List rule’ guidance For questions that require ***n*** responses (e.g. State **two** reasons …): • The response should be read as continuous prose, even when numbered answer spaces are provided. • Any response marked *ignore* in the mark scheme should not count towards ***n***. • Incorrect responses should not be awarded credit but will still count towards ***n***. • Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should **not** be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response. • Non-contradictory responses after the first ***n*** responses may be ignored even if they include incorrect science.

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6 Calculation specific guidance

Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'.

For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values.

For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme.

Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.

7 Guidance for chemical equations

Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme.

State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

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Annotations guidance for centres

Examiners use a system of annotations as a shorthand for communicating their marking decisions to one another. Examiners are trained during the standardisation process on how and when to use annotations. The purpose of annotations is to inform the standardisation and monitoring processes and guide the supervising examiners when they are checking the work of examiners within their team. The meaning of annotations and how they are used is specific to each component and is understood by all examiners who mark the component.

We publish annotations in our mark schemes to help centres understand the annotations they may see on copies of scripts. Note that there may not be a direct correlation between the number of annotations on a script and the mark awarded. Similarly, the use of an annotation may not be an indication of the quality of the response.

The annotations listed below were available to examiners marking this component in this series.

Annotations

Annotation	Meaning
A	Information missing or insufficient for credit
AE	Arithmetic error
BOD	Benefit of the doubt given
CON	Contradiction in response, mark not awarded
	Incorrect point or mark not awarded
ECF	Error carried forward applied
I	Ignore the response
MO	Mandatory mark not awarded
POT	Power of ten error
SEEN	Blank page seen
SF	Error in number of significant figures

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Annotation	Meaning
TE	Transcription error
✓	Correct point or mark awarded
XP	Incorrect physics
SC	Special Case

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Abbreviations

/	Alternative and acceptable answers for the same marking point.
( )	Bracketed content indicates words which do not need to be explicitly seen to gain credit but which indicate the **context** for an answer. The context does not need to be seen but if a context is given that is incorrect then the mark should not be awarded.
___	Underlined content must be present in answer to award the mark. This means either the exact word or another word that has the same technical meaning.

Mark categories

B marks	These are independent marks, which do not depend on other marks. For a **B** mark to be awarded, the point to which it refers must be seen specifically in the candidate's answer.
M marks	These are mandatory marks upon which **A** marks later depend. For an **M** mark to be awarded, the point to which it refers must be seen specifically in the candidate's answer. If a candidate is not awarded an **M** mark, then the later **A** mark cannot be awarded either.
C marks	These are compensatory marks which can be awarded even if the points to which they refer are not written down by the candidate, providing subsequent working gives evidence that they must have known them. For example, if an equation carries a **C** mark and the candidate does not write down the actual equation but does correct working which shows the candidate knew the equation, then the **C** mark is awarded. If a correct answer is given to a numerical question, all of the preceding **C** marks are awarded automatically. It is only necessary to consider each of the **C** marks in turn when the numerical answer is not correct.
A marks	These are answer marks. They may depend on an **M** mark or allow a **C** mark to be awarded by implication.

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Question	Answer	Marks	Guidance
1(a)	product of mass and velocity	**B1**	not speed for velocity
1(b)(i)	a = $\Delta v / (\Delta)t$ or $=(v - u) / (\Delta)t$ or = gradient of slope	**C1**	
	acceleration = $(-0.8 - 1.6) / (0.12 - 0.09)$ $= (-)80\ m\ s^{-2}$	**A1**	No tolerance.
1(b)(ii)	0.15×1.6 or 0.15×0.8	**C1**	No tolerance. ECF from **(b)(i)** for *v* read-offs
	change in momentum $= 0.15 \times (-0.8 - 1.6)$ $= (-)0.36\ kg\ m\ s^{-1}$	**A1**	Ignore sign
	OR $F = 0.15 \times (-)80$ $= (-)12$	**(C1)**	ECF *a* from **(b)(i)**
	Change in momentum $= (-)12 \times (0.12 - 0.09)$ $= (-)0.36\ kg\ m\ s^{-1}$	**(A1)**	
1(b)(iii)	$½ \times 1.60 \times 0.020$ or $1.6^2 / (2 \times 80)$ or $1.6 \times 0.02 - ½ \times 80 \times 0.02^2$ or $½ \times 80 \times 0.02^2$ =0.016	**C1**	ECF from **(b)(i)** for *v* read-offs ECF *a* from **(b)(i)**.
	OR $½ \times 0.80 \times 0.010$ or $0.8^2 / (2 \times 8)$ or $-0.8 \times 0.01 + ½ \times 80 \times 0.01^2$ or $½ \times 80 \times 0.01^2$ = 0.004	**(C1)**	
	distance $= 0.016 + 0.004$ $= 0.020$ m	**A1**	Allow 0.02 (1SF) treat finding displacement as XP e.g. $(-0.8)^2 - 1.6^2 / 2 \times 80 = 0.12$ m

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Question	Answer	Marks	Guidance
1(c)(i)	relative speed of approach = relative speed of separation	**B1**	Allow 'relative speed before collision = relative speed after collision' Allow $u_1 – u_2 = v_2 – v_1$ not relative <u>velocity</u> for speed, but allow relative velocity of approach = – relative velocity of separation
	(speed of separation = 1.6 m s^{-1} so) $v_Y – (–0.80) = 1.6$, so $v_Y = 0.80$ m s^{-1}	**B1**	Allow $v_Y + 0.80 = 1.6$ Allow 0.80 + 0.80 = 1.6 BOD Ignore $v_Y = 1.6 / 2$
1(c)(ii)	a straight horizontal line from (0, 0) to (0.09, 0) or a straight horizontal line from (0.12, 0.8) to (0.20, 0.8)	**B1**	tolerance ± ½ small square
	Both horizontal lines correct <u>and</u> straight diagonal line from (0.09, 0) to (0.12, 0.8)	**B1**	Allow horizontal line from (0, 0) to (0.09, 0) to be missing if diagonal line starts at (0.09, 0)
1(c)(iii)	$(0.15 \times 1.6) = (0.15 \times (–0.80)) + 0.80m$ $0.36 = 0.8m$	**C1**	ECF from **(b)(i)** for v read-offs
	OR $(½ \times 0.15 \times 1.6^2) = (½ \times 0.15 \times 0.80^2) + (½ \times m \times 0.80^2)$ $0.192 = 0.048 + 0.32m$	**(C1)**	Allow with the ½ factor missing from all three terms.
	OR $m = 0.15 \times 80 / [\,0.80 / (0.12 – 0.09)]$ $m = 12 / 26.7$	**(C1)**	
	mass = 0.45 kg	**A1**	

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Question	Answer	Marks	Guidance
2(a)	stress per unit strain	**B1**	Allow stress per (unit) strain Allow stress divided by strain Allow stress over strain Allow stress/strain Allow ratio of stress by strain (BOD) Allow ratio of stress and/to strain Ignore stress by strain or stress into strain Ignore symbols unless defined Ignore reference to 'object' or 'material' etc. Ignore any units Ignore 'stiffness'
2(b)(i)	$\rho = m / V$ $V = 0.018 / 960 = 1.875 \times 10^{-5}$	**C1**	Allow *d* or *D* for density
	(length =) $0.018 / (960 \times 2.6 \times 10^{-6}) = 7.2$ (m)	**A1**	Full substitution and answer needed.
2(b)(ii)	(σ =) F / A	**C1**	
	$= (0.018 \times 9.81) / (2.6 \times 10^{-6}) = 6.8 \times 10^{4}$ (Pa)	**A1**	Full substitution and answer needed. not $g = 10$ as leads to 6.9×10^{4}
2(b)(iii)	weight (of thread supported at a point) decreases as distance (from top of thread) increases	**B1**	Allow weight depends on distance / length Allow weight varies with distance / length Not weight is proportional to distance
	(uniform cross-section so) stress (at a point in the thread) is proportional to the weight (of thread below the point)	**B1**	Not stress is proportional to distance Allow 'force'/'mass' instead of 'weight' throughout
2(c)(i)	$\varepsilon = x / L$ or $E = \sigma / \varepsilon$ or $x = \sigma L / E$	**C1**	Allow *e* or ΔL for *x*
	extension $= 7.2 \times (6.8 \times 10^{4}) / (8.0 \times 10^{8})$ $= 6.1 \times 10^{-4}$ m	**A1**	

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Question	Answer	Marks	Guidance
2(c)(ii)	$E = ½Fx$ or $E = ½kx^2$ or $E = ½F^2 / k$ $= ½ \times 6.8 \times 10^4 \times 2.6 \times 10^{-6} \times 6.1 \times 10^{-4}$ or $= ½ \times 0.018 \times 9.81 \times 6.1 \times 10^{-4}$ or $= ½ \times (6.8 \times 10^4 \times 2.6 \times 10^{-6}/6.1 \times 10^{-4}) \times (6.1 \times 10^{-4})^2$ or $= ½ \times (0.018 \times 9.81/6.1 \times 10^{-4}) \times (6.1 \times 10^{-4})^2$ or $= ½ \times (6.8 \times 10^4 \times 2.6 \times 10^{-6})^2 / (6.8 \times 10^4 \times 2.6 \times 10^{-6} / 6.1 \times 10^{-4})$ or $= ½ \times (0.018 \times 9.81)^2 / (0.018 \times 9.81 / 6.1 \times 10^{-4})$	**C1**	Allow W or E_P or EPE for E ECF x from **(c)(i)**. $k = 6.8 \times 10^4 \times 2.6 \times 10^{-6}/6.1 \times 10^{-4} = 290$
	$= 5.4 \times 10^{-5}$ J	**A1**	
2(d)	(extension in **2(b)** is) less (than in **2(c)(i)**) because stress (in vertical thread) is less (than 6.8×10^4 Pa everywhere except the top)	**B1**	Allow ‘force’ for ‘stress’ ignore ‘weight’ / ‘mass’ for stress

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Question	Answer	Marks	Guidance
3(a)	(when two or more) waves meet / overlap (at a point)	**B1**	Allow 'waves intersect / merge' Ignore 'waves superpose / interfere / combine / interact' Allow answer given in the context of two waves forming a stationary wave. E.g. 'waves travelling in opposite directions'
	(resultant) displacement is sum of the individual displacements	**B1**	Allow 'resultant wave' or 'total displacement' for 'resultant displacement' Ignore just 'resultant displacement is the sum of the waves' Ignore 'sum of displacement is the sum of the individual displacements' but allow 'sum of resultant displacement is the sum of the individual displacements' Not 'amplitude' instead of 'displacement' ignore 'waves of same speed / frequency / wavelength' ignore the word 'algebraic' (as in 'algebraic sum') ignore 'coherent / identical waves'
3(b)(i)	constant phase difference	**B1**	Ignore 'same frequency/wavelength/speed' Ignore 'have the same phase difference' Ignore 'phase relationship' for 'phase difference' Ignore 'in phase'
3(b)(ii)	(the light) diffracts	**B1**	Ignore 'spreads'

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Question	Answer	Marks	Guidance
3(b)(iii)	waves (from each slit) meet / superpose (at the screen)	**B1**	Allow 'overlap / superimpose / interfere / combine' for 'superpose' Ignore 'interact / collide / coincide / intersect'
	waves have 0 path difference / 0 phase difference	**B1**	Allow 'in phase' Ignore 360°
	constructive interference occurs / (resultant wave has) maximum amplitude	**B1**	Ignore 'intensity is maximum' Ignore references to nodes and antinodes
3(b)(iv)	$\lambda = ax / D$	**C1**	
	$x = (560 \times 10^{-9} \times 0.43) / (1.4 \times 10^{-5})$ $= 0.017$	**C1**	
	distance $= 0.017 / 2$ $= 0.0086$ m	**A1**	
3(c)(i)	*either*: B marked above Y at same distance above Y as XY *or*: B marked below 0.43 m arrow, at a distance from X that is 2XY	**B1**	tolerance $\pm$ 3 mm
3(c)(ii)	D marked below X at distance below X equal to XY	**B1**	tolerance $\pm$ 3 mm
3(d)	$c = f\lambda$	**C1**	Allow v for c
	frequency $= (3.00 \times 10^{8}) / (560 \times 10^{-9})$ $= 5.4 \times 10^{14}$ Hz	**A1**	ECF POT error in the wavelength from **3(b)(iv)** if same POT error made

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Question	Answer	Marks	Guidance
4(a)	correct circuit symbol used for cell with internal resistance	**B1**	Allow just a cell symbol, without the resistor. Ignore presence or absence of dotted box around it. not a battery symbol
	correct symbol for LDR connected to terminals of cell	**B1**	
	correct symbol for voltmeter connected in parallel with LDR	**B1**	If symbol for LDR incorrect, allow voltmeter connected in parallel with component drawn, provided only one component drawn.
4(b)(i)	zero error / calibration error	**B1**	Allow 'contact resistance error in the connections'. Allow 'resistance of connecting wires' Allow 'not calibrated' BOD Apply List Rule
4(b)(ii)	$P = V^2 / R$ or $P = VI$ and $V = IR$ or $P = I^2R$ and $V = IR$ $= 4.78^2 / 230$ or $= 4.78 \times (4.78 / 230)$ or $= (4.78 / 230)^2 \times 230$	**C1**	
	$= 0.099$ W	**A1**	
4(b)(iii)	$E = V + Ir$ $r = (4.90 – 4.78) \times 230 / 4.78$ or $4.78 / 4.90 = 230 / (230 + r)$ or $(4.90 – 4.78) / 4.90 = r / (230 + r)$ or $(4.90 – 4.78) / 4.78 = r / 230$ or $4.90 / (4.78 / 230) = r + 230$	**C1**	Allow $E = I(R+ r)$
	internal resistance = 5.8 Ω	**A1**	

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Question	Answer	Marks	Guidance
4(b)(iv)	<table><tr><td></td><td>decreases</td><td>unchanged</td><td>increases</td></tr><tr><td>resistance of LDR</td><td></td><td></td><td>✓</td></tr><tr><td>current in cell</td><td>✓</td><td></td><td></td></tr><tr><td>p.d. across LDR</td><td></td><td></td><td>✓</td></tr></table>	**B1** **B1** **B1**	Allow symbols other than ticks if no ticks seen

Question	Answer	Marks	Guidance
5(a)	(a hadron consisting of) one quark and one antiquark	**B1**	Allow 'quark and antiquark' (assume 1 of each) Allow $q\bar{q}$ BOD Ignore named quarks whether correct or incorrect e.g. 'up quark and antidown quark' as general composition required. Ignore '2 quarks'
5(b)	quark charges have magnitudes of either ⅔(e) or ⅓(e)	**C1**	May be implied by the use of these magnitude of charge (and only these charges) in the reasoning below.
	–1e *: either*: –⅓(e), –⅓(e) and –⅓(e) *or*: –⅔(e), –⅔(e) and +⅓(e)	**A1**	Allow in terms of specific quarks provided magnitudes of charge are stated e.g. d has charge –1 / 3(e) so ddd. Not e^- for e
	+2e: +⅔(e), +⅔(e) and +⅔(e)	**A1**	Not e^- for e
5(c)(i)	P = 18 and R = 0	**A1**	
	Q = 7 and S = (+)1	**A1**	
5(c)(ii)	(electron) neutrino	**B1**	

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Question	Answer	Marks	Guidance
5(c)(iii)	up to down	**B1**	Allow u to d Allow uud to udd ignore any charges
5(c)(iv)	total energy emitted (in each decay) is constant	**B1**	
	one of : • particle Z takes some of the energy • the energy (emitted) is shared between Z and the β^+ particle • the way the total energy is distributed is different for each decay	**B1**	Ignore momentum arguments unless linked to KE Allow 'neutrino' for Z

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Cambridge International AS & A Level

CANDIDATE NAME

CENTRE NUMBER

CANDIDATE NUMBER

6218141275

PHYSICS **9702/24**

Paper 2 AS Level Structured Questions **May/June 2026**

1 hour 15 minutes

You must answer on the question paper.

No additional materials are needed.

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen. Do **not** use correction fluid or tape.
- Do **not** write on any bar codes.
- You may use a calculator.
- You should show all your working and use appropriate units.

INFORMATION

- The total mark for this paper is 60.
- The number of marks for each question or part question is shown in brackets [ ].

This document has **16** pages.

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Data

acceleration of free fall	g	=	$9.81\ \text{m s}^{-2}$
speed of light in free space	c	=	$3.00 \times 10^{8}\ \text{m s}^{-1}$
elementary charge	e	=	$1.60 \times 10^{-19}\ \text{C}$
unified atomic mass unit	$1\ \text{u}$	=	$1.66 \times 10^{-27}\ \text{kg}$
rest mass of proton	m_p	=	$1.67 \times 10^{-27}\ \text{kg}$
rest mass of electron	m_e	=	$9.11 \times 10^{-31}\ \text{kg}$
Avogadro constant	N_A	=	$6.02 \times 10^{23}\ \text{mol}^{-1}$
molar gas constant	R	=	$8.31\ \text{J K}^{-1}\ \text{mol}^{-1}$
Boltzmann constant	k	=	$1.38 \times 10^{-23}\ \text{J K}^{-1}$
gravitational constant	G	=	$6.67 \times 10^{-11}\ \text{N m}^2\ \text{kg}^{-2}$
permittivity of free space	ε_0	=	$8.85 \times 10^{-12}\ \text{F m}^{-1}$
	$\left(\frac{1}{4\pi\varepsilon_0}\right.$	=	$\left.8.99 \times 10^{9}\ \text{m F}^{-1}\right)$
Planck constant	h	=	$6.63 \times 10^{-34}\ \text{J s}$
Stefan–Boltzmann constant	σ	=	$5.67 \times 10^{-8}\ \text{W m}^{-2}\ \text{K}^{-4}$

Formulae

uniformly accelerated motion	s	=	$ut + \frac{1}{2}at^2$
	v^2	=	$u^2 + 2as$
hydrostatic pressure	Δp	=	$\rho g \Delta h$
upthrust	F	=	$\rho g V$
Doppler effect for sound waves	f_o	=	$\frac{f_\text{s} v}{v \pm v_\text{s}}$
electric current	I	=	$Anvq$
resistors in series	R	=	$R_1 + R_2 + \ldots$
resistors in parallel	$\frac{1}{R}$	=	$\frac{1}{R_1} + \frac{1}{R_2} + \ldots$

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1 **(a)** Underline **all** the vector quantities in the following list:

distance displacement speed velocity acceleration.

[1]

(b) A car of mass 1400 kg is initially travelling horizontally in a straight line at a constant speed of 15 m s^{-1}. The brakes are applied and the car decelerates uniformly to rest in a distance of 35 m.

Calculate:

(i) the initial kinetic energy of the car

kinetic energy = ... J [2]

(ii) the acceleration of the car

acceleration = ... m s^{-2} [2]

(iii) the resultant force on the decelerating car.

force = ... N [2]

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(c) The initial speed of the car in **1(b)** is now doubled to $30\,m\,s^{-1}$. The brakes of the car are then applied and the car comes to rest. The resultant force on the car remains the same as in **1(b)(iii)**.

Describe and explain the effect, if any, of doubling the initial speed on:

(i) the magnitude of the acceleration of the car

.....

.....

..... [1]

(ii) the distance travelled by the car in coming to rest.

.....

.....

.....

..... [2]

[Total: 10]

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2 **(a)** State Hooke's law.

.....

..... [1]

(b) In an experiment to determine the spring constant of a spring X, a student measures the original length of the spring. The student then suspends an object in equilibrium from the spring and measures the final length of the spring.

The spring does not extend beyond its limit of proportionality.

The measurements taken by the student are shown in Table 2.1.

Table 2.1

original length of spring X	(2.1 ± 0.1) cm
final length of spring X	(14.0 ± 0.1) cm
mass of suspended object	(0.300 ± 0.005) kg

Calculate:

(i) the extension, in cm, of spring X together with its absolute uncertainty

extension = (..... ± .....) cm [1]

(ii) the spring constant of spring X

spring constant = ... $N\,m^{-1}$ [3]

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(iii) the percentage uncertainty in the value of the spring constant calculated in **2(b)(ii)** due to the uncertainties in the measurements.

percentage uncertainty = ...% [2]

(c) An identical object to the one in **2(b)** is suspended in equilibrium from a spring Y that has a greater spring constant than spring X.

Spring Y does not extend beyond its limit of proportionality.

Describe and explain any difference in:

(i) the extension of the springs

.....

..... [1]

(ii) the elastic potential energy of the springs.

.....

.....

.....

..... [2]

[Total: 10]

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3 A stretched string is attached to two fixed points and a vibration generator as shown in Figure 3.1.

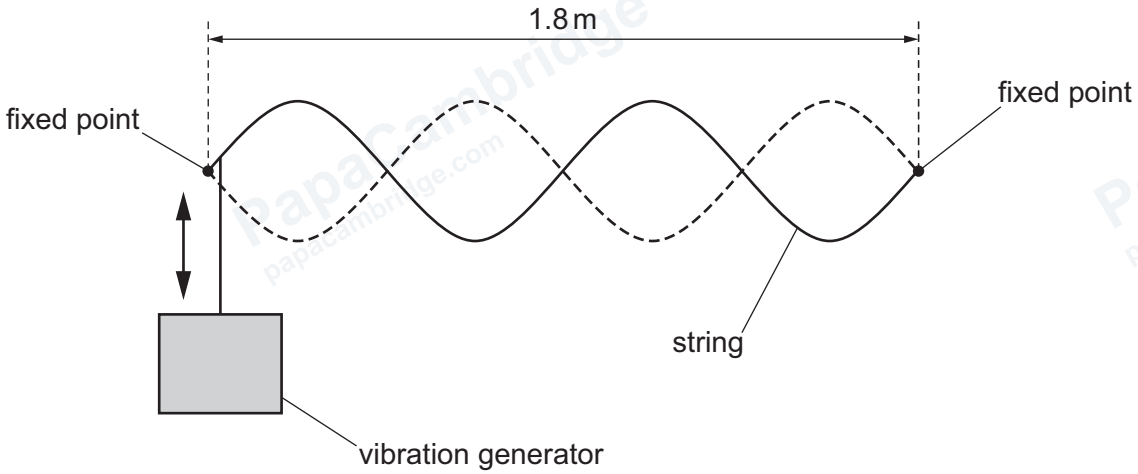


Figure 3.1

The vibration generator oscillates vertically and a stationary wave is formed on the string.

(a) Explain how the stationary wave is formed on the string.

.....

.....

.....

.....

..... [2]

(b) On Figure 3.1, draw **one** cross (×) at the position of a node. [1]

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(c) The distance between the fixed points is 1.8 m. The vibration generator oscillates at a frequency of 26 Hz.

Calculate the speed of progressive waves on the string.

speed = ... $m\,s^{-1}$ [3]

(d) The frequency of the vibration generator is now halved and a new stationary wave is formed on the string. The speed of progressive waves on the string is unchanged.

On Figure 3.2, draw the string at an instant when it has maximum displacement.

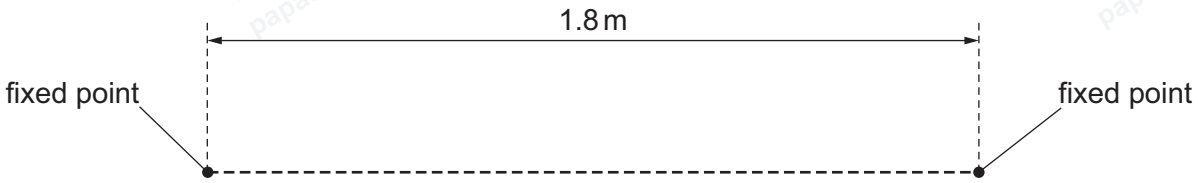


Figure 3.2

[2]

[Total: 8]

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4 An object is held at rest just below the surface of a liquid of uniform density as shown in Figure 4.1.

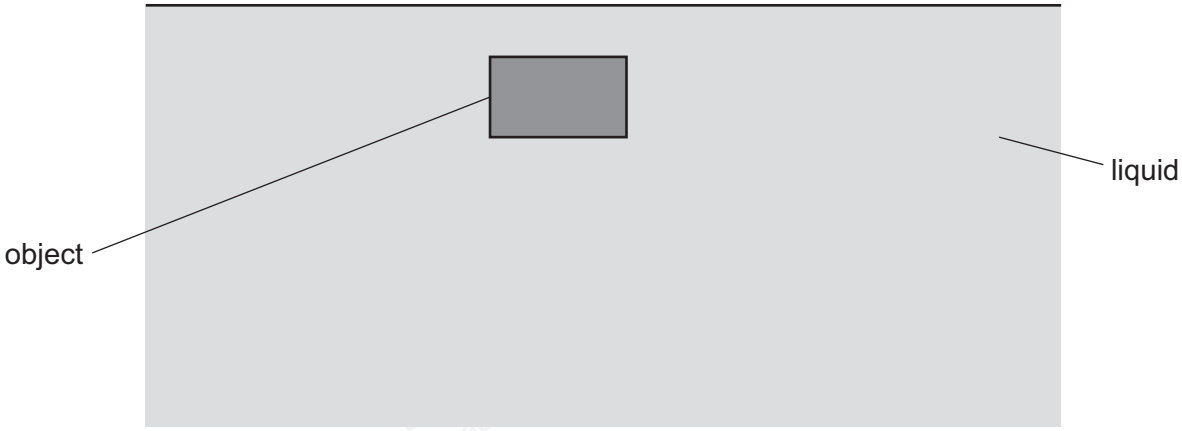


Figure 4.1

The object is released and accelerates downwards until it reaches terminal velocity.

The magnitude of the weight of the object is *W*.
The magnitude of the upthrust acting on the object is *U*.
The magnitude of the drag force acting on the object is *D*.

(a) By placing **one** tick (✓) in each row, complete Table 4.1 to show what happens to the magnitude of the three forces acting on the object as it accelerates.

Table 4.1

force	increases	stays the same	decreases
W			
D			
U			

[3]

(b) State an expression, in terms of *W*, *D* and *U*, that shows the relationship between the forces acting on the object when the object is at its terminal velocity.

[1]

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(c) The object has a mass of 0.070 kg and a volume of $1.0 \times 10^{-5}\,m^3$.

The density of the liquid is $1100\,kg\,m^{-3}$.

Calculate D when the object is at terminal velocity.

D = ... N [3]

[Total: 7]

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5 The I–V characteristic of an electrical component X is shown in Figure 5.1.

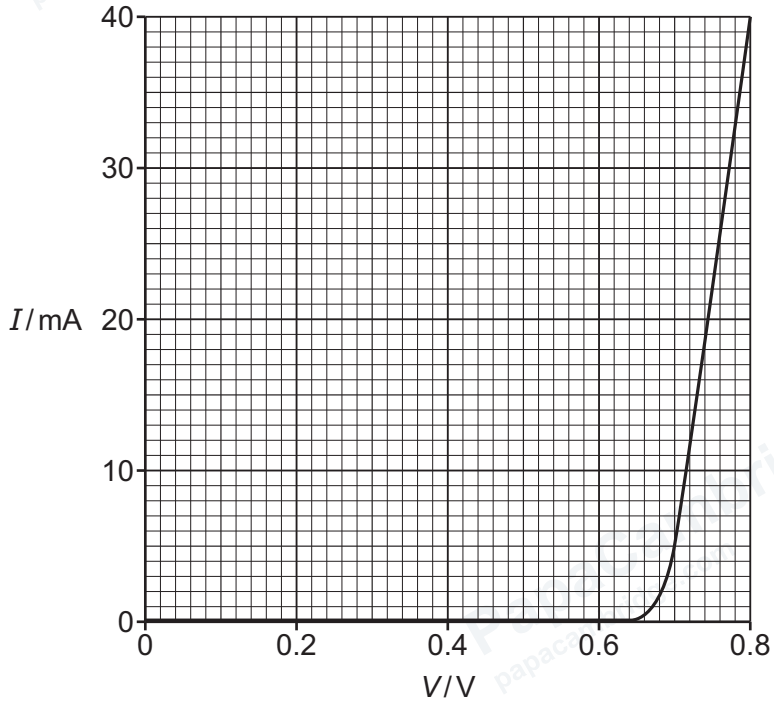


Figure 5.1

(a) State the name of electrical component X.

..... [1]

(b) A metallic conductor has a constant resistance of 40 Ω. On Figure 5.1, sketch the I–V characteristic of this conductor between $V = 0$ and $V = 0.80$ V. [2]

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(c) Component X is connected in a circuit with a resistor of resistance 20 Ω as shown in Figure 5.2.

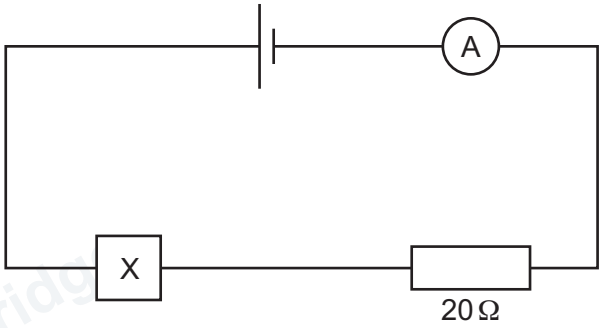


Figure 5.2

The current in the ammeter is 5.0 mA and the cell has negligible internal resistance.

Show that the electromotive force (e.m.f.) of the cell is 0.80 V.

[3]

(d) The terminals of the cell in **5(c)** are now reversed without making any other changes to the circuit.

State the reading on the ammeter.

reading on ammeter = ...A [1]

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(e) The resistor of resistance 20 Ω and component X are now connected in parallel to the cell in **5(c)** in the circuit shown in Figure 5.3.

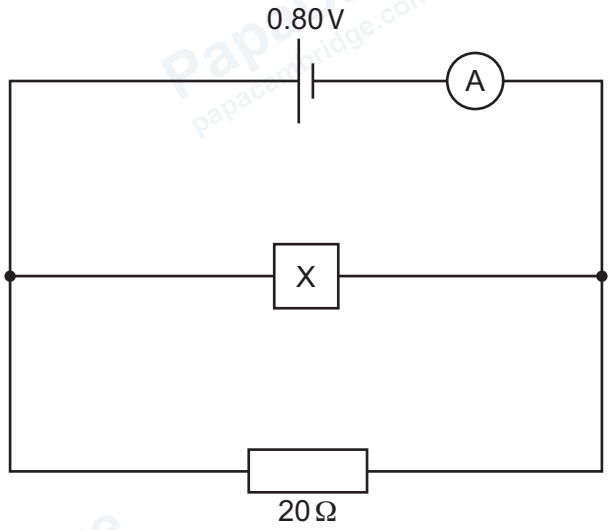


Figure 5.3

The current in component X is **not** negligible.

Use Figure 5.1 to determine the current in the ammeter.

current = ...A [3]

[Total: 10]

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6 A nichrome wire of cross-sectional area $5.7 \times 10^{-8}\,m^2$ is connected to a power supply. The potential difference (p.d.) across the wire is 230 V. The power dissipated in the wire is 1.0 kW.

(a) Show that the resistance of the wire is 53 Ω.

[2]

(b) The resistivity of nichrome is $1.3 \times 10^{-6}\,\Omega\,m$.

Calculate the length of the wire.

length = ... m [2]

(c) The average drift speed of the charge carriers (free electrons) in the nichrome wire is $5.4 \times 10^{-3}\,m\,s^{-1}$.

Calculate the number density of charge carriers of nichrome.

number density = ... m^{-3} [3]

[Total: 7]

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7 (a) (i) State what is meant by isotopes.

.....

.....

..... [2]

(ii) A thorium-232 nucleus ($^{232}_{90}$Th) undergoes radioactive decay to form another nucleus. A sequence of further decays occurs until a nucleus is formed that is another isotope of thorium. In each decay, either a single α-particle or a single β^- particle is emitted.

Determine the minimum numbers of α-particles and β^- particles that could be emitted during the sequence of decays to form the other isotope of thorium from thorium-232.

minimum number of α-particles = ...

minimum number of β^- particles = ...

[2]

(iii) Complete Figure 7.1 to show the nucleon number and proton number of the thorium isotope formed by the sequence of decays.

.....
..... Th

Figure 7.1

[2]

(b) (i) State the quark composition of a proton.

..... [1]

(ii) Describe the change to the quark composition of the proton when it undergoes β^+ decay.

.....

..... [1]

[Total: 8]

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CAMBRIDGE
International Education

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PHYSICS **9702/24**

Paper 2 AS Level Structured Questions **May/June 2026**

MARK SCHEME

Maximum Mark: 60

Published

This mark scheme contains the instructions and marking guidance that our examiners used to mark candidate responses for this component.

Before they start marking, examiners complete a standardisation process. They review a range of candidate responses and discuss the mark scheme in detail to agree which answers can and cannot be accepted. Examiners award marks where it is clear that candidate responses have met the requirements of the mark scheme. These requirements may vary between different components or different versions of the same component, depending on the exact requirements of the question and the candidate responses seen during the standardisation process.

Sometimes, our mark schemes may allow marks to be awarded to responses which contain elements that are factually incorrect or for content not covered by the syllabus. We do this in response to the demands of a specific question to support candidates and make sure that our assessments are fair. However, these allowances should not be used as a guide for teaching and learning.

We cannot discuss the mark scheme with schools, and we recommend that schools read the mark scheme together with the assessment material and the Principal Examiner Report for Teachers.

This document consists of **16** printed pages.

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General marking principles

All examiners must apply these marking principles when marking candidate responses.

1	Examiners must award marks for each response in line with: • the specific **content** defined in the mark scheme • the specific **skills** defined in the mark scheme or in the level descriptions • the **standard of response** required as exemplified by the standardisation scripts.
2	Examiners must award **whole marks** (not half marks, or other fractions).
3	Examiners must award marks **positively**. This means that: • marks are not deducted for errors or omissions • marks are awarded for correct/valid answers as defined in the mark scheme and also for **other valid answers** which go beyond the scope of the syllabus and mark scheme referring to your team leader as appropriate • the quality of spelling, punctuation and grammar are judged **only** when the mark scheme indicates that these features are **specifically assessed** in the question. However, the meaning of all responses must be unambiguous.
4	Examiners must **consistently** apply the requirements of the mark scheme and any rules for what to do when candidates have not followed instructions correctly.
5	Examiners must use the **full range of marks** defined in the mark scheme, unless limited by the quality of the candidate responses seen.
6	Examiners must award marks according to the mark scheme and **must not** award marks with grade thresholds or grade descriptions in mind.

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Science-Specific Marking Principles

1	Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.
2	The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.
3	Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).
4	The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.
5	<u>'List rule' guidance</u> For questions that require ***n*** responses (e.g. State **two** reasons …): • The response should be read as continuous prose, even when numbered answer spaces are provided. • Any response marked *ignore* in the mark scheme should not count towards ***n***. • Incorrect responses should not be awarded credit but will still count towards ***n***. • Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should **not** be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response. • Non-contradictory responses after the first ***n*** responses may be ignored even if they include incorrect science.

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6 Calculation specific guidance

Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'.

For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values.

For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme.

Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.

7 Guidance for chemical equations

Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme.

State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

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Annotations guidance for centres

Examiners use a system of annotations as a shorthand for communicating their marking decisions to one another. Examiners are trained during the standardisation process on how and when to use annotations. The purpose of annotations is to inform the standardisation and monitoring processes and guide the supervising examiners when they are checking the work of examiners within their team. The meaning of annotations and how they are used is specific to each component and is understood by all examiners who mark the component.

We publish annotations in our mark schemes to help centres understand the annotations they may see on copies of scripts. Note that there may not be a direct correlation between the number of annotations on a script and the mark awarded. Similarly, the use of an annotation may not be an indication of the quality of the response.

The annotations listed below were available to examiners marking this component in this series.

Annotations

Annotation	Meaning
A	Information missing or insufficient for credit
AE	Arithmetic error
BOD	Benefit of the doubt given
CON	Contradiction in response, mark not awarded
	Incorrect point or mark not awarded
ECF	Error carried forward applied
I	Ignore the response
MO	Mandatory mark not awarded
POT	Power of ten error
SEEN	Blank page seen
SF	Error in number of significant figures

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Annotation	Meaning
TE	Transcription error
✓	Correct point or mark awarded
XP	Incorrect physics
SC	Special Case

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Abbreviations

/	Alternative and acceptable answers for the same marking point.
( )	Bracketed content indicates words which do not need to be explicitly seen to gain credit but which indicate the **context** for an answer. The context does not need to be seen but if a context is given that is incorrect then the mark should not be awarded.
___	Underlined content must be present in answer to award the mark. This means either the exact word or another word that has the same technical meaning.

Mark categories

B marks	These are <u>independent</u> marks, which do not depend on other marks. For a **B** mark to be awarded, the point to which it refers must be seen specifically in the candidate's answer.
M marks	These are <u>mandatory</u> marks upon which **A** marks later depend. For an **M** mark to be awarded, the point to which it refers must be seen specifically in the candidate's answer. If a candidate is not awarded an **M** mark, then the later **A** mark cannot be awarded either.
C marks	These are <u>compensatory</u> marks which can be awarded even if the points to which they refer are not written down by the candidate, providing subsequent working gives evidence that they must have known them. For example, if an equation carries a **C** mark and the candidate does not write down the actual equation but does correct working which shows the candidate knew the equation, then the **C** mark is awarded. If a correct answer is given to a numerical question, all of the preceding **C** marks are awarded automatically. It is only necessary to consider each of the **C** marks in turn when the numerical answer is not correct.
A marks	These are <u>answer</u> marks. They may depend on an **M** mark or allow a **C** mark to be awarded by implication.

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Question	Answer	Marks	Guidance
1(a)	displacement, velocity and acceleration underlined only.	**A1**	Allow other methods of indicating such as circling, ticking, but not crossing through. All three need to be indicated by the same notation
1(b)(i)	$(E_{(K)})$ $= ½mv^2$ $= ½ \times 1400 \times 15^2$	**C1**	
	$= 1.6 \times 10^5$ J	**A1**	
1(b)(ii)	$0 = 15^2 + (2 \times a \times 35)$	**C1**	Allow any subject.
	$a = -3.2$ m s^{-2}	**A1**	
	OR $(35 = ½ \times 15 \times t$ so $t = 4.7$ so) $0 = 15 + (a \times 4.7)$ or $35 = -½ \times a \times 4.7^2$ or $35 = 15 \times 4.7 + (½ \times a \times 4.7^2)$	**(C1)**	
	$a = -3.2$ m s^{-2}	**(A1)**	

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Question	Answer	Marks	Guidance
1(b)(iii)	$F = ma$ $= 1400 \times 3.2$	**C1**	ecf a from **1(b)(ii)**
	$= 4500$ N	**A1**	Allow $\pm$ 4500 N
	OR $F = W / s$ $= 1.6 \times 10^5 / 35$	**(C1)**	Allow E / E_K / KE for W. Allow d or x for s
	$= 4600$ N	**(A1)**	Allow $\pm$ 4600 N ecf E_K from **1(b)(i)** **Allow** 4500 or 4600 for either method.
1(c)(i)	This will remain the same as ($F = ma$ and) the force and the mass are the same	**A1**	Allow no change / constant / equal for 'the same' allow F and m as symbols
1(c)(ii)	The (initial) kinetic energy is four times greater	**B1**	Allow kinetic energy is proportional to speed squared
	(So for the same applied force) the distance travelled will be four times greater	**B1**	Allow distance = 140 m
	OR acceleration is the same and using $s = (-)u^2 / 2a$	**(B1)**	Not $v^2 = u^2 + 2as$ unless $v = 0$ also given Allow acceleration the same and distance is proportional to initial speed squared
	The distance will be four times greater	**(B1)**	Allow distance = 140 m

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Question	Answer	Marks	Guidance
2(a)	Extension is proportional to (applied) force	**B1**	Ignore symbols unless defined. allow 'force is proportional to extension' Allow force $\propto$ extension allow load / tension for force ignore reference to the elastic limit ignore force = spring constant $\times$ extension
2(b)(i)	(11.9 ± 0.2) cm	**A1**	
2(b)(ii)	$k = F / x$	**C1**	Allow $e / \Delta x / \Delta L$ for x
	$= (0.300 \times 9.81) / 0.119$	**C1**	ecf of extension from **2b(i)** Do not allow $g = 10$
	$= 24.7$ N m^{-1}	**A1**	Allow 25
2(b)(iii)	0.2 / 11.9 or 0.005 / 0.300 % uncertainty $= [(0.2 / 11.9) + (0.005 / 0.300)] \times 100$	**C1**	ecf of extension from **2b(i)** 1.68% and 1.67% imply the C mark
	$= 3\%$	**A1**	
	OR $k_{max} = (0.305 \times 9.81) / 0.117$ or $k_{min} = (0.295 \times 9.81) / 0.121$ $= 25.6$ $= 23.9$	**(C1)**	Allow answers to two or more SF that are correct when rounded by the examiner to 1SF.
	% uncertainty $= [(25.6 - 23.9) / 2] / 24.7$ $= 3\%$	**(A1)**	allow $(25.6 - 24.7) / 24.7$ and $(24.7 - 23.9) / 24.7$ which lead to 4% and 3% respectively.
2(c)(i)	The extension will be less for Y as the spring constant is greater (and the applied force is the same).	**B1**	or vice-versa Assume refers to Y unless otherwise stated Allow syllabus symbols not defined, $x / k / F$ Allow the extension is less as extension is inversely proportional to spring constant Allow $x \downarrow$ and $k \uparrow$

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Question	Answer	Marks	Guidance
2(c)(ii)	(elastic potential energy) will be less for Y	**B1**	or vice-versa. Assume refers to Y unless stated otherwise Allow *EPE* / *E* / ΔE / E_P
	As the extension is less (for the same force) and $E_P = (\frac{1}{2}Fx) / (\frac{1}{2}mgx)$	**B1**	Allow syllabus symbols *x* / *k* / *F* not defined
	OR As the spring constant is greater (for the same force) and $E_P = \frac{1}{2}F^2 / k$	**(B1)**	Allow $k \uparrow$ Allow $E_P = \frac{1}{2}(mg)^2 / k$ ignore $E = \frac{1}{2}kx^2$

Question	Answer	Marks	Guidance
3(a)	wave reflects at fixed point	**B1**	Ignore rebound / bounces back
	incident and reflected waves superpose	**B1**	Allow 'wave from (vibration) generator / initial / original wave' for 'incident wave' Allow 'two progressive / travelling waves in opposite directions' for 'incident and reflected waves' (BOD), but ignore just 'different directions' Allow 'overlap / superimpose / interfere / combine' for 'superpose' Allow 'destructive / constructive interference' to imply 'superpose' Ignore 'meet / interact / collide / coincide / intersect' Ignore reference to phase difference Ignore reference to coherence
3(b)	cross vertically aligned with a node	**A1**	within one dash length

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Question	Answer	Marks	Guidance
3(c)	λ = 1.80 / 2 = 0.90	C1	The C marks are independent.
	v = $f\lambda$ = 26 × 0.90	C1	Allow use of $T = 1/f$ and $v = \lambda / T$
	= 23 m s^{-1}	A1	
3(d)	Line touches both fixed points and crosses the dotted line once in the middle	B1	Middle judged by eye below the 1.8 m label Allow line to be ±2 mm from fixed point ignore amplitude
	Line with sinusoidal shape from fixed point to fixed point	B1	Ignore a second correct solid or dashed line (candidate mirroring presentation in Fig 3.1). *Allow a sinusoid to include a 'half-loop' for the fundamental mode*

Question	Answer	Marks	Guidance
4(a)	<table><tr><th>force</th><th>increases</th><th>stays the same</th><th>decreases</th></tr><tr><td>*W*</td><td></td><td>✓</td><td></td></tr><tr><td>*D*</td><td>✓</td><td></td><td></td></tr><tr><td>*U*</td><td></td><td>✓</td><td></td></tr></table>	B1 B1 B1	Allow any other consistent ways of indicating choice provided no other marks seen. One tick on each line only.
4(b)	$W = D + U$	A1	Allow any subject.

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Question	Answer	Marks	Guidance
4(c)	W = 0.070 × 9.81 = 0.69	**C1**	treat use of g = 10 as TE unless already noted (**2(b)(ii)** ecf for second and third marks (gives 0.59 N)
	U = 1100 × 9.81 × 1.0 × 10^{-5} = 0.11	**C1**	C1 marks are independent
	D =0.69 – 0.11 = 0.58 N	**A1**	No ecf for incorrect expression in **4(b)**

Question	Answer	Marks	Guidance
5(a)	(semiconductor) diode	**A1**	Allow any other type of diode. E.g. LED
5(b)	straight line through the origin	**B1**	Tolerances ±½ small square for both marks
	a line passing through (0.8, 20)	**B1**	Ignore line continuing outside graph past 0.80 V Check for a candidate's graph drawn in space below Fig. 5.1. Allow one mark for straight line from a labelled origin (0,0)

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Question	Answer	Marks	Guidance
5(c)	(p.d. across component X =) 0.70 (from graph)	**C1**	Allow 0.7
	(p.d. across 20 Ω resistor = 0.005 × 20) = 0.10	**C1**	Allow 0.1 C1 marks are independent
	(e.m.f. =) 0.70 + (0.005 × 20) = 0.80 (V)	**A1**	Allow 0.8 Need 0.005 or 5×10^{-3} not 5 mA
	OR (resistance of diode) = 140	**(C1)**	
	total resistance = 20 + 140	**(C1)**	
	(e.m.f. =) $(0.7 / 5 \times 10^{-3} + 20) \times 5 \times 10^{-3} = 0.80$ V	**(A1)**	Allow 0.8
5(d)	0 A	**A1**	Allow 'approximately zero' but not '-'
5(e)	(current in resistor =) 0.80 / 20 = 0.040	**C1**	C1 marks are independent
	(current in component X) = 0.040 (A) / 40 (mA) (from graph)	**C1**	Allow 0.04 A
	current in ammeter = 0.080 A	**A1**	Allow 0.08 A
	or resistance of diode is $0.8 / 40 \times 10^{-3} = 20\ \Omega$	**(C1)**	
	total resistance in parallel is 10 Ω	**(C1)**	
	current = 0.8 / 10 = 0.080 A	**(A1)**	Allow 0.08 A

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Question	Answer	Marks	Guidance
6(a)	$P = V^2 / R$	C1	Allow any subject. Must see full working and answer in all cases. Not 1 kW need $1000 / 10^3$
	so $R = (V^2 / P) = 230^2 / 1000 = 53\,\Omega$	A1	
	Or $P = IV$ **and** $V = IR$	(C1)	
	$R = 230 / (1000 / 230) = 53\,\Omega$	(A1)	
	Or $P = IV$ **and** $P = I^2R$	(C1)	
	$R = 1000 / (1000 / 230)^2 = 53\,\Omega$	(A1)	
6(b)	$R = \rho L / A$ $L = (53 \times 5.7 \times 10^{-8}) / 1.3 \times 10^{-6}$	C1	Allow any subject.
	$= 2.3$ m	A1	
6(c)	$n = I / qAv$ $I = 1000 / 230$ *or* $230 / 53$ $= 4.35$	C1	Allow current of 4.3 to imply the C mark
	$n = 4.35 / (1.6 \times 10^{-19} \times 5.7 \times 10^{-8} \times 5.4 \times 10^{-3})$	C1	
	$= 8.8 \times 10^{28}\,m^{-3}$	A1	

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Question	Answer	Marks	Guidance
7(a)(i)	(different forms of the same) element / atom / nuclei with the same proton number	**B1**	Allow atomic number Ignore particle Ignore reference to electron number Allow mass number Allow number to be included once only
	and different nucleon number / number of neutrons	**B1**	
7(a)(ii)	minimum number of α particles = 1	**A1**	
	minimum number of β^- particles = 2	**A1**	
7(a)(iii)	$^{228}_{90}\mathrm{Th}$	**A1** **A1**	No ecf from **7(a)(ii)**
7(b)(i)	up, up, down	**B1**	Allow u, u, d Allow any combination of words and symbols Allow arrows Ignore charges
7(b)(ii)	an up quark changes to a down quark	**B1**	Allow u → d Allow uud → udd Allow it changes to udd if **7b(i)** is correct Ignore neutrino / antineutrino / beta-plus / beta-minus Allow arrows.

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